Debt Capacity and the Anatomy of Sovereign Crises*

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November 2026 (`63dcbaf`)

Abstract

Sovereign defaults are rarely complete: governments withhold part of their debt service, the missed payments accumulate as arrears, and borrowing continues. We study the partial default model of Arellano, Mateos-Planas, and Ríos-Rull (2023) in continuous time and find that it changes the account of how sovereign debt crises begin, unfold, and end. Withholding debt service raises funds at a price fixed by the recovery on arrears, so a sovereign's debt capacity is set by the cost of defaulting, in closed form, and not by its patience, its risk aversion, or how much it values market access. Default episodes are triggered by shocks that overwhelm a precautionary buffer, and they begin with a jump whose size the default costs pin down. Once an episode has begun, it is a financing regime: the sovereign keeps borrowing, and the regime ends when output recovers rather than when the debt is repaid. Pessimism alone can bring on a slow-moving default, without a run. The model is tractable because the sovereign never loses market access: given the bond price, the default decision is static. Calibrated to the same targets, it fits the data at least as well as the discrete-time original.

Keywords: sovereign debt, partial default, arrears, debt capacity, self-fulfilling crises
JEL classification: E44, F34, F41, H63

*We are thankful to Narayana Kocherlakota, Galo Nuño, Pontus Rendahl, and seminar participants at Ohio State and the University of Rochester for many insightful questions and comments. Claude (Fable 5.1, Anthropic) was employed extensively in the production and revision of the code and of the text; the authors verified every result and take full responsibility for the content. An earlier version of this paper circulated under the title "Sovereign Partial Default in Continuous Time." *Contact information:* mihalache@gmail.com (corresponding author)

1 Introduction

Sovereign defaults are rarely complete. Governments in distress typically continue to service some of their obligations while suspending payments on others, the missed payments accumulate as arrears that add to the stock of debt, and countries often emerge from a default episode owing more than when it began. Arellano, Mateos-Planas, and Ríos-Rull (2023) document these facts and build a theory of *partial default* around them. In their theory, a sovereign chooses, period by period, what share of its debt service to withhold, at an output cost that rises with the share withheld. It never loses access to the market, where the price of its bonds reflects the default intensity lenders expect. Their framework replaces the binary default decision of the standard model, introduced by Eaton and Gersovitz (1981) and quantified by Arellano (2008), with a continuous one, and exclusion from financial markets with continued access. In their data, emerging economies are in partial default about a third of the time and withhold about forty percent of the debt service when they are. Their calibrated model reproduces the frequency, the depth, and the dynamics of these episodes. The framework has since been applied, for instance to the debt crises that the COVID-19 pandemic set off in emerging markets (Arellano, Bai, and Mihalache 2024). We regard partial default as the right description of sovereign distress and the natural baseline for the quantitative theory of sovereign debt.

This paper asks what the partial default model teaches about the economics of sovereign crises, and finds four lessons. The limit on a sovereign's debt is set by the cost of defaulting, not by the value the sovereign attaches to market access. Defaults begin with a jump, whose size the cost structure fixes, and they are triggered by shocks that overwhelm a precautionary buffer rather than by borrowing into default. Once a crisis episode has begun, the economy is in a financing regime in which the sovereign keeps borrowing, and the episode ends when output recovers rather than when the debt is repaid. Pessimism alone can produce a slow crisis, without a run or loss of market access. We draw our lessons from three versions of the model. The first has a constant endowment, where a sharp characterization in closed form is possible. The second puts the endowment on a Markov chain, which adds precautionary behavior. The third has a diffusion calibrated to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023). We write the model in continuous time, where the default intensity is a flow decision and arrears accrue continuously. We find it especially tractable: the sovereign never loses market access, so there is no stopping problem, and given the bond price the default decision is static. Tractability does not come at the cost of quantitative performance. The calibrated diffusion model fits the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023) at least as well as their discrete-time model, so the lessons we draw from the closed forms carry over to a model that matches the data.

The first lesson concerns the determinants of the sovereign's debt limit. In the binary model, the limit is the outcome of the whole dynamic problem. It is the debt at which the value of staying in the market no longer covers repayment, and it depends on the sovereign's patience, on its risk aversion, and on the punishment that follows a default. Under partial default, the limit reflects a static comparison. Withholding a unit of debt service raises consumption by one minus

the market value of the arrears it creates, at the cost of lost output (Lemma 1). Withholding is a source of funds whose price is fixed by the recovery on arrears and not by the market, with the haircut falling on existing creditors. Our analysis makes exact the description by Arellano, Mateos-Planas, and Ríos-Rull (2023) of partial default as a flexible way to raise funds. It also explains why arrears are attractive precisely when market prices are low: the terms of this source of funds do not deteriorate with the sovereign's creditworthiness. Given debt, the endowment, and the price, the share withheld maximizes a resource function that is concave away from zero default and depends neither on the value function nor on the direction in which debt is moving (Proposition 1). So the regions of no, partial, and full default are explicit in the state and the price (Proposition 2 and Figure 2). In the deterministic model the impatient sovereign borrows at the risk-free bond price up to the largest debt it is willing to service in full and stops there, because one more unit would start a default that the price would charge for. Its debt capacity is the no-default threshold at the risk-free price (Theorem 1). This capacity is a closed-form function of the default costs, of the recovery on arrears, and of the maturity of the debt, independent of the sovereign's impatience, its risk aversion, and the borrowing limit. Preferences decide whether a sovereign goes to its limit, but the cost of defaulting decides where the limit is.[1]

The second lesson is about how defaults begin. A sovereign never defaults a little. Although the intensity is a continuous choice, a fixed cost makes entry into default lumpy: the first default is a jump whose size the endowment and the cost parameters alone determine (Proposition 2). At the calibrated parameters it is half of the debt service. The fixed cost makes full repayment possible, since without it the first unit withheld has zero marginal cost and the sovereign defaults a little at every positive debt. A steep variable cost then makes the jump large. Explaining what triggers the jump requires uncertainty, which adds a precautionary motive and a kink in the price schedule. In a state that can switch to a worse one, the sovereign can hold its debt below its threshold when the loss from arriving in a worse state with more debt is large enough. Its buffer is then set by an Euler equation that balances the impatience gain from borrowing against that loss (Proposition 5). Otherwise, and always in the state with the lowest consumption, the sovereign borrows to its edge. A second force works through prices. Beyond the threshold of a worse state that one switch can reach, a switch triggers an immediate default, so the price schedule of the better state has a kink there. The kink can hold the better state at or below the worse state's threshold, so that a switch does not land the sovereign in default. Default occurs when a shock moves the threshold below the sovereign's current debt, further than its buffer guards against. Such a shock is a rare jump to a much worse state in the chain model, or the continuum of shocks of the diffusion model, where the threshold moves continuously with the endowment and the sovereign is caught on the wrong side of it. How often a sovereign defaults depends on how the tail of its shocks compares with the buffer it holds, not on a willingness to

1. Amador and Phelan (2021, 2023) develop the reputational theory of sovereign default. The latter does so in a setting with partial default, the closest theoretical treatment of a partial default decision we know of. Tsoy (2026) adds reputation to the model of DeMarzo, He, and Tourre (2023). There, a government builds a reputation for debt discipline by reducing its debt after large negative shocks, not by repaying on time.

borrow into default.

The third lesson speaks to the nature of default episodes. In the binary model a sovereign past its limit defaults, and the adjustment is a jump into exclusion or renegotiation. Under partial default the adjustment is gradual, and in the deterministic model we characterize it exactly. Above its capacity the sovereign is in partial default, the price is below par, and it repays, but at a speed that vanishes like the cube root of the excess debt, while the price leaves par with a vertical tangent (Theorem 2). A sizable excess takes a few years to work off, but even a small excess is visibly priced at once. So a debt overhang is worked off without exclusion or restructuring, and the spread reacts to it before the debt itself moves. From below, the sovereign approaches its capacity along a square-root law and reaches it in finite time (Lemma 2). Bornstein (2020) identifies a painful deleveraging effect specific to settings in which the stock of debt moves slowly. In the partial default model the law of motion of the adjustment is in closed form. With shocks the episode acquires a feature the deterministic model cannot show. Once a bad shock has moved the threshold below the current debt, the fixed cost is being paid regardless and no longer restrains borrowing. The sovereign's stationary debt is then set by the dilution of its own price, the force that Hatchondo, Martinez, and Sosa-Padilla (2016) show drives default risk with long-term debt. The stationary debt is higher than in good times, so the sovereign in partial default borrows. In the calibrated diffusion model, after a fall in output of about one standard deviation the default intensity jumps and stays elevated for a decade, and debt keeps rising for six years (Figure 1). New issues add several times as much to the debt as arrears. Continued market access is the larger cushion against the shock and arrears the smaller, which puts a mechanism and an accounting behind the facts that debt rises during episodes and that countries exit them owing more. The episode ends when output recovers and the threshold rises back above the debt, not when the debt is repaid: default is lumpy at entry and sticky thereafter. Hurtado, Nuño, and Thomas (2023) study the closest mechanism in continuous time, a sovereign that lowers the real value of its local-currency debt through inflation. Inflation acts on the whole stock, whereas partial default acts on the flow of debt service and raises the stock through arrears.

The fourth lesson is that pessimism alone can produce a crisis, and that it unfolds slowly. If lenders expect the sovereign to settle at a debt typically above its capacity and to withhold a constant share of its debt service there forever, they price the bond accordingly. At the lower price withholding is cheaper, so the sovereign starts defaulting at debt levels at which it would otherwise have repaid. It can also borrow past its old capacity, because the default is priced and the funds it raises cost it only the risk-free rate. With curvature it drifts into a permanent partial default over years, through new issues and arrears, and the arrears validate the price. The crisis is slow because debt is a stock that moves at the rate of the deficit and of the arrears. This is the slow-moving crisis of Lorenzoni and Werning (2019), with arrears in the place of rollover. The idea that an expected repudiation raises the interest rate, and that the higher rate justifies the repudiation, goes back to Calvo (1988). Unlike the rollover crises of Cole and Kehoe (2000), there is no run and no loss of market access. The crisis is closer to the self-fulfilling dilution of

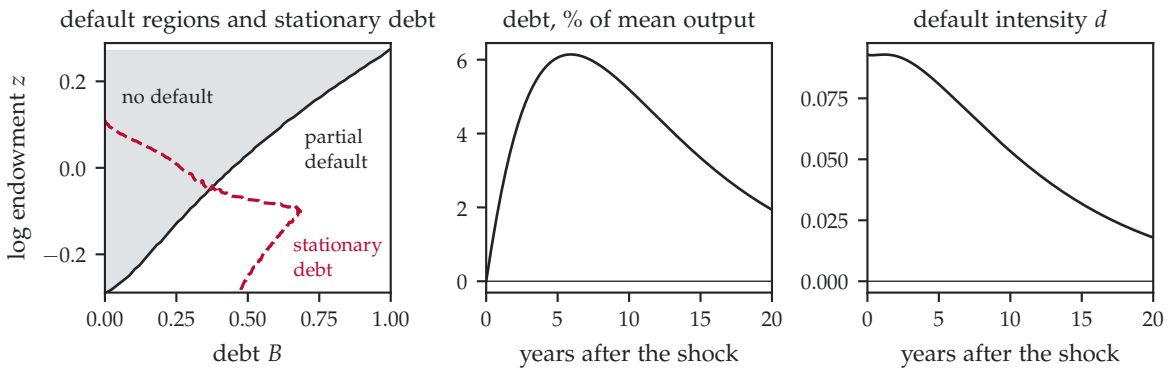


Figure 1: The anatomy of a crisis in the calibrated model. Left: the no-default region (shaded), bounded by the default threshold, and the stationary debt, the locus where the drift of debt vanishes. In good times the stationary debt lies inside the no-default region, and in bad times inside the partial-default region. Center and right: the responses of debt and of the default intensity to a fall of 0.1 in the log endowment at date zero, from the ergodic distribution (Section 4.3).

Aguiar and Amador (2020) and to Aguiar et al. (2022), in which the government keeps issuing at depressed prices, here with a continuous default decision. The difference from a rollover crisis is where the sovereign's willingness to participate enters. A sovereign in a rollover crisis must roll over, whereas ours can stay out of the default region. So a pessimistic expectation is self-fulfilling only if borrowing into default at a discount beats staying out. The candidates form a continuum indexed by the intensity of the permanent default, and whether one is an equilibrium depends on the whole path from zero debt: the sovereign must be willing to borrow into default all the way. For a risk-neutral sovereign we nearly solve it: a self-fulfilling partial default can be reached from zero debt only if the gain from front-loading the borrowed funds covers the output cost of the cheapest default. It is an equilibrium when, in addition, the steady state is stable and the price schedule continues regularly above it. When the default cost is not too convex, the threshold is that the impatience wedge exceed the spread implied by the first default (Proposition 4). Risk aversion works against the crisis, because a sovereign that values smooth consumption gains less from front-loading. In a numerical example a relative risk aversion of two roughly triples the impatience that a pessimistic expectation needs to be self-fulfilling, and the requirement falls toward its risk-neutral value as the curvature falls. Uncertainty works against the crisis too. With shocks, the range of impatience that supports a self-fulfilling partial default narrows as volatility rises, and it closes at moderate volatility (Appendix B.5, Figure 14). A lower bound on the price of new issues rules such equilibria out whenever the spread implied by the first default exceeds the spread at the bound (Proposition 3). At the calibrated parameters it does, narrowly, and the impatience wedge is in any case far below the threshold.

The quantitative section calibrates the diffusion model to the moments that Arellano, Mateos-Planas, and Ríos-Rull (2023) target, with their nine free parameters and their weights. The model reproduces the frequency and the depth of partial default, the debt service and debt due ratios, and the output process. The thresholds and the debt capacity of the theory are visible in the

policies: the no-default region is a wedge that rises with the endowment, and the default intensity jumps to the first default at its edge. The ergodic distribution concentrates on the locus where debt is stationary, and the shape of that locus, the stationary debt of Figure 1, reflects the first three lessons at once. In good times the stationary debt lies below capacity, held there by precaution. As output falls the stationary debt rises, since the sovereign is in partial default and borrows against its diluted price. In the worst states, more than one standard deviation below the mean, the price of new issues reaches the bound and the stationary debt falls back. Spreads are low most of the time and high in the tail, and they track output and debt more tightly than in the data, since the model has no shocks on the lenders' side. Defaulted coupons in default are lower and less volatile than in the data, as in the discrete-time model.

Two further contributions are technical. The first is computational. The default intensity is a continuous control, so the computed decision is the model's own and needs no smoothing, unlike in discrete-time models with long-term debt, which rely on i.i.d. shocks, to income or to the default and borrowing choices, to make the price schedule converge (Chatterjee and Eyigungor 2012; Dvorkin et al. 2021; Mihalache 2020); see Mihalache (2025) for a survey. A solution of the calibrated model takes minutes, fast enough to calibrate all nine free parameters by direct search, and the long-run distribution and the impulse responses come exactly from the generator of the discretized process.[2] The second concerns the model at high debt. Where default carries no fixed cost, a sovereign in full default faces a marginal cost of debt close to zero while lenders still pay a positive price, so it borrows up to whatever borrowing limit the model imposes. Quantitative models leave the limit implicit, as the top of the debt grid, so a choice about the computation becomes part of the model. We make the limit explicit and adopt an underwriting standard, a lower bound on the price of new issues as in Chatterjee and Eyigungor (2015), under which the level of the limit is irrelevant (Section 3.2). Hatchondo, Martinez, and Roch (2022) study a related rule, a spread brake that caps the deficit when the sovereign spread exceeds a threshold. At the calibrated parameters the main visible effect of the standard is to set the stationary debt in the worst states.

2 The Partial Default Model

The economy consists of a small open economy governed by a risk-averse sovereign and of a unit measure of competitive, risk-neutral international lenders. Time is continuous. We describe the two problems in turn and define the equilibrium.

2. Bornstein (2020) finds that computation is faster in continuous time in the binary model. Rendahl (2022) discusses the extent to which similar results can be obtained in discrete time by adapting the sparse matrix methods at the core of the continuous-time solution. Tourre (2017) studies the binary model in continuous time with recursive utility and a rich pricing kernel. Aguiar et al. (2019) study default and debt maturity in continuous time, and DeMarzo, He, and Tourre (2023) a sovereign that cannot commit to a debt path. These papers solve optimal stopping problems with HJB variational inequalities, because default is a discrete transition into exclusion. The partial default model has no such transition, which is the source of its tractability.

2.1 The Sovereign's Problem

The sovereign receives an endowment stream, borrows abroad with long-term bonds, and chooses at every instant what fraction of its scheduled debt service to pay. The part of the debt service it does not pay accrues as arrears and adds to the stock of debt.

Preferences and endowment. The sovereign maximizes

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt,$$

where ρ is its discount rate and u is increasing and strictly concave. The log endowment z_t follows a Markov process with generator $\mathcal{L}_z$, which we leave general in this section: in Section 3 it is constant or a finite-state chain, in Section 4 an Ornstein–Uhlenbeck diffusion. We call the model with a general Markov process the *general model*, and its three special cases the *deterministic model* (constant endowment), the *chain model* (a finite-state Markov chain), and the *diffusion model* (an Ornstein–Uhlenbeck diffusion). Output is e^{z_t} when the sovereign is current on its debt. When it defaults, output is $\phi(d_t, z_t)\, e^{z_t}$, where the cost function ϕ is described below.

Debt. The stock of debt is B_t. Each unit of debt pays a coupon λ and matures at rate δ, so that the scheduled debt service is $(\delta + \lambda)B_t$ per unit of time.[3] The sovereign chooses a *default intensity* $d_t \in [0, 1]$, the share of the scheduled debt service it withholds. Of every unit withheld, a fraction κ is added to the stock of debt as arrears. The fraction $1 - \kappa$ is the haircut suffered by lenders on defaulted payments. The sovereign also issues ℓ_t units of new debt at the market price q_t (when $\ell_t < 0$ it buys debt back). The budget constraint and the law of motion of debt are

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \tag{1}$$

$$\dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t\, B_t. \tag{2}$$

The three terms in (1) are income net of the default cost, the part of the debt service that is paid, and the proceeds from issuance. The three terms in (2) are issuance, amortization, and the accrual of arrears. It is convenient to write

$$a(d) := \delta - \kappa(\delta + \lambda)\, d, \qquad a := a(1) = \delta - \kappa(\delta + \lambda),$$

so that $\dot{B}_t = \ell_t - a(d_t)B_t$: absent issuance, debt decays at rate $a(d)$, which is lowest in full default. We assume $a > 0$, so that even a sovereign in full default sees its debt shrink, slowly, when it does not issue. Debt is bounded, $B_t \in [0, \bar{B}]$: the sovereign does not lend, and, as in every incomplete-markets model, a borrowing limit $\bar{B}$ rules out Ponzi schemes. The restriction to

3. This is the continuous-time counterpart of the long-term bond of Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012). The maturing fraction δ of the stock is repaid at par, and a single state variable summarizes the debt.

borrowing follows Arellano, Mateos-Planas, and Ríos-Rull (2023), and Appendix C.2 shows that relaxing it leaves the behavior of the model qualitatively unchanged. We make the limit explicit because, without the underwriting standard of Section 2.4, its level matters.

The cost of default. Following Arellano, Mateos-Planas, and Ríos-Rull (2023) the cost of default has a variable and a fixed component,

$$\phi(d,z) = (1-\gamma_0 d^{\gamma_1})\,F(z) \;\text{ for } d>0, \qquad \phi(0,z)=1, \qquad F(z) = 1-\gamma_2\,(e^z-y^*)\,\mathbb{1}\{e^z \geq y^*\}, \tag{3}$$

with $\gamma_0 \in (0,1)$, $\gamma_1 > 1$, and $\gamma_2 \geq 0$. The variable cost $\gamma_0 d^{\gamma_1}$ is convex in the default intensity with zero marginal cost at $d=0$. The fixed cost $(1-F(z))e^z$ is paid whenever the sovereign defaults at all, however little, and only when output e^z exceeds the threshold y^*. The fixed cost is larger in better times. Below the threshold default carries only the variable cost, and we call these states the *no-fixed-cost region*. The jump of ϕ at $d=0$ above the threshold creates a region of no default: a small default is not worth its fixed cost.

Substituting issuance ℓ_t out of the budget constraint (1) using the law of motion (2), we write the drift of debt as a function of the state, the price, and the two controls,

$$S(B,z,c,d,q) = \frac{c-\phi(d,z)e^z+(1-d)(\delta+\lambda)B}{q} - a(d)\,B,$$

and we treat consumption c and the default intensity d as the controls. Equivalently, for a given drift S,

$$c = \mathcal{R}(d;B,z,q) + q\,S, \qquad \mathcal{R}(d;B,z,q) := \phi(d,z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)\,B. \tag{4}$$

We call $\mathcal{R}$ the *resource function*: it is consumption at zero drift, and it is all that the default decision affects for a given drift. The characterization of Section 3 rests on this decomposition.

Arrears as borrowing. The decomposition has a second reading, which makes exact the description of partial default by Arellano, Mateos-Planas, and Ríos-Rull (2023) as a flexible way to raise funds. Let $n_t := \ell_t + \kappa(\delta+\lambda)d_t B_t$ be the total addition to the stock of debt, from issuance and from arrears, so that $\dot{B}_t = n_t - \delta B_t$. Substituting into (1),

$$c_t = \phi(d_t,z_t)e^{z_t} - (\delta+\lambda)B_t + q_t\,n_t + (1-\kappa q_t)(\delta+\lambda)B_t\,d_t. \tag{5}$$

Holding the path of debt fixed, withholding a unit of debt service raises consumption by $1-\kappa q_t > 0$: the unit saved, net of the market value of the κ units of arrears that replace market issuance one for one. Equivalently, withholding raises funds at the implicit price $1/\kappa$ per unit of arrears, which exceeds the market price and does not depend on it, paid for with the output cost $(1-\phi)e^z$ of default. Section 3.1 draws the consequences.

The HJB equation. Given a bond price function $q(B,z)$, the sovereign's value $V(B,z)$ solves

$$\rho V(B,z) = \max_{c,\ d\in[0,1]} \{u(c) + S(B,z,c,d,q(B,z))\, V_B(B,z) + \mathcal{L}_z V(B,z)\}, \tag{6}$$

subject to the state constraints $0 \le B \le \bar{B}$ and the underwriting standard of Section 2.4. Subscripts denote partial derivatives. Unlike in models with a binary default decision, there is no variational inequality and no second regime: the sovereign never loses market access, and a single value function with possibly discontinuous policies describes it at all states.

At an interior drift, the first-order condition for consumption is

$$u'(c)\, q(B,z) = -V_B(B,z), \tag{7}$$

the price of a bond times the marginal utility of the resources it brings equals the marginal cost of debt in utility units. By (4) the default intensity enters the objective only through $\mathcal{R}$, which is strictly concave on $(0,1]$ with

$$\mathcal{R}_d(d) = -\gamma_0\gamma_1 F(z) e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B.$$

The marginal gain from raising the default intensity is $(1-\kappa q)(\delta+\lambda)B$: the payment saved, net of the market value κq of the arrears it creates. The interior maximizer is

$$d_{\text{int}}(B,z,q) = \min\left\{1,\ \left[\frac{(1-\kappa q)(\delta+\lambda)B}{\gamma_0\gamma_1 F(z) e^z}\right]^{1/(\gamma_1-1)}\right\}, \tag{8}$$

and the optimal default intensity is either 0 or d_{int}, whichever gives the larger resource function. Section 3.1 makes this precise and shows that it holds for every maximizer of (6) wherever the underwriting standard is slack.

2.2 International Lenders

A unit measure of deep-pocketed, competitive, risk-neutral lenders with outside option r buy and hold the sovereign's bonds. A unit of debt held at t yields the flow $(1-d_t)(\delta+\lambda)$ in cash and $\kappa(\delta+\lambda)d_t$ in new units (the arrears), and δ of it matures. The break-even price is the expected discounted value of the cash flow, where the discount rate is adjusted for the growth of the position through arrears,

$$q(B,z) = \mathbb{E}_{B,z} \int_0^\infty \exp\left(-\int_0^t \xi(d_s)\, ds\right) (1-d_t)(\lambda+\delta)\, dt, \qquad \xi(d) := r + \delta - \kappa(\delta+\lambda)\, d. \tag{9}$$

The effective discount rate $\xi(d)$ combines the lenders' opportunity cost r, the amortization δ, and the accrual of arrears, which lowers it. By the Feynman–Kac formula, q solves the linear equation

$$\xi(d(B,z))\, q(B,z) = (1 - d(B,z))(\lambda + \delta) + \tilde{S}(B,z)\, q_B(B,z) + \mathcal{L}_z q(B,z), \tag{10}$$

where $\tilde{S}(B,z)$ is the equilibrium drift of debt and $d(B,z)$ the equilibrium default policy. The evolution of the stock of debt is governed by $\tilde{S}$, which includes issuance. The cash flow of a single unit is governed by $\xi(d)$, which does not.

The price of a bond that never defaults, the *risk-free price*, is $q^{\mathrm{rf}} = (\delta + \lambda)/(r + \delta)$. We set the coupon equal to the risk-free rate, $\lambda = r$, so that $q^{\mathrm{rf}} = 1$ and the face value of a bond equals its risk-free value. We keep the symbol q^{rf} wherever a price is the risk-free one. The yield to maturity implied by a price q is $(\delta + \lambda)/q - \delta$, so the spread over the risk-free yield is

$$\mathrm{spread}(B,z) = \frac{\delta + \lambda}{q(B,z)} - (\delta + r),$$

and the risk-free Macaulay duration of the bond is $1/(\delta + r)$.

2.3 Equilibrium and the Ergodic Set

Definition 1. A *Markov equilibrium* consists of a value function $V(B,z)$, policies $c(B,z)$ and $d(B,z)$, and a bond price $q(B,z)$ such that

1. given q, the value V solves the HJB equation (6) and (c,d) attain the maximum on its right-hand side, subject to the state constraints $0 \le B \le \bar{B}$ and the underwriting standard of Section 2.4;

2. given the policies, q solves the pricing equation (10), equivalently (9).

Given an equilibrium, let $g(B,z)$ be the density of the state. Given the equilibrium drift $\tilde{S}$ and the generator $\mathcal{L}_z$ with adjoint $\mathcal{L}_z^*$, the density solves the Kolmogorov forward equation

$$\partial_t g = -\partial_B \big[\tilde{S}(B,z)\, g\big] + \mathcal{L}_z^* g, \tag{11}$$

and the stationary distribution solves $0 = -\partial_B[\tilde{S} g] + \mathcal{L}_z^* g$. The forward equation is the adjoint of the HJB equation, which is how we compute the long-run distribution (Appendix B). The stationary distribution can have atoms: at $B = 0$ when the sovereign repays in full at a high endowment, and at stationary points of debt in the deterministic and chain models. We call its support the *ergodic set*. Its shape, in particular how far it extends into the default region, is a central object of the paper.

2.4 The Underwriting Standard

Without a further restriction the borrowing limit $B \leq \bar{B}$ is not innocuous. In the states where default carries no fixed cost, a sovereign in full default faces almost no marginal cost of debt while lenders still pay a positive price, because output will recover. So, as we argue in Section 3.2, the sovereign issues at any positive price and borrows up to the limit, whatever its level. We therefore impose an *underwriting standard*: the banks that underwrite the sovereign's issues in the financial centers do not bring a new issue to market at a price below a floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}. \tag{12}$$

The restriction is on placements, not on holdings: the outstanding bonds continue to trade at the price q that competitive lenders set, buybacks remain allowed, and partial default and the accrual of arrears are unchanged. A lower bound of this kind on the price of new issues appears in Chatterjee and Eyigungor (2015). In yield terms, with $\lambda = r$ the floor corresponds to a spread of $(\delta + \lambda)(1/\underline{q} - 1)$. For the value $\underline{q} = 0.6$ we use, this is about 1,280 basis points at the calibrated maturity rate. The level is a judgment: at the calibrated parameters it lies far above the highest price at which the unconstrained sovereign issues in full default, about 0.3 (Appendix C). The restriction is on the price rather than on the quantity of debt because the accumulation is located by the price. The accumulation begins at the full-default line in the states where default carries no fixed cost, at debt levels that differ across states. A floor on the price is a state-dependent debt limit expressed in the one variable both parties observe, the underwriters' counterpart of the spread brakes that Hatchondo, Martinez, and Roch (2022) study as fiscal rules. The standard makes the full-default region transient and the level of the borrowing limit $\bar{B}$ irrelevant, assuming the limit exceeds the full-default thresholds. It also bounds the support of debt, and where it binds it makes the sovereign apply its default rule at a shadow price below the market price. It binds only in deep distress: at the calibrated parameters, on just under one percent of the ergodic mass, all of it more than one standard deviation below the mean endowment.

No trade at a zero price. A second, milder convention rules out the autarky equilibrium in which lenders expect full default forever, the price is zero, and the sovereign keeps itself above the full-default line by issuing at a zero price. We impose that no bonds change hands at a zero price. Section 3.1 shows that the price is then strictly positive at every state.

3 Characterization

This section collects what can be established analytically, organized around the questions behind the four lessons of the introduction. Section 3.1 contains the results for the general model, any Markov endowment process: the arrears identity, the static default decision and the thresholds, and the bond price and the value. Section 3.2 turns to high debt, where the model needs the bor-

rowing limit and the underwriting standard. Section 3.3 specializes to the deterministic model, where the closed-form results live: debt capacity in closed form, the laws of motion on either side of it, and self-fulfilling partial default. Section 3.4 treats the chain model. Proofs, and the more technical results, are in Appendix A. The numerical illustrations of the deterministic and chain models borrow the parameters of the diffusion model's calibration (Table 1), with the endowment held at its mean or at a few fixed levels. The exception is Figure 5, a numerical example with parameter values of its own. Throughout we maintain the following.

Assumption 1.

(i) $a = \delta - \kappa(\delta + \lambda) > 0$, so that also $\xi(1) = r + a > 0$.

(ii) $\lambda = r$, so that $q^{\mathrm{rf}} = 1$; $\rho > r$.

(iii) u is C^2, strictly increasing, and strictly concave; $\gamma_1 > 1$, $0 < \gamma_0 < 1$, and $F(z) > 0$.

(iv) The drift is bounded above, $S \leq \bar{S}$, consumption is bounded below, $c \geq c_{\min} > 0$, which bounds buybacks, and $B \in [0, \bar{B}]$.

3.1 Results for any Markov endowment process

For any Markov endowment process, the default decision is static and its thresholds are explicit. This makes debt capacity a closed form.

Lemma 1 (arrears raise funds at a fixed price)**.** *For any path of the total addition to debt n_t, from issuance and from arrears, consumption satisfies (5). That is, a unit of debt service withheld raises one unit of funds against κ units of arrears, at the implicit price $1/\kappa$, independent of the market price, in exchange for the output cost of default. In particular, for a given drift of debt the default intensity affects consumption only through the resource function $\mathcal{R}(d; B, z, q) = \phi(d, z)e^z - (1 - d)(\delta + \lambda)B + q\,a(d)B$, and $\mathcal{R}_d = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1 - \kappa q)(\delta + \lambda)B$ for $d > 0$.*

The lemma is an identity, and it organizes the model. The sovereign has two ways of raising funds, the market at the price q and arrears at the price $1/\kappa$. The second is always the cheaper per unit of debt but always costs output. The results below follow from the lemma, starting with the fact that the choice between the two is static.

By the first-order condition $\mathcal{R}_d = 0$, the sovereign withholds until the marginal output cost of default per unit of debt service withheld, $\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1}/((\delta + \lambda)B)$, equals $1 - \kappa q$. The right-hand side is the unit of cash it saves net of the market value of the κ units of arrears it creates. Everything the default policy does follows from this margin. A lower price makes the arrears cheaper, so the sovereign withholds more: intensity rises as the price falls, in the default region of the calibrated model (Section 4.3) and under a pessimistic expectation (Section 3.3). A larger debt raises the cash saved per unit of intensity while the marginal output cost does not depend on debt, so intensity rises with debt. Where the underwriting standard binds, the

sovereign values its arrears at its own shadow price, below the market price, and withholds more still (Proposition A.6). The fixed cost decides only whether the sovereign is on this margin at all.

Proposition 1 (the default decision is static)**.** *For every state (B, z), price $q \geq 0$, and value function V, every maximizer (d^*, c^*) of the right-hand side of (6), subject to constraints on the drift that do not depend on d (in particular wherever the underwriting standard is slack), satisfies*

$$d^* \in \arg\max_{d \in [0,1]} \mathcal{R}(d; B, z, q).$$

The default policy depends on $(B, z, q(B, z))$ only, and not on V, on the sign of the drift, or on the constraints on the drift, as long as those constraints do not depend on d.

The qualification matters for the underwriting standard, whose constraint $\ell = S + a(d)B \leq 0$ does depend on d. Section 3.2 gives the rule in that case (Proposition A.6). Everywhere else the default decision is a static maximization of the function $\mathcal{R}$, concave on $(0, 1]$, which yields explicit thresholds.

Proposition 2 (default thresholds)**.** *In the general model, fix (B, z, q) with $B > 0$ and $\kappa q < 1$. The optimal default intensity is $d^* = 0$ if the fixed cost exceeds the gain from defaulting,*

$$(1 - F(z))\, e^z \;\geq\; \Gamma(B, z, q) := (1 - \kappa q)(\delta + \lambda) B\, d_{\text{int}} - \gamma_0 F(z) e^z d_{\text{int}}^{\gamma_1}, \tag{13}$$

and $d^* = d_{\text{int}}(B, z, q)$ of (8) otherwise. Γ is continuous, strictly increasing in B, strictly decreasing in q, and zero at $B = 0$. Consequently:

(a) there is a threshold $B_{\text{pd}}(z, q)$, nondecreasing in q, with $d^* = 0$ for $B \leq B_{\text{pd}}$ and $d^* = d_{\text{int}} > 0$ for $B > B_{\text{pd}}$;

(b) in the no-fixed-cost region ($F(z) = 1$), $B_{\text{pd}}(z, q) = 0$: any positive debt is partially defaulted on;

(c) if $F(z) < 1$ and $d_{\text{first}}(z) < 1$ (below), the default intensity jumps at B_{pd} from 0 to the first default intensity

$$d_{\text{first}}(z) = \left[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\right]^{1/\gamma_1}, \tag{14}$$

which depends only on the endowment and the cost parameters, not on debt, the price, or preferences;

(d) full default, $d^* = 1$, occurs for $B \geq B_{\text{fd}}(z, q)$ and $B > B_{\text{pd}}(z, q)$, where

$$B_{\text{fd}}(z, q) := \frac{\gamma_0 \gamma_1 F(z) e^z}{(1 - \kappa q)(\delta + \lambda)};$$

when $d_{\text{first}}(z) \geq 1$ the intensity jumps from 0 directly to 1 at B_{pd};

(e) the threshold has a closed form: when $d_{\text{first}} < 1$,

$$B_{\text{pd}}(z,q)^{\frac{\gamma_1}{\gamma_1-1}} = \frac{\gamma_1}{\gamma_1-1}\,\frac{(1-F(z))\,e^z\,G^{\frac{1}{\gamma_1-1}}}{K^{\frac{\gamma_1}{\gamma_1-1}}}, \qquad G := \gamma_0\gamma_1 F(z)e^z, \quad K := (1-\kappa q)(\delta+\lambda), \tag{15}$$

and when $d_{\text{first}} \geq 1$, $B_{\text{pd}}(z,q) = (1-(1-\gamma_0)F(z))e^z/K$.

Two monotonicity properties follow, and both reflect the default margin at work. First, $d^*(B,z,q)$ is nondecreasing in B and nonincreasing in q, in γ_0, in κ, and in $F(z)e^z$ holding the fixed cost $(1-F(z))e^z$ fixed. Second, the thresholds B_{pd} and B_{fd} are increasing in γ_0, in κ, and in q, decreasing in $\delta+\lambda$, and B_{pd} is increasing in the fixed cost $1-F(z)$ (using $\gamma_0 < 1$). The thresholds make the regions of default explicit in the primitives and the price, which the discrete-time model obtains only numerically.

The thresholds are stated at a given price. In equilibrium the no-default region is $\{B \leq B_{\text{pd}}(z, q(B,z))\}$, a fixed-point condition, which is explicit at the risk-free price. Two implications follow directly. The first concerns the size of the first default, the jump d_{first}, which depends only on the endowment and the cost parameters. At the calibrated parameters it is half of the debt service, $d_{\text{first}} \approx 0.52$ at the average endowment. The jump is large not because the fixed cost is large but because the variable cost is steep. The jump d_{first} is the ratio of the fixed cost, $1-F$, to $\gamma_1 - 1$ times the variable cost at full default, $\gamma_0 F$ (both per unit of output), raised to the power $1/\gamma_1$ with $\gamma_1 = 3.1$. Here that ratio is 0.13. The fixed cost is what makes full repayment possible rather than what produces the jump. With $F(z) = 1$ the variable cost has zero marginal cost at $d = 0$, so $\Gamma(B,z,q) > 0$ and (13) fails at every $B > 0$. Then the sovereign defaults a little at every positive debt level, and there is no region of full repayment. Any positive fixed cost restores that region, and a jump at its edge is then the only way to leave it. Second, at the risk-free price the no-default region at the average endowment ends at $B_{\text{pd}}(0, q^{\text{rf}}) = 0.485$, which is where the steady state of the deterministic model sits (Theorem 1). So the debt capacity of the partial default model is an explicit function of the primitives. Figure 2 draws both implications at the calibrated parameters.

Four further properties of the bond price q and the value V hold in the general model. They are routine, and their statements and proofs are in Appendix A.1. The bond price lies between 0 and q^{rf}. It equals q^{rf} only along paths without default, and 0 only along paths in full default forever (Proposition A.1). The price can jump in debt only at points where debt stops moving and that are not reached in finite time from both sides, and is continuous elsewhere (Proposition A.2). This locates the possible price cliffs of computed solutions. An equilibrium in which lenders expect full default forever and the price is zero exists only on a knife edge. The convention against trade at a zero price removes it, so that the price is strictly positive at every state (Proposition A.3). Finally, when the price is nonincreasing in debt the value is too, with buybacks and with the underwriting standard (Proposition A.4).

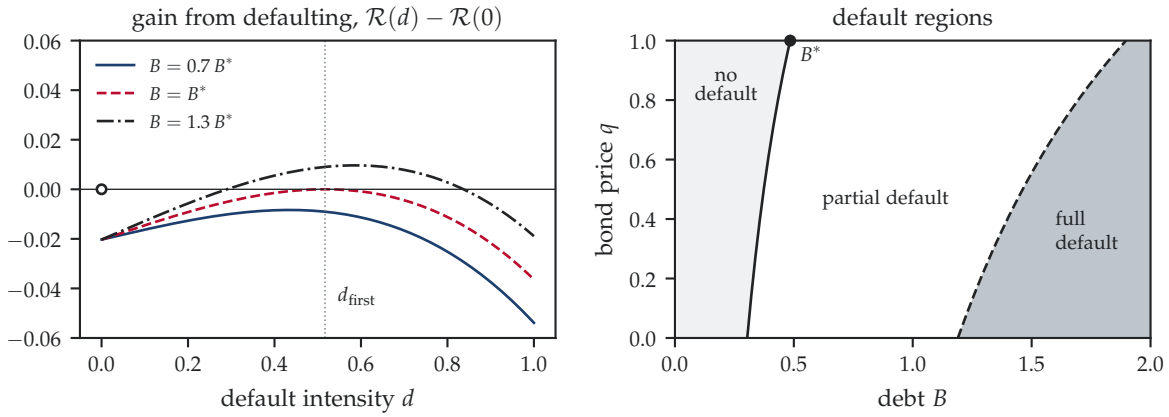


Figure 2: The default decision at the mean endowment, from the closed forms of Proposition 2 at the calibrated parameters. Left: the gain from defaulting at intensity d, $\mathcal{R}(d) - \mathcal{R}(0)$, at the risk-free price and three debt levels. At $d = 0$ the gain is zero (open circle). Any positive intensity pays the fixed cost, so the gain jumps down. At $B = B^*$ the best default, d_{first}, ties with full repayment. Right: the thresholds $B_{\text{pd}}(0, q)$ and $B_{\text{fd}}(0, q)$ divide the (B, q) plane into the regions of no, partial, and full default. The no-default region shrinks as the price falls, and its edge at the risk-free price is the debt capacity $B^* = B_{\text{pd}}(0, q^{\text{rf}})$.

3.2 High Debt and the Underwriting Standard

In the no-fixed-cost region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt close to zero, while lenders still pay a positive price because output will recover. By (7) the sovereign issues at any positive price. Without a borrowing limit debt could not stay above every full-default threshold forever, since debt that is never serviced again is worthless (Proposition A.5). With a limit the sovereign borrows up to it, whatever its level, so the level of the limit shapes equilibrium behavior at high debt. The results are in Appendix A.2.

The underwriting standard removes the accumulation. With issuance switched off below $\underline{q}$, debt shrinks in full default, the full-default region is transient, and any borrowing limit above the full-default thresholds is an exact truncation, so that its level is irrelevant (Proposition A.7). The long-run distribution of debt is confined to the debt levels at which the price can still reach the floor (Proposition A.7(f)). The thresholds of Proposition 2, as functions of the state and the price, are unaffected. Where the standard binds, its constraint $\ell \leq 0$ involves the default intensity through $\ell = S + a(d)B$, so the separation of Proposition 1 fails. The sovereign then applies the rule (8) at its own shadow price of debt, $\hat{q} = -V_B/u'(c)$, instead of the market price (Proposition A.6). The shadow price is below the market price, because the sovereign would like to issue and cannot. So it values its arrears less and defaults more. This is the default margin of Section 3.1 at the shadow price.

3.3 The Deterministic Model

The deterministic model gives a sovereign's debt capacity, and the motion of its debt on either side of it, in closed form. For a risk-neutral sovereign it also answers whether lenders' pessimism alone can carry debt past that capacity. With z fixed, normalized to 0, the equilibrium is a pair of ordinary differential equations. Writing S^*, c^*, d^* for the policies, (6) and (10) read

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*}, \tag{16}$$

with $d^* = d^*(B, q(B))$ from Proposition 2, $c^* = (u')^{-1}(-V'/q)$, and $S^* = (c^* - \mathcal{R}^*)/q$. The system is singular where the drift vanishes, and the stationary points of debt organize the solution. We call a zero of the drift a *steady state*. Theorem 1 pins down one of them within the class of no-default-basin equilibria, the no-default steady state B^*. A zero inside the partial-default region would be a self-fulfilling partial-default steady state, discussed at the end of this subsection.

Definition 2. An equilibrium of the deterministic model is of *no-default-basin type* with steady state $B^* \in (0, \bar{B})$ if $d \equiv 0$ on $[0, B^*]$, $S > 0$ on $[0, B^*)$, $S(B^*) = 0$, and $S < 0$ on $(B^*, \bar{B}]$. We say that debt reaches B^* from above in finite time if from $B^* + x$ the state reaches B^* in a time $\tau(x) \to 0$ as $x \downarrow 0$.

Theorem 1 (the deterministic steady state in closed form)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium in which debt reaches B^* from above in finite time. Then $q \equiv q^{\mathrm{rf}}$ on $[0, B^*]$, $V(B^*) = u(1 - rB^*)/\rho$, and*

$$B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}}),$$

the largest debt the sovereign is willing to service in full at the risk-free price, given explicitly by Proposition 2(e) with $q = q^{\mathrm{rf}}$.

The steady state is where the impatient sovereign, borrowing at the risk-free price, runs into its own default threshold. We call it the sovereign's *debt capacity*. It is a closed form in the primitives, because the default decision that defines it is a static comparison of the gain from the first default with the fixed cost. Capacity is increasing in the default costs γ_0 and $1 - F$, in income, and in the recovery κ, since a higher recovery makes withholding more expensive. It is decreasing in $\delta + \lambda$, and independent of ρ (as long as $\rho > r$), of the curvature of u, of the drift cap, and of the borrowing limit. Table 3 in Appendix B.2 confirms the comparative statics on computed solutions. Steady-state consumption is $1 - rB^*$ and the first partial default beyond B^* has intensity d_{first}. On the basin $[0, B^*)$ the sovereign faces the constant risk-free price and borrows up to the steady state, which it reaches in finite time. The approach is governed by a square-root law.

Lemma 2 (the square-root law)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium with V twice continuously differentiable on $(0, B^*)$ and S continuous at B^*, and write $\psi := u'/|u''|$ for*

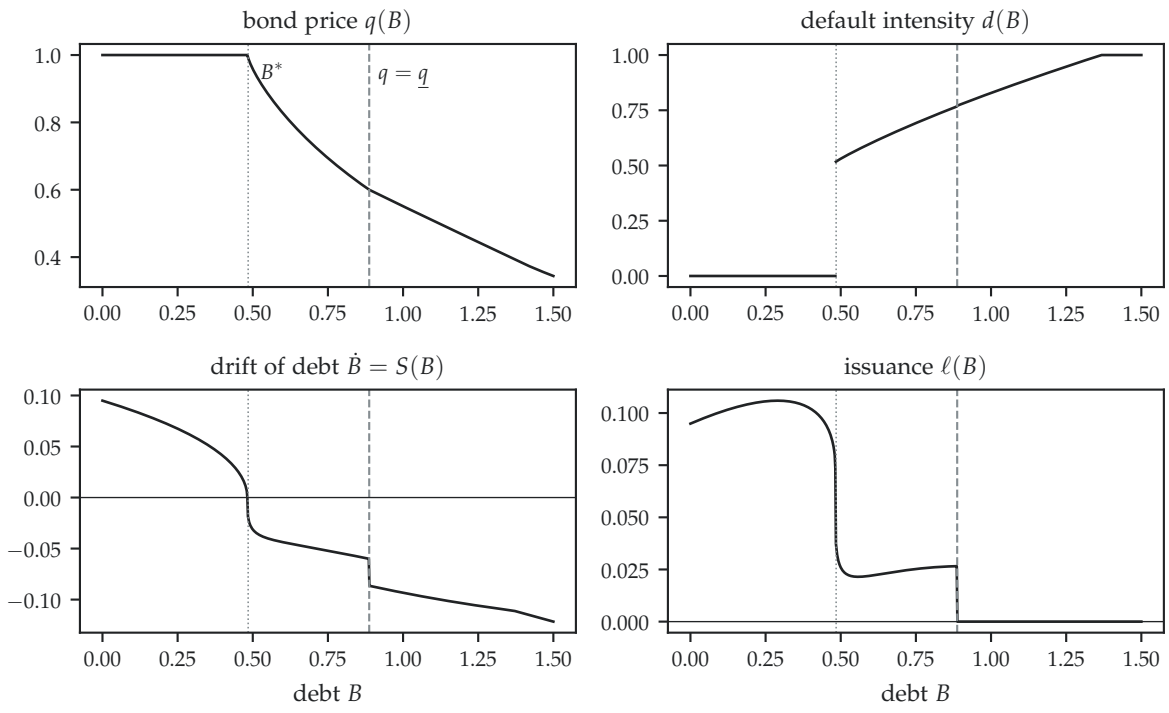


Figure 3: The deterministic model: bond price, default intensity, drift of debt, and issuance as functions of debt. Parameters of Table 1 with the endowment at its mean. Dotted vertical line: the steady state $B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$. Dashed vertical line: the debt at which the price falls through the floor $\underline{q}$.

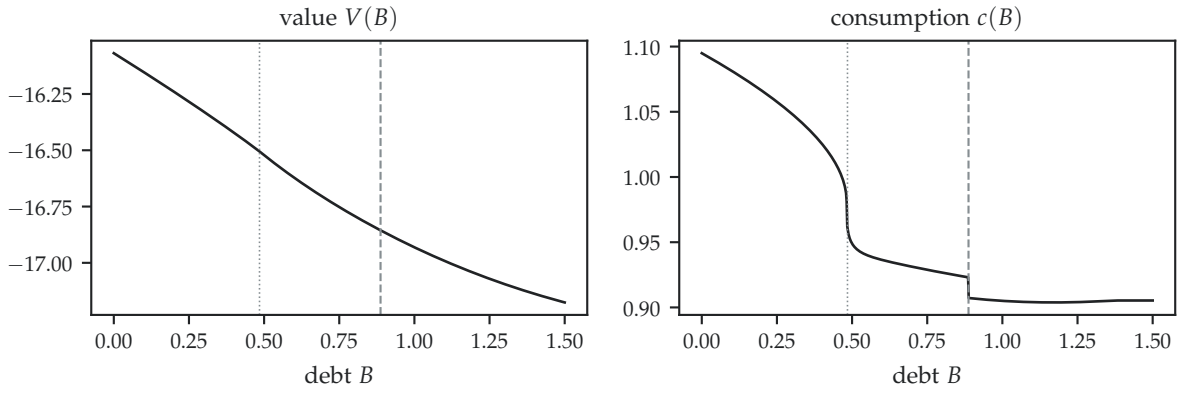


Figure 4: The deterministic model: value and consumption as functions of debt. The value is C^1 at the steady state (Theorem 2). Consumption falls steeply but continuously at the first default, with the vertical tangent of the cube-root law, and is nearly flat in the default region, with a small step where the underwriting standard starts to bind (dashed vertical line).

the absolute risk tolerance and $\bar{c} = 1 - rB^*$. *Then as* $x \downarrow 0$

$$S(B^* - x) = \left(2(\rho - r)\,\psi(\bar{c})\right)^{1/2} x^{1/2}\,(1 + o(1)),$$

and the state reaches B^* *from the left in finite time.*

Proof. On the basin $q \equiv q^{\mathrm{rf}}$ and $d \equiv 0$, so $c = 1 - rB + S$ and the HJB equation reads $\rho V = u(c) + SV'$ with $V' = -u'(c)$. Differentiating in B and substituting the first-order condition gives $SS' = rS - (\rho - r)\,\psi(c)$. As $B \uparrow B^*$ the drift vanishes and $c \to \bar{c}$, so $(S^2)' \to -2(\rho - r)\psi(\bar{c})$, which integrates to the stated law. The time to reach B^* from $B^* - x$ is $\int_0^x dy/S(B^* - y) \propto x^{1/2}$, which is finite. □

This square-root law characterizes the impatient sovereign's approach to a limit it cannot cross at the risk-free price. Beyond B^* the sovereign is in partial default, the price is below q^{rf}, and the drift is negative, so the state returns to B^* from above.

Theorem 2 (the cube-root law at the steady state)**.** *In the deterministic model, let* (V, q) *be a no-default-basin equilibrium in which debt reaches* B^* *from above in finite time, with* $d_{\mathrm{first}} < 1$, *and suppose that on a right neighborhood of* B^* *the drift is interior and* V *is* C^1. *Let* u *be real-analytic near* $\bar{c}$, *as CRRA utility is (for the second-order terms and the last statement). Write* $x = B - B^*$, $\bar{c} = 1 - rB^*$, *and*

$$\Pi_0 := (1 - \kappa)(\delta + \lambda)\,d_{\mathrm{first}}, \qquad \beta := \left(3\,a(d_{\mathrm{first}})\,B^*\,\Pi_0\,\frac{u'(\bar{c})}{|u''(\bar{c})|}\right)^{1/3}, \qquad \alpha := \frac{3\Pi_0}{2\beta}.$$

Then as $x \downarrow 0$,

$$S(B^* + x) = -\beta\,x^{1/3}\,(1 + o(1)), \qquad c(B^* + x) = \bar{c} - \beta\,x^{1/3}\,(1 + o(1)),$$
$$q^{\mathrm{rf}} - q(B^* + x) = \alpha\,x^{2/3}\,(1 + o(1)),$$

and $V'(B^{*+}) = -u'(\bar{c}) = V'(B^{*-})$. *More precisely, with* $\psi(c) := u'(c)/|u''(c)|$,

$$S(B^* + x) = -\beta\,x^{1/3} - \Sigma_1\,x^{2/3} + O(x), \qquad q^{\mathrm{rf}} - q(B^* + x) = \alpha\,x^{2/3} + Q_1\,x + O(x^{4/3}),$$
$$\Sigma_1 = -\frac{1}{4}\left[\beta^2\,\frac{u'''(\bar{c})}{|u''(\bar{c})|} + \frac{3\psi(\bar{c})\,(\rho - r + \Pi_0)}{\beta}\right], \qquad Q_1 = -\frac{\Pi_0\,\Sigma_1}{\beta^2},$$

where $u'''(\bar{c})/|u''(\bar{c})| = (\nu + 1)/\bar{c}$ *for CRRA utility with relative risk aversion* ν. *Both terms in the bracket are positive when* u *is prudent and* $\rho > r$, *so* $\Sigma_1 < 0$. *Thus* V *is* C^1 *but not* C^2 *at the steady state, the drift leaves zero with a vertical tangent, and the price has a vertical tangent from the right. Moreover the solution above capacity (the right branch of* (16)*) exists on a right neighborhood of* B^*, *is unique among solutions that stay bounded in the scaled variables* $(q^{\mathrm{rf}} - q)/x^{2/3}$ *and* $-qS/x^{1/3}$ *as* $x \downarrow 0$ *(hence among those with these asymptotics), and is an analytic function of* $(B - B^*)^{1/3}$ *(Appendix A).*

The cube-root law describes gradual deleveraging: a sovereign over its capacity works its debt down at a speed that falls as it nears capacity, and it arrives in finite time, to leading order

$(3/(2\beta))\,x^{2/3}$. In the illustration below this is about 0.4 years from an excess of one percent of output and 1.9 years from ten percent. Even a small excess, however, lowers the price visibly at once. With a binary default decision the sovereign would jump into default instead. Above capacity the HJB equation reads $\rho V = u(c) + w\,u'(c)$, with $w = -qS \geq 0$ the resources devoted to repayment. Letting $B \downarrow B^*$ forces $c \to \bar{c}$ and hence $S \to 0$, and differentiating gives $w' \approx M/w^2$ with $M := a(d_{\text{first}})B^*\Pi_0\,\psi(\bar{c})$, whence $w^3 \approx 3Mx$. The price gap then integrates $q' \approx -\Pi_0/w$ to the two-thirds power. The constant Π_0 is the flow of debt service lost at the first default, net of the arrears it creates. The price has a vertical tangent to the right of the steady state because a small excursion into default costs lenders this flow at a rate that does not vanish, over a time that is only $O(x^{2/3})$.

The equilibrium with these properties exists. Below capacity it is the unique solution of the basin equation of Lemma 2 that reaches B^*, and above capacity it is the solution of (16) that leaves B^* along the cube-root law. Proposition A.8 in Appendix A shows that together they form a no-default-basin equilibrium, the only one with these asymptotics, whenever the solution above capacity stays regular up to full default. The condition is on an explicit initial-value problem, not on an equilibrium, so it can be checked at any parameters. Its main requirement is that the drift above capacity have no second zero before full default. The computed solution in the illustration satisfies it. A second zero would be a self-fulfilling partial-default steady state.

Numerical illustration. Figure 3 shows the deterministic equilibrium in the illustration, with the endowment fixed at its mean and the underwriting standard at $\underline{q} = 0.6$ (Appendix B.2). The drift vanishes at the steady state $B^* = B_{\text{pd}}(0, q^{\text{rf}}) = 0.485$ of Theorem 1, where the stationary distribution has its atom, and the default intensity jumps to $d_{\text{first}} = 0.52$ there, as the closed forms say. The underwriting standard binds from $B = 0.888$, where the price falls through 0.6 (dashed vertical line). Beyond that point issuance is zero and the drift is $-a(d)B$. From $B = 1.37$ default is complete, and the price follows the power law of Proposition A.8. Figure 4 shows the value and consumption. The value is C^1 through the steady state, as Theorem 2 says. Consumption falls steeply at the first default, with the vertical tangent of the cube-root law, but it is continuous there. At the threshold the resource function takes the same value at $d = 0$ and at d_{first}, and the drift vanishes on both sides. With two terms, the expansion of Theorem 2 matches the computed drift and price gap to within a few percent outside the first grid steps (Appendix B.2).

Self-fulfilling partial default. In the equilibrium of Theorem 1, lenders expect full repayment and the price is q^{rf}. Suppose instead that lenders expect the sovereign to settle at a debt B°, typically above its capacity, and to withhold a constant share d° of its debt service there forever. They then pay at B° the price of a bond that defaults at rate d° forever,

$$q_r(d) := \frac{(1-d)(\lambda+\delta)}{\xi(d)},$$

and less than q^{rf} at every lower debt, since every path leads to B°. Two consequences validate the expectation. First, default starts earlier. At a lower price the arrears created by withholding a unit of debt service are worth less, so withholding costs the sovereign less (Lemma 1). The threshold $B_{\text{pd}}(0, q)$ falls with the price (Proposition 2), and the sovereign defaults at debt levels at which it would repay in full at the risk-free price. Second, the sovereign can borrow past its old limit. Because the default is priced, the sovereign pays only the risk-free rate on the funds it raises: at the steady state its net payments to lenders, debt service less issuance, are $rq^\circ B^\circ$ per unit of time. Since it discounts at $\rho > r$, it wants to borrow. It stops at B° only because there each further unit of debt lowers the price at which it must roll over the rest, and the impatience wedge balances this dilution exactly (Lemma A.2). At B° the sovereign withholds the share that lenders expected. There is no run and no loss of market access. With curvature the sovereign drifts into a permanent partial default over years, paying an output cost that exists only because lenders expected the default.

Lemma A.2 in Appendix A gives the local conditions at such a point: the price is $q_r(d^\circ)$, its slope to the right is pinned by the impatience wedge, $-q'(B^{\circ+})/q^\circ = (\rho - r)/(a(d^\circ)B^\circ)$, and the point is stable from above (attracting) only if additional debt dilutes the price strongly enough through additional default. These conditions admit a continuum of candidates, one for every intensity $d^\circ \geq d_{\text{first}}$. Whether a candidate is an equilibrium depends on the whole path from zero debt (a two-point boundary-value problem): the sovereign must be willing to borrow into partial default, at prices below q^{rf}, all the way from zero debt. Two cases can be settled analytically, with the underwriting standard and with a risk-neutral sovereign. Both are governed by the stationary spread at the first default intensity, the spread lenders charge a sovereign that withholds forever the smallest share it ever withholds,

$$s_{\text{first}} := (\lambda + \delta)\left(\frac{1}{q_r(d_{\text{first}})} - 1\right) = \frac{(1-\kappa)(\lambda+\delta)\, d_{\text{first}}}{1 - d_{\text{first}}}.$$

Proposition 3 (the standard and self-fulfilling partial default)**.** *In the deterministic model with the underwriting standard, every partial-default steady state has a price at or above the floor, $q_r(d^\circ) \geq \underline{q}$. Since q_r is decreasing and $d^\circ \geq d_{\text{first}}$, a self-fulfilling partial-default steady state can exist only if $q_r(d_{\text{first}}) \geq \underline{q}$, that is, only if the stationary spread at the first default does not exceed the spread at the floor,*

$$s_{\text{first}} \leq (\lambda + \delta)\left(\frac{1}{\underline{q}} - 1\right).$$

The condition involves neither the discount rate nor the utility function.

Standing still requires issuance. At a steady state the sovereign must replace the debt that matures and is not replaced by arrears, $\ell = a(d^\circ)B^\circ > 0$, and the standard forbids issuance below the floor.

A risk-neutral sovereign. With linear utility the sovereign does not smooth consumption, and the existence question has a nearly complete answer. We let issuance be unrestricted in speed, so that the sovereign can make a discrete issuance, moving its debt at once to a higher level at the price of the state it moves to. DeMarzo, He, and Tourre (2023) study a closely related problem in continuous time: an impatient, risk-neutral sovereign that cannot commit to a debt path and borrows from competitive lenders. Define for $d \in [d_{\text{first}}, 1)$ the stationary spread $s(d) := (1-\kappa)(\lambda+\delta)d/(1-d)$ and

$$\Theta(d) := s(d)\,\frac{1+(\gamma_1-1)(d_{\text{first}}/d)^{\gamma_1}}{\gamma_1},$$

the output cost of withholding the share d forever, per unit of the funds the sovereign raises by issuing at once the debt of the steady state with that intensity (Proposition A.9).

Proposition 4 (self-fulfilling partial default with a risk-neutral sovereign)**.** *Let $u(c) = c$ and $\rho > r$ in the deterministic model, with issuance unrestricted in speed.*

(a) In any equilibrium in which debt goes from zero to a partial-default steady state, $\rho - r \geq \Theta^* := \min_{d \geq d_{\text{first}}} \Theta(d)$.

(b) $\Theta(d_{\text{first}}) = s_{\text{first}}$, and if $\gamma_1 \leq 2$, Θ is increasing and $\Theta^* = s_{\text{first}}$.

(c) If $(\gamma_1 - 1)\xi(d_{\text{first}}) < \delta$ and $q_r(d_{\text{first}}) > \underline{q}$, there is a bound $\bar{\Omega} > s_{\text{first}}$, in closed form in Proposition A.9, such that for every wedge $\rho - r$ in $(s_{\text{first}}, \bar{\Omega})$ there are self-fulfilling partial-default equilibria, in which the sovereign issues the debt of a steady state at once from zero debt and withholds a constant share there forever, provided the price schedule continues regularly above the steady state.

The condition $\rho - r \geq \Theta(d)$ is a comparison that a sovereign at zero debt can make directly. Issuing at once the debt of the steady state with intensity d raises its market value. From then on the sovereign pays lenders the risk-free rate on the funds raised, because the default is priced, and it loses $1 - \phi(d, 0)$ of output to the default. Relative to staying debt-free forever, front-loading gains the impatience wedge times the funds raised, per unit of time. A discrete issuance up to the pessimistic steady state pays when this gain covers the output cost. By part (a), no route from zero debt reaches any partial-default steady state when the wedge is below Θ^*, the threshold of the cheapest discrete issuance. For a given steady state the test is sufficient but not necessary, since a smaller discrete issuance followed by gradual borrowing into default can also reach it. Part (b) identifies the threshold with the spread implied by the first default. At the first default the output cost equals the payments saved net of the arrears created, and at the stationary price this equals the spread times the value of the debt. With a more convex variable cost, deeper defaults cost less relative to their spread, and the threshold can lie below s_{first}. Every intensity that passes the test gives an equilibrium under the stability and continuation conditions of part (c), so when one self-fulfilling steady state exists a continuum does, and lenders' expectations alone select among them. The bound $\bar{\Omega}$ in part (c) is a stability condition: at a larger wedge the

steady state is no longer stable from above. In the numerical example of Figure 5, where $\gamma_1 = 1.5$ and $(\gamma_1 - 1)\xi(d_{\text{first}}) = 0.075 < \delta = 0.12$, the threshold is $s_{\text{first}} = 47$ basis points and $\bar{\Omega} = 80$ basis points.

Risk aversion and uncertainty. Both work against self-fulfilling partial default (Appendix B.5). With CRRA utility the sovereign cannot jump, and it borrows into default gradually. In a numerical example with $\nu = 2$ and a wedge of 210 basis points, a self-fulfilling partial-default steady state at $B = 1$ exists next to the no-default steady state at $B = 0.7$ (Figure 5). In every case we computed, such a steady state is reached from zero debt only above a threshold that exceeds Θ^* and rises with the curvature of utility, about 139 basis points for $\nu = 2$ against 47 for a risk-neutral sovereign. With shocks and a risk-neutral sovereign, for steady debts that do not move with the endowment, volatility narrows the range of supporting wedges from both sides, and at a standard deviation of the log endowment of about 7 percent no equilibrium of this kind remains (Figure 14). Section 4.3 returns to the question at the calibrated parameters.

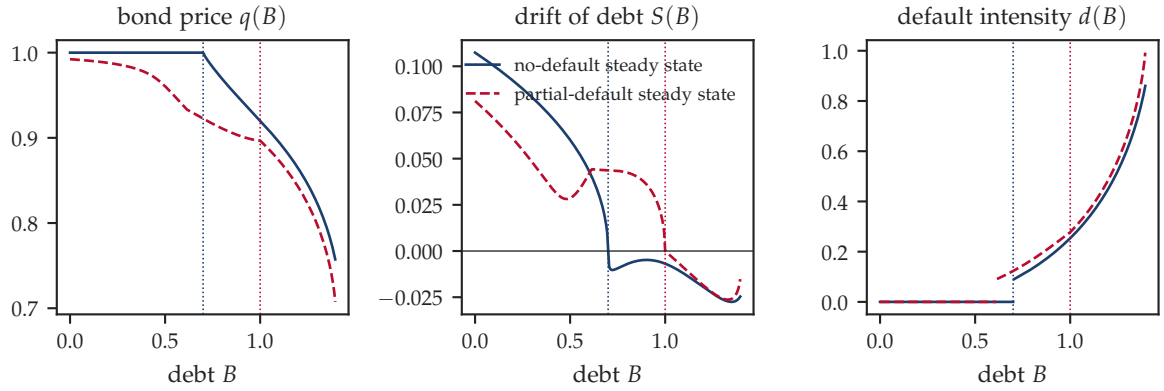


Figure 5: A numerical example of self-fulfilling partial default. Two equilibria of the deterministic model at the same parameters: the no-default steady state and a self-fulfilling partial-default steady state (dotted lines). Parameters: $\gamma_0 = 0.075$, $\gamma_1 = 1.5$, $F(0) = 0.999$, $\rho = 0.06$, $\delta = 0.12$, $\kappa = 0.7$, $r = \lambda = 0.039$, $\nu = 2$, and the standard at 0.7. The impatience wedge is 210 basis points.

Self-fulfilling partial default is the continuous-time form of the slow-moving debt crises of Lorenzoni and Werning (2019). In their model a pessimistic price raises the cost of rolling over the debt, and the sovereign borrows into a crisis that validates the price. The crisis unfolds slowly because debt is a stock that moves at the rate of the deficit. The idea that an expected repudiation raises the interest rate, and that the higher rate justifies the repudiation, goes back to Calvo (1988). In our model the same logic runs through arrears. A pessimistic price raises the default intensity the sovereign chooses, the higher intensity justifies the price, and debt moves at the rate of the deficit and of the arrears. Unlike the rollover crises of Cole and Kehoe (2000), self-fulfilling partial default involves no run. It is closer to the crises of Aguiar et al. (2022), in which the government keeps issuing at depressed prices. The difference from Lorenzoni and Werning (2019) is where the sovereign's willingness to participate enters. In their setting the

sovereign must roll over. In ours it can stay out of the default region, and a candidate survives only if borrowing into default at the pessimistic price beats staying out (Proposition 4).

3.4 The Chain Model

Uncertainty adds a precautionary motive and a kink in the price schedule, which together decide how defaults begin. Let the log endowment take finitely many values z_j and switch from j to k at rate η_{jk}. The HJB and pricing equations are

$$\rho V_j(B) = \max_{c,\,d} \left\{ u(c) + S\, V_j'(B) + \sum_{k \neq j} \eta_{jk} \big[V_k(B) - V_j(B) \big] \right\},$$
$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \sum_{k \neq j} \eta_{jk} \big[q_k(B) - q_j(B) \big].$$

Proposition 1 and the thresholds of Proposition 2 hold state by state with $q = q_j(B)$. The precautionary motive can hold a steady state at the risk-free price short of capacity.

Proposition 5 (steady states at the risk-free price in the chain model)**.** *In the chain model, suppose $[0, \underline{B}]$ is invariant under the equilibrium dynamics in every state and no state defaults on it. Then $q_j \equiv q^{\mathrm{rf}}$ on $[0, \underline{B}]$ for every j and $\underline{B} \leq \min_j B_{\mathrm{pd}}(z_j, q^{\mathrm{rf}})$. If state j has a steady state at an interior point B° of $[0, \underline{B}]$, at which V_j is twice differentiable, every V_k is differentiable, and the drift cap and the consumption floor are slack, then*

$$(\rho - r)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left(u'(c_k) - u'(c_j) \right) \qquad \text{at } B^\circ. \tag{17}$$

In particular, a state whose consumption at B° is the lowest across states has no steady state in the interior of $[0, \underline{B}]$.

Condition (17) is the Euler equation at a stationary point: the impatience gain from borrowing a little balances the precautionary loss from arriving in a worse state with more debt. In the state with the lowest consumption the precautionary term is nonpositive and nothing offsets the impatience term, so the sovereign cannot rest inside the default-free set and borrows to its edge. In a better state the two terms compete, and a second force works through prices. Beyond the threshold of a state that can be reached in one switch, a switch triggers an immediate first default. So the price schedule of the better state has a kink there, and the marginal revenue of issuance drops. Both forces can hold the better state at or below the threshold of the worse one, and a switch then does not land the sovereign in default. In the three-state illustration of Appendix B.1 (Figure 11), where a rare jump to the lowest state leads to default, no default-free invariant set contains the steady states, and all of them sit at default thresholds. The ergodic distribution has atoms at two of them, and default occurs on 0.07 of the ergodic mass, along the paths that follow a jump. Unlike in the calibrated diffusion model (Section 4.3), debt does not rise during these episodes: in the low state the sovereign repays from the moment it defaults.

Existence and the price cliffs. Given policies, the prices are the Feynman–Kac values (9) and are continuous except possibly at points where the drift vanishes or changes sign that are not reached in finite time from both sides (Proposition A.2). In our computations without the underwriting standard the chain model has such a point in the lowest state at the full-default threshold, and the price jumps there. With the standard, every stationary point of the computed solution is reached from both sides and the prices are continuous. A general existence theorem for the coupled equilibrium is not available: the best-response map from prices to prices is discontinuous (thresholds move with q) and not monotone (a higher price lowers default today but raises borrowing), so neither Schauder's nor Tarski's theorem applies directly. We show in Appendix A (Proposition A.11) that the best response does preserve the class of prices nonincreasing in debt, when the induced debt flow is order preserving and the standard is slack.

Open questions. Beyond existence, one question remains open: whether self-fulfilling partial-default steady states exist. For it we have analytic answers in the deterministic model only: a necessary condition under the standard (Proposition 3), and, for a risk-neutral sovereign, conditions that are necessary and, up to a stability and a continuation condition, sufficient (Proposition 4). With curvature and with shocks we have numerical findings.

4 Quantitative Analysis

This section calibrates the diffusion model to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023) and shows the results of Section 3 at work in this quantitative model. The regions of default and the debt capacity appear in the policies, the stationary debt in the shape of the ergodic set, and the fixed cost in the response to a shock.

4.1 The Diffusion Model

The log endowment follows an Ornstein–Uhlenbeck process, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$. Its generator is $\mathcal{L}_z = -\mu z\,\partial_z + \frac{1}{2}\sigma^2\partial_{zz}$, so the HJB and pricing equations are

$$\begin{aligned} \rho V &= \max_{c,\,d} \left\{ u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz} \right\}, \\ \xi(d)\,q &= (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}, \end{aligned} \tag{18}$$

with Neumann conditions $V_z = q_z = 0$ at $\pm\bar{z}$, and the stationary density solves

$$0 = -\partial_B\left[\tilde{S}\,g\right] + \partial_z[\mu z\,g] + \tfrac{1}{2}\sigma^2\,\partial_{zz}g,$$

with zero-flux conditions $\mu z g + \frac{1}{2}\sigma^2 g_z = 0$ at $z = \pm\bar{z}$ and $\tilde{S}g = 0$ at $B = \bar{B}$, and an atom at $B = 0$ that collects the inflow where $\tilde{S}(0^+, z) < 0$. The process is the continuous-time counterpart of the annual AR(1) of Arellano, Mateos-Planas, and Ríos-Rull (2023): an AR(1) with persistence ρ_z and

innovation standard deviation σ_ϵ observed at annual frequency corresponds to $\mu = -\ln \rho_z$ and a stationary standard deviation $\sigma_\epsilon / \sqrt{1-\rho_z^2} = \sigma / \sqrt{2\mu}$. Given a continuous price, (18) is a degenerate elliptic equation with a state constraint in B. The value function is its unique constrained viscosity solution and is Lipschitz in B (Soner 1986; Capuzzo-Dolcetta and Lions 1990; Fleming and Soner 2006), and the upwind scheme of Appendix B.3 converges to it (Barles and Souganidis 1991). We compute the coupled equilibrium as the fixed point of the two equations.

4.2 Calibration and Computation

Utility is CRRA, $u(c) = c^{1-\nu}/(1-\nu)$, and the default cost is (3), the level form of the fixed cost used by Arellano, Mateos-Planas, and Ríos-Rull (2023). We follow their calibration as closely as the continuous-time formulation allows. We fix risk aversion at $\nu = 2$ and the risk-free rate at $r = \ln 1.04$. Our free parameters are the nine they estimate: the default cost parameters $\gamma_0, \gamma_1, \gamma_2, y^*$, the discount rate ρ, the recovery κ, the maturity rate δ, and the endowment parameters ρ_z, σ_ϵ mapped as above. The targets are the eleven moments of their Table 4, with their weights (three on the mean of debt due to output, 1.25 on the standard deviation of the spread, one on the rest). The loss is the weighted sum of squared proportional deviations. We set the underwriting standard at $\underline{q} = 0.6$ (Section 2.4).

We compute the model's moments with the annual accounting of Arellano, Mateos-Planas, and Ríos-Rull (2023): exactly from the stationary distribution for functions of the state, and from simulated annual data for path functionals (Appendix B.3).

Table 1 reports the calibrated parameters next to their counterparts in Arellano, Mateos-Planas, and Ríos-Rull (2023), and Appendix B.3 describes the search. Relative to their values, the recovery is lower, and the variable default cost is higher and more convex. The fixed cost rises less steeply with output but starts at a lower output level. The discount rate and the maturity rate are somewhat higher, and the endowment is somewhat less persistent, with larger innovations.

We compute the equilibrium by the implicit upwind finite difference scheme of Achdou et al. (2022) on a grid of 201 nodes in debt on $[0, 1]$ and 75 nodes in z on ± 2.5 stationary standard deviations. Every result in the paper uses the calibrated parameters and this grid. No taste shocks or other smoothing of the decision problem are needed, unlike in discrete-time models with long-term debt (Chatterjee and Eyigungor 2012; Mihalache 2025; Section 1). The default intensity is a continuous control, and the monotone three-candidate scheme handles the only non-concavity, the fixed cost's jump at $d = 0$, directly. Appendix B describes the scheme, the treatment of the non-concave Hamiltonian, the initial guess, the convergence criteria, the accuracy of the solution, and the exact reverse-shooting solution of the deterministic model that we use as a check.

4.3 Quantitative Properties

Table 2 reports the targeted moments in the data, in the model of Arellano, Mateos-Planas, and Ríos-Rull (2023), and in ours, and the untargeted moments of their Table 4. The loss at the

Table 1: Calibration

		AMR (2023)	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
λ	coupon, $= r$	0.039	0.039
q	underwriting standard	–	0.6
Calibrated			
γ_0	variable default cost, level	0.0476	0.0759
γ_1	variable default cost, curvature	2	3.07
γ_2	fixed default cost, slope	0.12	0.082
y^*	fixed default cost, threshold (level)	0.938	0.751
ρ	discount rate (factor $e^{-\rho}$)	0.047	0.062
κ	recovery of defaulted coupons	0.7	0.373
δ	maturity rate (survival $1-\delta$)	0.12	0.153
ρ_z	annual persistence of the endowment	0.875	0.819
σ_ϵ	annual innovation s.d.	0.052	0.066

Notes: The discrete-time parameters of Arellano, Mateos-Planas, and Ríos-Rull (2023) (Table 3) are mapped to continuous time by $\rho = -\ln(\text{discount factor})$, $\delta = 1 - \text{survival rate}$ (which matches their debt service per unit of debt), $\mu = -\ln \rho_z$, $\sigma = \sigma_\epsilon \sqrt{2\mu/(1-\rho_z^2)}$. We choose the free parameters to minimize the weighted distance to the targets of Table 2, as in their calibration.

calibrated point is 0.096, against 0.228 for their model under the same weights, most of which comes from their standard deviation of debt to output. The continuous-time model reproduces the frequency of partial default, the mean and dispersion of the default intensity in default, the debt service and debt due ratios, and the output process. It carries somewhat less debt than the data and generates a spread volatility below the target. Among the untargeted moments, the relative volatility of consumption is close to the data. Defaulted coupons in default are lower than in the data, as in their model, and less volatile. The spread is more strongly correlated with output and with debt than in the data, as in their model and more so. With no shocks on the lenders' side, the spread is a function of debt and the endowment alone, so it tracks the state more tightly than a spread that also carries global factors.

Figures 6 and 7 show the value, the policies, the price, and the ergodic distribution, as functions of debt at four endowment levels and as contour maps over the state space. The value is decreasing in debt and increasing in the endowment. The default policy has the structure of Proposition 2. At the lowest endowment shown, $z = -0.29$, just below the fixed-cost threshold $\ln y^* = -0.286$, the sovereign defaults at any positive debt, as Proposition 2(b) says. This no-fixed-cost region carries about 0.1 percent of the ergodic mass. There the default intensity rises with debt, along the default margin of Section 3.1, to nearly full default at the highest debt levels. At average and high endowments the sovereign does not default until debt reaches the threshold $B_{\text{pd}}(z, q)$, where the default intensity jumps to the first default intensity and rises thereafter. The threshold rises with the endowment, so the no-default region is a wedge in the (B, z) plane. Full

Table 2: Moments in Data and Model

Moment	Weight	Data	AMR (2023)	This paper
Targeted				
Partial default: frequency	1	0.36	0.37	0.38
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.41
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.22
Debt to output: mean	1	0.32	0.32	0.27
Debt to output: s.d.	1	0.18	0.25	0.19
Debt service to output: mean	1	0.036	0.035	0.037
Debt service to output: s.d.	1	0.021	0.022	0.020
Debt due to output: mean	3	0.049	0.050	0.052
Spread: s.d.	1.25	0.041	0.037	0.035
Output: persistence	1	0.89	0.88	0.88
Output: s.d.	1	0.10	0.12	0.11
Loss			0.228	0.096
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.041
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.028
Spread: mean		0.053	0.016	0.029
Spread: correlation with output		-0.17	-0.38	-0.82
Spread: correlation with debt		0.24	0.56	0.88
Consumption s.d. relative to output s.d.		1.00	0.91	0.87

Notes: Data and the discrete-time model from Arellano, Mateos-Planas, and Ríos-Rull (2023), Table 4. The loss is the weighted sum of squared proportional deviations from the data.

default is not reached at any endowment level. The price is close to q^{rf} in the no-default region at a high endowment and lower at a low endowment, where default is expected after future shocks (between 0.89 and 0.99 at zero debt). The price falls at the default threshold and declines in the default region, through the floor at high debt. Its contours in Figure 7 track the default thresholds. Consumption falls steeply at the default threshold and is nearly flat in debt in the default region, where a higher debt is met with a higher default intensity rather than with lower consumption. The visible steps at a low endowment are where the price crosses the floor and issuance stops.

The drift of debt is positive at low debt and negative at high debt in every endowment state below about one standard deviation above the mean, with a single zero-drift locus, the stationary debt. Down to about one standard deviation below the mean endowment the locus slopes down, so the stationary debt is higher in bad times than in good times. In good times it lies below the no-default threshold, where precaution holds the sovereign short of the fixed cost. In bad times the threshold has fallen below the current debt, and the sovereign in partial default pays the fixed cost regardless. Its stationary debt is then set by the dilution of its own price, inside the default region. The episode ends when output recovers and the threshold rises back above the debt, not when the debt is repaid, so the sovereign borrows while it lasts. In worse states the stationary debt is the debt at which the price reaches the floor and issuance stops. That debt falls with the endowment, so the locus bends back, from $B = 0.67$ to 0.48 at the lowest endowment node. Above about one standard deviation above the mean endowment the drift is negative at every debt level, and the sovereign repays all the way to zero debt, the atom of the ergodic distribution at $B = 0$. The ergodic set is the region swept by the zero-drift locus as the endowment moves. The density in Figure 7 concentrates along the locus, with most of the mass between zero debt and the deterministic debt capacity $B_{\mathrm{pd}}(0, q^{\mathrm{rf}}) = 0.485$ and a tail that extends into the partial-default region at a low endowment. The equilibrium threshold at the average endowment is lower, 0.445, because the price there is 0.85 rather than par. The density thins out above about $B = 0.64$, well before the borrowing limit and short of the peak of the locus at 0.67, which the state approaches only slowly. In the worst states the floor locus, where issuance stops, bounds the ergodic set. Spreads are therefore low most of the time and high in the tail, which is where the model's spread volatility comes from.

Figure 8 shows the response of the economy to a negative shock to the endowment. Starting from the ergodic distribution, we lower z by 0.1 at date zero, about one stationary standard deviation and a ten percent fall in output, and report the expected path of each variable minus its ergodic mean. We compute the expectation with the forward equation (Appendix B.3). Output falls on impact and recovers at the rate μ. Consumption falls by less than output and recovers more slowly, because the sovereign smooths the shock by issuing new debt and withholding debt service. The default intensity jumps on impact and stays elevated for a decade. Debt *rises* for about six years after the shock, by some six percent of annual output at the peak, before it is worked off. Up to the peak, new issues add 8.5 percent of annual output to the debt and

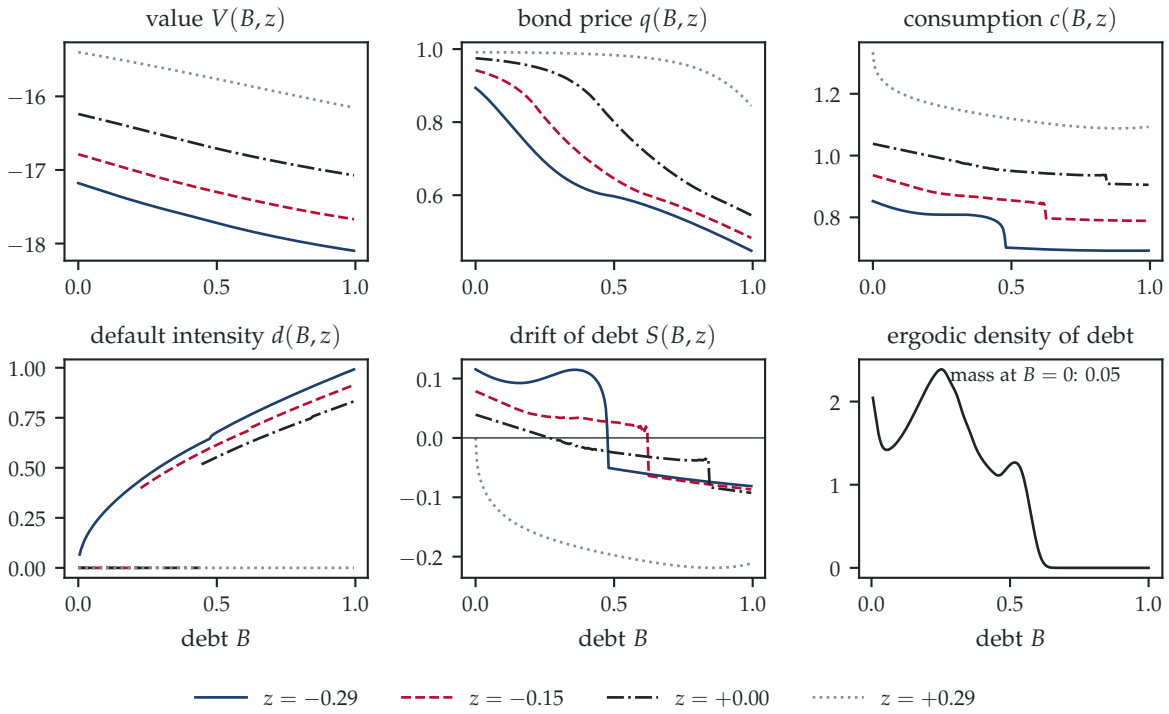


Figure 6: The diffusion model: value, bond price, consumption, default intensity, and drift of debt as functions of debt at four endowment levels, and the ergodic marginal density of debt (the atom at $B = 0$ is reported in the panel).

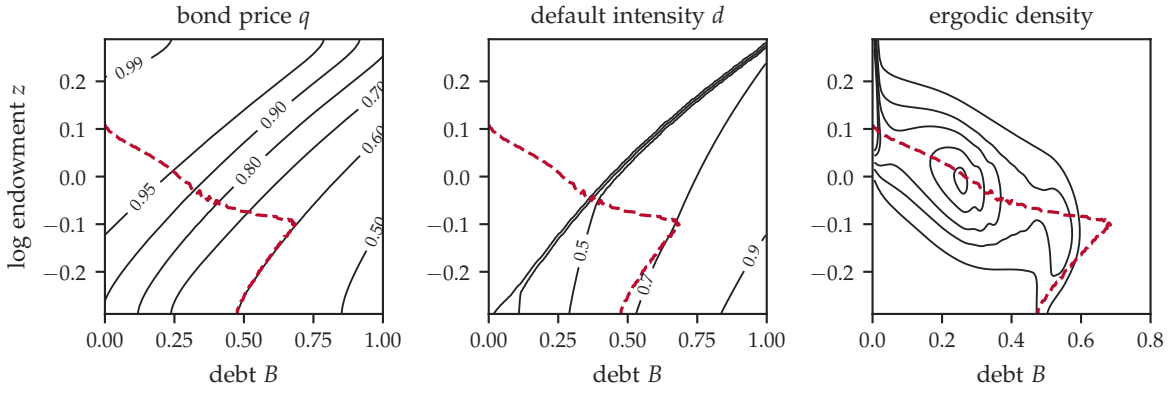


Figure 7: The diffusion model over the state space: contours of the bond price, of the default intensity, and of the ergodic density. Dashed: the zero-drift locus, where debt is stationary.

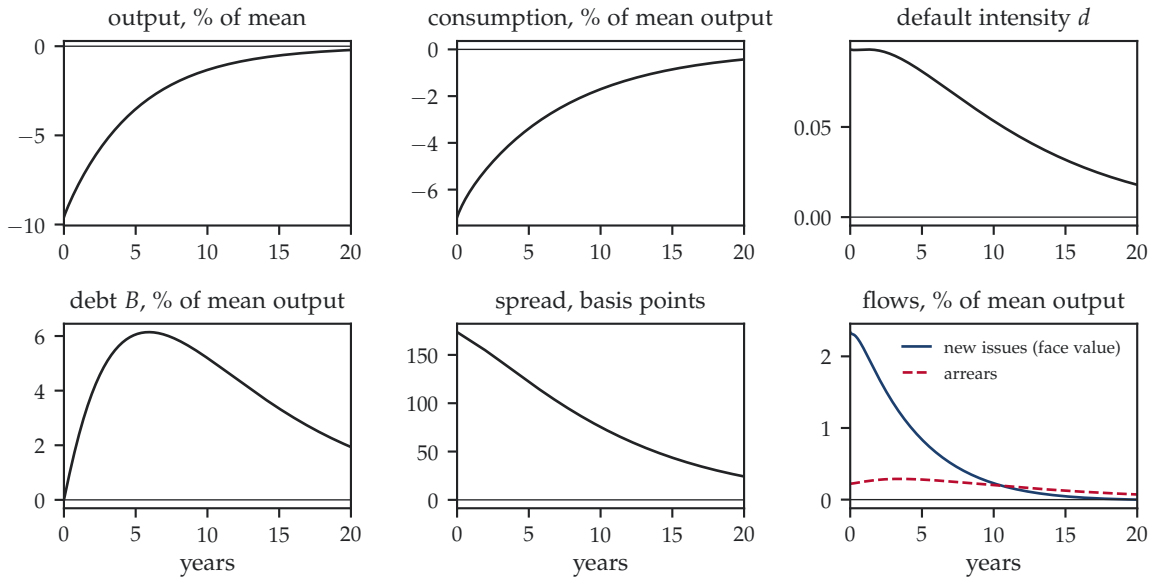


Figure 8: Impulse responses to a negative shock of 0.1 to the log endowment at date zero, from the ergodic distribution: expected path under the forward equation minus the ergodic mean. Flows are in percent of mean annual output, and the spread is in basis points.

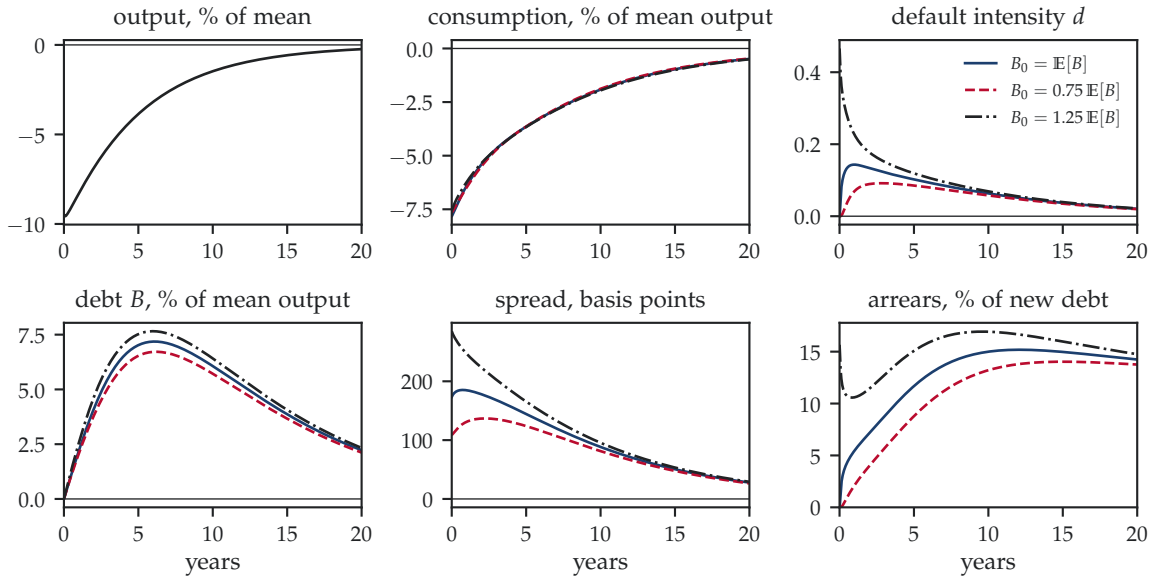


Figure 9: The same shock from three states. The economy is at the mean endowment with debt at its ergodic mean (solid), 25 percent below it (dashed), or 25 percent above it (dash-dotted): expected path under the forward equation from that state minus the path from the same state without the shock. The last panel shows arrears as a share of the gross additions to debt, new issues plus arrears, along the shocked path.

arrears 1.6 percent, the rest of the difference being amortization. The spread rises on impact by about 174 basis points and declines as output recovers. The last panel shows the two flows behind the rise in debt, new issues in face value and arrears. New issues are the larger flow for ten and a half years and decline as debt rises and the price falls. Arrears decline more slowly, because the default intensity stays elevated. As sources of funds the two flows are closer than the panel shows. Withholding raises the full defaulted coupons, of which only the fraction κ adds to the debt, and the proceeds of the issues are below their face value. These are the dynamics of partial default episodes documented by Arellano, Mateos-Planas, and Ríos-Rull (2023): default is protracted, debt rises during the episode, and the economy exits the episode with more debt than it entered it. The sovereign in partial default remains a net issuer, and the recovery of output, not repayment, ends the episode.

Figure 9 applies the same shock at the mean endowment from three debt levels, the ergodic mean and 25 percent below and above it, and reports each response against the path from the same state without the shock. The response of debt is nearly the same from the three states: it rises by some seven percent of output and peaks after six years. The responses of the default intensity and of the spread are not. From above the ergodic mean the shock moves the threshold below the debt at once, the intensity jumps to the first default and declines from there, and the spread rises by more than twice as much as from below. From the ergodic mean the economy enters default over the first year, as the endowment keeps moving and the threshold with it, and from below it enters later and less deeply. This state dependence echoes Fernández-Villaverde, Hurtado, and Nuño (2023), in whose economy the same shock has larger effects when leverage is high. Along all three paths arrears are a small share of the new debt, about 15 percent after the first years, so new issues remain the larger source of funds whatever the state.

The calibration rules out the self-fulfilling partial default of Section 3.3 in the deterministic model with the endowment at its mean. The stationary spread at the first default, 1,286 basis points, exceeds the floor spread of 1,278, so the standard excludes partial-default steady states (Proposition 3), if narrowly. The impatience wedge, 225 basis points, is in any case far below the risk-neutral threshold, 1,286 basis points with these default costs, so a risk-neutral sovereign would not reach such a steady state from zero debt (Proposition 4(a)). Curvature raises the threshold further (Section 3.3). The no-default-basin equilibrium of Theorem 1 is not the only equilibrium at these parameters. The cube-root solution of Theorem 2 continues below capacity, where the sovereign defaults at the first default intensity over the last 0.26 percent of output of borrowing, at a spread of 18 basis points, and Appendix A.4 shows that this is the only other equilibrium along the path from zero debt. Whether the diffusion model admits a self-fulfilling equilibrium is open.

4.4 A Neural-Network Solution

The number of nodes in a tensor grid grows with the power of the number of states. We therefore also solve the diffusion model without a grid, with one neural network of (B, z) for the value

and the price, a physics-informed neural network (Raissi, Perdikaris, and Karniadakis 2019; Sirignano and Spiliopoulos 2018; Duarte, Duarte, and Silva (2024) for continuous-time models with endogenous prices, and Maliar, Maliar, and Winant (2021) and Azinovic, Gaegauf, and Scheidegger (2022) in discrete time). The derivatives come from automatic differentiation, and the policies from the same Hamiltonian maximization as in the grid scheme, applied at random points of the state space. In two dimensions the method is less precise than the grid scheme, but its cost does not grow with a grid, so it extends to models with more states, such as reserves, a maturity structure, or shocks on the lenders' side.

Minimizing the stationary residuals alone fails in this model: the training converges to functions that are not the equilibrium, with the wrong drift over much of the state space. The residual loss does not select the constrained viscosity solution of Section 4.1, a known failure of these methods on equations with transport terms and kinked solutions (Krishnapriyan et al. 2021). We therefore train the network by the implicit pseudo-time iteration of the grid scheme, with the grid replaced by quasi-random collocation points (Appendix B.4).

The network solution is close to the grid solution (Figure 13 and Table 4 in Appendix B.4). The default thresholds agree to within a grid step, and the drift of debt has the same sign on 98.2 percent of the grid. Mean debt is 0.257 against 0.257, the targeted moments agree to within a hundredth, and the calibration loss is 0.102 against 0.096. The price differs by at most 0.006, in the partial-default region at middle endowments, where it falls steeply in debt.

5 Conclusion

In the standard model of sovereign default, the value of market access limits how much a sovereign can borrow, and the limit is the outcome of the whole dynamic problem. In the partial default model of Arellano, Mateos-Planas, and Ríos-Rull (2023) the limit is a static comparison. Withholding part of the debt service raises funds at a price fixed by the recovery on arrears, on terms that do not deteriorate with the sovereign's creditworthiness, against the output cost of default. The debt capacity that follows is a closed form in the default costs, the recovery, and the maturity of the debt, and it does not depend on the sovereign's preferences. The same structure fixes the anatomy of a crisis. The fixed cost makes the first default a jump, and a shock that moves the threshold below the sovereign's buffer triggers it. Once the fixed cost is being paid, only the dilution of the sovereign's own price restrains borrowing, so the sovereign keeps issuing, and the episode ends when output recovers. A sovereign above its capacity instead works its debt down along a cube-root law. Pessimism alone can start a crisis, when the gain from front-loading borrowed funds covers the output cost of the cheapest default, and the crisis unfolds over years without a run. Curvature raises that bar, and the calibrated model is far from it.

These results come from a model that is tractable and quantitative at once. Calibrated to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023), with their weights, the continuous-time model fits them at least as well as the discrete-time original. Because the sovereign never loses

market access, there is no stopping problem. A single Hamilton–Jacobi–Bellman equation and a single linear pricing equation describe the equilibrium at every state, the default rule is explicit given the price, and the deterministic model reduces to a pair of ordinary differential equations whose behavior at capacity we characterize in closed form. The underwriting standard makes the borrowing limit irrelevant and binds on less than one percent of the ergodic mass.

Several avenues remain. On the theoretical side, self-fulfilling partial default is nearly settled for a risk-neutral sovereign, up to a stability and a continuation condition. With curvature or shocks we have computed the threshold but not proved it, and a proof would complete the account of slow-moving debt crises with arrears. With shocks we have also studied only steady debts that do not move with the endowment, and we have studied curvature and uncertainty only one at a time. Existence of the coupled equilibrium in the stochastic models is open. On the quantitative side, a tensor grid becomes expensive with a third state and infeasible with a fourth. The neural-network solution of Section 4.4 reproduces the grid solution in two dimensions, and extending it to models with reserves, a maturity structure, or shocks on the lenders' side is the natural next step. Such methods can exploit the closed-form policies and the concentration of the stationary distribution around the thresholds.

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Appendix to "Debt Capacity and the Anatomy of Sovereign Crises"

by Sangdong Kim and Gabriel Mihalache

A Additional Results and Proofs

A.1 Results and Proofs for Section 3.1

We also state and prove here the four properties of the price and the value summarized in Section 3.1.

Proof of Lemma 1. With $\ell = n - \kappa(\delta+\lambda)dB$, (1) gives $c = \phi e^z - (1-d)(\delta+\lambda)B + qn - \kappa q(\delta+\lambda)dB = \phi e^z - (\delta+\lambda)B + qn + (1-\kappa q)(\delta+\lambda)Bd$. At a given drift S, $n = S + \delta B$, so $c = \mathcal{R}(d) + qS$ with $\mathcal{R}$ as stated, and $\mathcal{R}_d$ follows from (3). □

Proof of Proposition 1. Fix any admissible drift S. The objective $u(\mathcal{R}(d) + qS) + SV_B$ depends on d only through $\mathcal{R}(d)$, u is increasing, and the feasibility constraint $c \geq c_{\min}$ is relaxed by a larger $\mathcal{R}(d)$. Hence for every S the best d maximizes $\mathcal{R}$, and so does the d-component of the joint maximizer. □

Proof of Proposition 2. On $(0,1]$, $\mathcal{R}'(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B$ is strictly decreasing with $\mathcal{R}'(0^+) > 0$, so $\mathcal{R}$ is strictly concave with maximizer d_{int} of (8). The gain from defaulting is $\mathcal{R}(d_{\text{int}}) - \mathcal{R}(0) = \Gamma(B,z,q) - (1-F(z))e^z$, which gives (13). Since d_{int} maximizes $(1-\kappa q)(\delta+\lambda)Bd - \gamma_0 F(z)e^z d^{\gamma_1}$ on $[0,1]$, the envelope theorem gives $\partial_B \Gamma = (1-\kappa q)(\delta+\lambda)d_{\text{int}} > 0$ and $\partial_q \Gamma = -\kappa(\delta+\lambda)Bd_{\text{int}} < 0$ for $B > 0$, whether d_{int} is interior or capped, and $\Gamma = 0$ at $B = 0$; (a), (b), and (d) follow. At $B = B_{\text{fd}}(z,q)$, where $d_{\text{int}} = 1$, $\Gamma = (\gamma_1 - 1)\gamma_0 F(z)e^z$, so $B_{\text{pd}} < B_{\text{fd}}$ if and only if $1 - F(z) < (\gamma_1 - 1)\gamma_0 F(z)$, that is $d_{\text{first}} < 1$. Eliminating B between $\Gamma = (1-F)e^z$ and (8) gives (14), and eliminating d_{int} gives (15); when $d_{\text{first}} \geq 1$ the jump is from 0 to 1, and $\mathcal{R}(1) = \mathcal{R}(0)$ gives the second formula. □

Proposition A.1 (the bond price)**.** *In the general model, fix measurable policies (d,S), let (B_t, z_t) be the induced process, and define q by (9), the bounded solution of (10) wherever that equation holds. Then $0 \leq q \leq q^{\text{rf}}$; $q(B,z) = q^{\text{rf}}$ if and only if $d_t = 0$ for almost every t, almost surely, along the path from (B,z); and $q(B,z) = 0$ if and only if $d_t = 1$ for almost every t, almost surely.*

Proof of Proposition A.1. (9) is the Feynman–Kac representation of (10) with discount rate $\xi(d_t) \geq \xi(1) > 0$. The bound: $(1-d)(\lambda+\delta) \leq q^{\text{rf}}\xi(d)$ for all $d \in [0,1]$ is equivalent to $d\,\xi(1) \geq 0$; integrating, $q \leq q^{\text{rf}}\,\mathbb{E}\int_0^\infty \xi(d_t)e^{-\int_0^t \xi}dt = q^{\text{rf}}$, with equality iff $d_t = 0$ for almost every t, almost surely. The integrand of (9) is nonnegative and vanishes exactly where $d_t = 1$, so $q = 0$ if and only if $d_t = 1$ for almost every t, almost surely. □

Lemma A.1 (the price is nonincreasing in the default path)**.** *In the general model, let d and d' be two adapted default processes with $d'_t \le d_t$ for all t, almost surely. Then the Feynman–Kac values* (9) *satisfy $q(d') \ge q(d)$.*

Proof. For $\theta \in [0,1]$ let $d^\theta = d' + \theta(d - d')$ and $Q(\theta)$ the corresponding value. Differentiating under the integral and collecting the effect of d^θ_t on the payment at t and on the discounting of the stream after t (through $\xi'(d) = -\kappa(\delta + \lambda)$), $Q'(\theta) = \mathbb{E}\int_0^\infty e^{-\int_0^t \xi(d^\theta)}(\lambda + \delta)[\kappa q^\theta_t - 1](d_t - d'_t)\,dt \le 0$, since $\kappa q^\theta_t \le \kappa q^{\mathrm{rf}} \le 1$. □

Proposition A.2 (continuity of the price along crossing flows)**.** *In the deterministic, chain, and diffusion models, fix a state (B_0, z_0) and suppose that on a neighborhood of it the equilibrium drift is negative and bounded away from zero, so that from every nearby state debt moves down through B_0 in a time that vanishes with the distance to B_0. Then $q(\cdot, z_0)$ is continuous at B_0: with no further condition in the deterministic and chain models, and in the diffusion model whenever $q(B, \cdot)$ is continuous at z_0 uniformly in B on a neighborhood of B_0. The same holds with positive drift. Hence q can be discontinuous in B only at points where the drift vanishes or changes sign that are not reached in finite time from both sides (and, in the diffusion model, where q is not continuous in z).*

At a stationary point of debt that is approached from below and left from above, the paths on the two sides go to different places, Proposition A.2 does not apply, and the price can jump.

Proof of Proposition A.2. With the drift bounded away from zero, from $(B_0 + x, z_0)$ debt reaches B_0 at a time $\tau(x) \to 0$ on every path along which z stays in the neighborhood. By (9) and the strong Markov property, $q(B_0 + x, z_0) = \mathbb{E}\left[\int_0^\tau e^{-\int_0^t \xi}(1 - d)(\lambda + \delta)\,dt + e^{-\int_0^\tau \xi} q(B_\tau, z_\tau)\right]$. The integral is $O(\tau)$ and the discount factor tends to one. In the deterministic model $(B_\tau, z_\tau) = (B_0, z_0)$. For a chain, z switches before τ with probability $O(\tau)$, and on the complement $(B_\tau, z_\tau) = (B_0, z_0)$; since q is bounded, the expectation tends to $q(B_0, z_0)$. For the diffusion, z leaves the neighborhood before τ with vanishing probability and otherwise $B_\tau = B_0$ with $z_\tau - z_0 = O_p(\tau^{1/2})$, so the expectation tends to $q(B_0, z_0)$ when $q(B_0, \cdot)$ is continuous at z_0. This gives continuity from the right. From the left, start at (B_0, z_0) itself: debt reaches $B_0 - x$ at a time $\tau(x) \to 0$, and the same decomposition gives $q(B_0, z_0) = O(\tau) + \mathbb{E}\left[e^{-\int_0^\tau \xi} q(B_0 - x, z_\tau)\right]$. In the deterministic and chain models $z_\tau = z_0$ with probability $1 - O(\tau)$, so $q(B_0 - x, z_0) \to q(B_0, z_0)$. In the diffusion model $q(B_0 - x, z_\tau) - q(B_0 - x, z_0) \to 0$ in probability by the uniform continuity in z, and the same conclusion follows. With positive drift the two sides exchange roles. □

Proposition A.3 (autarky and its removal)**.** *In the general model with z in a bounded set: under a zero price, full default is optimal if and only if B is at least $B_{\mathrm{fd}}(z, 0)$ and above $B_{\mathrm{pd}}(z, 0)$, and, without the underwriting standard, $(q \equiv 0, d \equiv 1)$ is an equilibrium of the model restricted to any invariant set contained in $\{B \ge \max\{B_{\mathrm{fd}}(z, 0), B_{\mathrm{pd}}(z, 0)\}\}$, a knife edge. Invariance requires the sovereign to issue at least aB at a zero price. Under the convention that no bonds are traded at a zero price, $q > 0$ at every state in every equilibrium.*

Proof of Proposition A.3. The default rule under $q = 0$ is Proposition 1 with $q = 0$. With $q = 0$ issuance does not enter the budget, so any ℓ is optimal. With $d = 1$ no payment is made and $q = 0$ is the lenders' valuation. Invariance is needed for self-consistency, and with $d = 1$ debt decays at rate a unless $\ell \geq aB$. For the removal: suppose $q(B_0, z_0) = 0$. By Proposition A.1, $d = 1$ a.e. along the path, so $q = 0$ along the path, so $\ell = 0$ along the path and $B_t = B_0 e^{-at}$. Since $B_{\mathrm{fd}}(z, 0) = \gamma_0\gamma_1 F(z)e^z/(\delta + \lambda)$ is bounded below by a positive constant on a bounded set of z, so is $\max\{B_{\mathrm{fd}}(z, 0), B_{\mathrm{pd}}(z, 0)\}$, and in finite time B_t falls below $\inf_z \max\{B_{\mathrm{fd}}(z, 0), B_{\mathrm{pd}}(z, 0)\}$. There full default is not optimal. With no issuance at a zero price, consumption is $\phi(d, z)e^z - (1 - d)(\delta + \lambda)B$ and debt decays at rate $a(d)$. Lowering d for an instant, slightly below 1 if $B < B_{\mathrm{fd}}(z, 0)$, where $\mathcal{R}_d(1; B, z, 0) < 0$, or to 0 if $B < B_{\mathrm{pd}}(z, 0)$, raises consumption now and lowers debt, while returning to $d = 1$ afterwards gives the same consumption $(1 - \gamma_0)F(z)e^z$ whatever the debt. The deviation is strictly better, a contradiction. □

The next result gives the sign of V_B that Proposition A.6 uses.

Proposition A.4 (the value is nonincreasing in debt)**.** *In the general model, if for every z the price $q(\cdot, z)$ is nonincreasing and strictly positive, and the drift cap is slack along the optimal path and along the mimicking path of the proof, then $V(\cdot, z)$ is nonincreasing, with or without buybacks and with or without the underwriting standard.*

Proof of Proposition A.4. Let $B' < B$ and let (d_t, ℓ_t) be the realized default and issuance path of the optimal policy from (B, z) along the realized z-path, with debt B_t. The sovereign at B' runs the same default path and the issuance $\ell'_t = \ell_t$ if $\ell_t \geq 0$, $\ell'_t = \ell_t\, q(B_t, z_t)/q(B'_t, z_t)$ if $\ell_t < 0$ (the same bonds when issuing, the same expenditure when buying back). The override $\ell'_t = 0$ applies if $B'_t = 0$ and $\ell_t < 0$. First, $B'_t \geq 0$ at all dates: at $B'_t = 0$ the rule gives $\ell'_t \geq 0$, hence $\dot{B}'_t = \ell'_t - a(d_t)B'_t \geq 0$. Next, the gap $b_t = B_t - B'_t$ solves $\dot{b} = (\ell_t - \ell'_t) - a(d_t)b$, with $b_0 > 0$. If b were negative on some interval $(t_2, t_1]$ with $b(t_2) = 0$, then on it $B'_t > B_t \geq 0$, so the override is inactive, and $q(B'_t, z_t) \leq q(B_t, z_t)$ because the price is nonincreasing. Hence $\ell'_t = \ell_t$ when $\ell_t \geq 0$ and $\ell'_t \leq \ell_t$ when $\ell_t < 0$, so $\ell_t - \ell'_t \geq 0$. With $-a(d_t)b > 0$ this gives $\dot{b} > 0$ on the interval and $b(t_1) > 0$, a contradiction. Hence $0 \leq B'_t \leq B_t$: the state constraints hold, and the standard is respected because $q(B'_t, z_t) < \underline{q}$ implies $q(B_t, z_t) < \underline{q}$, hence $\ell_t \leq 0$ and $\ell'_t \leq 0$. Consumption satisfies $c'_t - c_t = (1 - d_t)(\delta + \lambda)b_t + [q(B'_t, z_t)\ell'_t - q(B_t, z_t)\ell_t] \geq 0$. The bracket is 0 when $\ell_t < 0$, $\ell_t[q(B'_t, z_t) - q(B_t, z_t)] \geq 0$ when $\ell_t \geq 0$, and $-q(B_t, z_t)\ell_t > 0$ when the override is active ($B'_t = 0$ and $\ell_t < 0$). The default costs coincide, so the mimicking path gives the sovereign at B' at least the utility of the optimal path from B. □

A.2 The Model at High Debt

At high debt the model has the knife-edge autarky equilibrium of Proposition A.3, which a convention removes. Where default carries no fixed cost, debt also accumulates into the borrowing limit, and the underwriting standard removes this accumulation.

Accumulation without a fixed cost. In the no-fixed-cost region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt $-V_B$ close to zero, while the price is positive because output will recover. By (7) it issues at any positive price. Even without the borrowing limit it could not keep doing so forever: debt that stayed above every full-default threshold would never be serviced again and would be worthless, which the convention against trade at a zero price rules out (Proposition A.3).

Proposition A.5 (debt is not trapped above the full-default thresholds)**.** *In the general model with z in a bounded set Z and without the underwriting standard, drop the borrowing limit, letting debt range over $[0, \infty)$, impose the convention that no bonds are traded at a zero price, and let*

$$\hat{B} := \sup_{z \in Z} \max\{B_{\mathrm{fd}}(z, q^{\mathrm{rf}}), B_{\mathrm{pd}}(z, q^{\mathrm{rf}})\}.$$

In every equilibrium, from every state (B, z), debt falls to $\hat{B}$ or below at some date with positive probability.

In the deterministic model, where paths are monotone, debt never exceeds $\max\{B_0, \hat{B}\}$. Without the borrowing limit, whether an equilibrium exists, and what it looks like at high debt, is open. With the limit, the accumulation runs into it. Where default carries no fixed cost, the sovereign in full default borrows up to $\bar{B}$, whatever its level. The limit stops the growth and keeps the price at high debt positive, so its level, normally innocuous, shapes the equilibrium at high debt. In models that leave the limit implicit, the top of the debt grid plays this role, and the equilibrium at high debt depends on where the grid ends. The entry point of the accumulation is a stationary point approached from below (the sovereign borrows up to a self-imposed limit just below the full-default line) and left from above, so Proposition A.2 does not apply there.

The underwriting standard. With issuance switched off below $\underline{q}$, the drift beyond the full-default line is at most $-a(d)B < 0$, and the full-default line is crossed downward from both sides. In the deterministic model the price is continuous across it whenever the drift is also negative just below it (Proposition A.2). In every model, any $\bar{B}$ above the full-default thresholds is an exact truncation (Proposition A.7). With the standard, the level of the borrowing limit therefore does not matter. The standard also affects the default decision, because its constraint, $\ell \leq 0$, depends on d through $\ell = S + a(d)B$. We therefore re-examine the separation of Proposition 1 where the standard binds.

Proposition A.6 (the default decision under the standard)**.** *In the general model with the underwriting standard, fix a state with $q = q(B, z) < \underline{q}$, $B > 0$, and $V_B < 0$, and let (d^*, S^*) be the unconstrained maximizer with issuance $\ell^* = S^* + a(d^*)B$.*

(a) If $\ell^* \leq 0$ the standard is slack and Proposition 1 holds verbatim.

(b) If $\ell^* > 0$ and $d^* > 0$ (the unconstrained optimum is in partial default) the constrained maximum has

$\ell = 0$, *hence* $S = -a(d)B$, *and the default intensity maximizes*

$$J(d) := u(\phi(d,z)e^z - (1-d)(\delta+\lambda)B) - a(d)\,B\,V_B.$$

At a maximizer $d^{**} \in (0,1]$, *the corner* $d^{**} = 1$ *included, with* $c^{**} := \phi(d^{**},z)e^z - (1-d^{**})(\delta+\lambda)B$,

$$d^{**} = d_{\text{int}}(B,z,\hat{q}), \qquad \hat{q} := \frac{-V_B}{u'(c^{**})} \in (0,q) :$$

the unconstrained rule (8) *evaluated at the sovereign's own shadow price of a unit of debt instead of the market price. Since* d_{int} *is nonincreasing in the price,* $d^{**} \geq d_{\text{int}}(B,z,q)$, *strictly whenever* $d_{\text{int}}(B,z,q) < 1$.

(c) *If* $\ell^* > 0$ *and* $d^* = 0$ *(the unconstrained optimum repays in full while issuing), repaying in full forces* $\ell = 0$ *and gives* $J(0)$, *while defaulting gives at least* $J(d^{**})$, *with* d^{**} *the maximizer of* J *on* $(0,1]$ *and* $\hat{q} := -V_B/u'(c^{**})$; *a default may also come with a buyback, if it frees enough resources. The constrained optimum defaults whenever* $B > B_{\text{pd}}(z,\hat{q})$, *since* J *is the objective at the shadow price, and only if* $B > B_{\text{pd}}(z,q_0)$, *with* $q_0 := -V_B/u'(e^z - (\delta+\lambda)B)$ *the shadow price at full repayment. When the default branch also wants to issue,* $\hat{q} < q$ *and the threshold falls with the price, so the standard can induce a default that the market-price rule does not; when it would buy back,* $\hat{q} > q$ *and the sufficient condition is not met.*

Proposition A.7 (the standard removes the accumulation)**.** *In the general model with* z *in a bounded set* Z *and the underwriting standard, in every equilibrium in which* $q < \underline{q}$ *at every state where the sovereign is in full default:*

(a) *in full default the drift is* $S \leq -aB < 0$*: the full-default region is transient, and in the deterministic model, if* $V_B \leq 0$, *it is left in time at most* $a^{-1}\ln\big(B/\max\{B_{\text{fd}}(0,0), B_{\text{pd}}(0,0)\}\big)$;

(b) *in the deterministic model the drift has no zero in* $\{q < \underline{q},\ B > 0\}$; *a stationary point on the floor locus* $\{q = \underline{q}\}$ *has* $q = q_r(d^\circ) = \underline{q}$, *the equality case of Proposition 3, so it exists only if* $q_r(d_{\text{first}}) \geq \underline{q}$, *and the price is continuous across it whenever it is reached in finite time from both sides;*

(c) *any borrowing limit* $\bar{B} > \hat{B}$, *with* $\hat{B}$ *as in Proposition A.5, is an exact truncation: the drift points inward at* $\bar{B}$ *in every endowment state and the stationary distribution puts no mass on it;*

(d) *the thresholds of Proposition 2, as functions of the state and the price, are unaffected; in the deterministic no-default-basin equilibrium in which debt reaches* B^* *from above in finite time, where* $q \equiv q^{\text{rf}}$ *on* $[0,B^*]$, *so are the no-default region and the steady state of Theorem 1, whenever* $\underline{q} < q^{\text{rf}}$. *In the stochastic models a state without default can be priced below the floor, since default is expected after future shocks, and there the standard can change the default decision (Proposition A.6(c));*

(e) *the level of the borrowing limit does not affect the equilibrium on the states below it: for any two limits* $\bar{B}_2 > \bar{B}_1 > \hat{B}$, *the restriction to* $[0,\bar{B}_1]$ *of an equilibrium with limit* $\bar{B}_2$ *is an equilibrium with limit*

$\bar{B}_1$, and every equilibrium with limit $\bar{B}_1$ in which buybacks are not optimal in full default above $\bar{B}_1$, $-V_B \le qu'(c)$ there, extends to one with limit $\bar{B}_2$;

(f) for any limit $\bar{B} \ge B^\sharp := \max\Big\{0,\ \sup\{B :\ q(B,z) \ge \underline{q} \text{ for some } z\}\Big\}$, and without the hypothesis on the full-default prices: whenever $B_t > B^\sharp$ the drift satisfies $S_t \le -aB_t$, the set $[0, B^\sharp] \times Z$ is invariant, and every stationary distribution is supported in it.

Under the hypothesis on the full-default prices, $B^\sharp \le \hat{B}$, so by (e) neither $B^\sharp$ nor the equilibrium on $[0, B^\sharp]$ depends on a limit above $\hat{B}$. The computations use $\bar{B} = 1$, below the full-default thresholds at the risk-free price in every state, so there the irrelevance of the limit is a numerical finding (Appendix C).

Proofs.

Proof of Proposition A.5. Suppose that from (B_0, z_0) debt stays above $\hat{B}$ at every date almost surely. Without the standard the default decision is the static rule of Propositions 1 and 2 at every date. The thresholds B_{pd} and B_{fd} are nondecreasing in the price and $q \le q^{\text{rf}}$ (Proposition A.1), so $B_t > \hat{B} \ge \max\{B_{\text{fd}}(z_t, q_t), B_{\text{pd}}(z_t, q_t)\}$ and the sovereign is in full default at every date (Proposition 2(d)). No payment is ever made along the path, and (9) gives $q(B_0, z_0) = 0$, which contradicts Proposition A.3, whose argument does not use the upper bound $\bar{B}$. $\hat{B}$ is finite because the thresholds are continuous in z on the bounded set Z. □

Proof of Proposition A.6. Parametrize the choice by (d, ℓ). The objective is $\mathcal{G}(d, \ell) = u(\tilde{\mathcal{R}}(d) + q\ell) + (\ell - a(d)B)V_B$ with $\tilde{\mathcal{R}}(d) = \phi(d, z)e^z - (1 - d)(\delta + \lambda)B$ the resource function at a zero price, and the constraint is $\ell \le 0$, which no longer involves d. On $(0, 1] \times \mathbb{R}$, $\tilde{\mathcal{R}}$ is concave in d, u is concave and increasing, and the last term is linear, so $\mathcal{G}$ is jointly concave. (a) is immediate. For (b), let (d_c, ℓ_c) be a constrained maximizer on the $d > 0$ branch with $\ell_c < 0$. The segment from it to (d^*, ℓ^*) crosses $\{\ell = 0\}$ at a point at which, by concavity, $\mathcal{G}$ is at least $\min\{\mathcal{G}(d_c, \ell_c), \mathcal{G}(d^*, \ell^*)\} = \mathcal{G}(d_c, \ell_c)$, so that point is a constrained maximizer with $\ell = 0$ on the $d > 0$ branch. The $d = 0$ branch is treated separately, since $\mathcal{G}$ is not concave across the fixed-cost jump at $d = 0$. Its own unconstrained maximizer has issuance $\ell_0^* > \ell^* > 0$, because $d^* > 0$ maximizes $\mathcal{R}$ and $a(d^*) < \delta$. So the same comparison along $\{d = 0\}$ gives $\ell = 0$ there as well, and the constrained optimum is the better of $J(d^{**})$ and $J(0)$. It is $J(d^{**})$: the concavity bound in the proof of (c) below gives $J(d^{**}) - J(0) \ge u'(c^{**})[\mathcal{R}(d^{**}; B, z, \hat{q}) - \mathcal{R}(0; B, z, \hat{q})] > 0$, since $B > B_{\text{pd}}(z, q) \ge B_{\text{pd}}(z, \hat{q})$. On $\{\ell = 0\}$, $c = \tilde{\mathcal{R}}(d)$ and $S = -a(d)B$, which gives J. At an interior maximizer $J'(d) = u'(c)\tilde{\mathcal{R}}'(d) + \kappa(\delta + \lambda)BV_B = 0$, that is $\tilde{\mathcal{R}}'(d) = \kappa(\delta + \lambda)B\hat{q}$, the first-order condition of $\mathcal{R}(\cdot; B, z, \hat{q})$. At the corner $d^{**} = 1$, $J'(1) \ge 0$, that is $\tilde{\mathcal{R}}'(1) \ge \kappa(\delta + \lambda)B\hat{q}$, which is $d_{\text{int}}(B, z, \hat{q}) = 1$. In both cases $d^{**} = d_{\text{int}}(B, z, \hat{q})$, and the comparison with $d_{\text{int}}(B, z, q)$ follows from $\hat{q} < q$ and the monotonicity of (8), which is strict below the cap. Finally $\partial_\ell \mathcal{G}(d^{**}, 0) = qu'(c^{**}) + V_B > 0$ because the constraint binds strictly, which is $\hat{q} < q$, and $\hat{q} > 0$ because $V_B < 0$ by hypothesis. For (c), on the $d = 0$ branch the unconstrained issuance is $\ell^* > 0$, so the comparison of (b) along $\{d = 0\}$ gives $\ell = 0$ and the value

$J(0)$. On the $d > 0$ branch the unconstrained issuance can have either sign. If it is positive, the argument of (b) gives $\ell = 0$ and the branch value $J(d^{**})$. If it is nonpositive, the constraint is slack on the branch, and its value $\max_{d\in(0,1],\ell\le 0}\mathcal{G}(d,\ell)$ is at least $\mathcal{G}(d^{**},0) = J(d^{**})$. Either way the best default is worth at least $J(d^{**})$. By the concavity of u, $J(d^{**}) - J(0) \ge u'(c^{**})\big[\tilde{\mathcal{R}}(d^{**}) - \tilde{\mathcal{R}}(0)\big] + \kappa(\delta+\lambda)d^{**}BV_B = u'(c^{**})[\mathcal{R}(d^{**};B,z,\hat{q}) - \mathcal{R}(0;B,z,\hat{q})]$. Since $d^{**} = d_{\text{int}}(B,z,\hat{q})$ maximizes $\mathcal{R}(\cdot;B,z,\hat{q})$ on $(0,1]$, the bracket is positive whenever $B > B_{\text{pd}}(z,\hat{q})$ (Proposition 2), so the constrained optimum defaults there. The condition is sufficient, not necessary: the bound uses the concavity of u, and the default region under the standard can extend slightly below $B_{\text{pd}}(z,\hat{q})$. For the necessary condition, let $c_0 := e^z - (\delta+\lambda)B$ and $q_0 := -V_B/u'(c_0)$. The unconstrained issuance at $d = 0$ is positive, so $q > q_0$. By the concavity of u at c_0, $\mathcal{G}(d,\ell) - \mathcal{G}(0,0) \le u'(c_0)[\mathcal{R}(d;B,z,q_0) - \mathcal{R}(0;B,z,q_0)] + \ell\,u'(c_0)(q-q_0)$, whose last term is nonpositive for $\ell \le 0$, so a default can be optimal only if $B > B_{\text{pd}}(z,q_0)$. In the buyback case the default branch's optimum has $\ell < 0$, so $\partial_\ell\mathcal{G}(d^{**},0) < 0$, that is $\hat{q} > q$. □

Proof of Proposition A.7. (a) With $d = 1$ and $\ell \le 0$, $S = \ell - aB \le -aB$; integrate. Where the standard is slack, full default requires $B \ge B_{\text{fd}}(0,q)$ and $B > B_{\text{pd}}(0,q)$; where it binds, any default requires $B > B_{\text{pd}}(0,q_0)$ and full default requires $B \ge B_{\text{fd}}(0,\hat{q})$ (Proposition A.6), with $q_0, \hat{q} \ge 0$ because $V_B \le 0$. All these thresholds are nondecreasing in the price, so in the deterministic model debt has left the full-default region by the time it falls below $\max\{B_{\text{fd}}(0,0), B_{\text{pd}}(0,0)\}$. (b) In $\{q < \underline{q}\}$ the drift is $S \le -a(d)B \le -aB < 0$ for every d, so there is no zero there. A stationary point B° on the floor locus has $q = \underline{q}$ by definition and $q = q_r(d^\circ)$ by (9) with constant d°, and Proposition 3 gives $q_r(d_{\text{first}}) \ge q_r(d^\circ)$. If the paths from the two sides reach it in finite time, the Feynman–Kac value (9) from either side converges to the value at the point, as in the proof of Proposition A.2. (c) At $\bar{B}$ the sovereign is in full default, hence $S < 0$ there in every state, and a state with inward drift carries no stationary mass. (d) The thresholds of Proposition 2 are functions of (B,z,q) and do not involve the restriction. In the no-default-basin equilibrium $q \equiv q^{\text{rf}}$ on $[0,B^*]$ by Theorem 1, and the restriction binds only where $q < \underline{q} < q^{\text{rf}}$, so it is slack on the basin and on a right neighborhood of B^*, where q is continuous. (e) Above $\hat{B}$ the sovereign is in full default in every state at any price $q \le q^{\text{rf}}$. Where the standard is slack, this follows from Proposition 2(d), since the thresholds are nondecreasing in the price. Where it binds, it follows from Proposition A.6, since d_{int} is decreasing in the price, the shadow price is below q, and defaulting beats repaying at the shadow price above $B_{\text{pd}}(z,q^{\text{rf}})$. In full default the price is below the floor by hypothesis, so $\ell \le 0$ and every feasible drift satisfies $S \le -aB < 0$. Hence from any state with $B \le \bar{B}_1$ no feasible path, under either limit, rises above $\max\{B,\hat{B}\} \le \bar{B}_1$, and the constraint $B \le \bar{B}_1$ never restricts the sovereign. Given the price on $[0,\bar{B}_1]$, the feasible paths from them, and with them the value and the optimal policies, are the same under both limits. The price on $[0,\bar{B}_1]$, the Feynman–Kac value (9) along paths that stay in $[0,\bar{B}_1]$, is the same as well. So the restriction of an equilibrium with limit $\bar{B}_2$ is an equilibrium with limit $\bar{B}_1$. Conversely, given an equilibrium with limit $\bar{B}_1$, define the value on $(\bar{B}_1,\bar{B}_2]$ as that of the problem in which the sovereign, in full default, chooses $\ell \le 0$ until debt reaches $\bar{B}_1$, which it does in time

at most $a^{-1}\ln(\bar{B}_2/\bar{B}_1)$, and then continues with the value of the given equilibrium. Define the price there by (9) along the optimal paths, $q(B,z) = \mathbb{E}\left[e^{-\xi(1)\tau}q(\bar{B}_1, z_\tau)\right]$, where $\tau = a^{-1}\ln(B/\bar{B}_1)$ is the time at which debt reaches $\bar{B}_1$, since under the proviso the sovereign does not buy back and no payment is made before τ. Without the proviso, the value and the price on the extension would have to be solved jointly. This price is below the floor because $q(\bar{B}_1, \cdot) < \underline{q}$, so the standard forbids issuance there, as the construction assumes, and the extended pair is an equilibrium with limit $\bar{B}_2$. (f) Debt rises only when $\ell > 0$, which the standard permits only at $q \geq \underline{q}$, hence only at $B \leq B^\sharp$. Above $B^\sharp$, $\ell \leq 0$ and $S = \ell - a(d)B \leq -aB$. The rest follows from the continuity of B_t. □

A.3 Results and Proofs for Section 3.3

Proof of Theorem 1. From any $B \leq B^*$ the path rises to B^* and stays, with $d \equiv 0$. Then (9) gives $q = q^{\mathrm{rf}}$, and consumption at B^* is $1 - (\delta+\lambda)B^* + \delta B^* = 1 - rB^*$ forever. At B^* the price is q^{rf} and $d = 0$ is optimal, so $B^* \leq B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$. Suppose $B^* < B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$. Since debt reaches B^* from above in finite time, (9) gives $q(B^*+x) \geq e^{-(r+\delta)\tau(x)}q^{\mathrm{rf}} \to q^{\mathrm{rf}}$, and $B_{\mathrm{pd}}(0,\cdot)$ is continuous and increasing, so for small x, $B^*+x < B_{\mathrm{pd}}(0, q(B^*+x))$ and $d = 0$ at B^*+x. Along the path down to B^*, then, $d \equiv 0$ and $q(B^*+x) = q^{\mathrm{rf}}$ exactly. Thus $q \equiv q^{\mathrm{rf}}$ and $d \equiv 0$ on a neighborhood of B^*, where the sovereign faces the constant risk-free price with no default. There, staying at B^* is not optimal because $\rho > r$. The round trip that borrows at rate $m = \iota\tau$ for a time τ and repays at the same rate for the next τ perturbs consumption by $\Delta c_t = m - rmt$ on $[0,\tau]$ and $-m - rm(2\tau - t)$ on $[\tau, 2\tau]$, with first-order gain $u'(\bar{c})\int_0^{2\tau} e^{-\rho t}\Delta c_t\, dt = (\rho - r)u'(\bar{c})\iota\tau^3 + O(\iota\tau^4)$ and second-order loss $O(\iota^2\tau^3)$. For small ι the gain dominates. Since V is the value of the sovereign's problem, staying at B^* is then not optimal, contradicting $S(B^*) = 0$. □

Proof of Theorem 2. Step 1: the drift vanishes. On the right branch the first-order condition is $V' = -qu'(c)$, and with $S = (c - \mathcal{R})/q$, $\mathcal{R}(B) := \max_d \mathcal{R}(d; B, 0, q(B))$, the HJB reads

$$\rho V = u(c) + w\,u'(c), \qquad w := \mathcal{R} - c = -qS \geq 0. \tag{19}$$

Let $x \downarrow 0$: $V \to u(\bar{c})/\rho$, $q \to q^{\mathrm{rf}}$ (because debt reaches B^* in finite time, by (9), as in the proof of Theorem 1) and $\mathcal{R} \to \bar{c}$, because at $B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$ the sovereign is indifferent between $d = 0$ and d_{first}. Along any sequence on which $c \to c^+$, (19) gives $u(\bar{c}) = u(c^+) - (c^+ - \bar{c})u'(c^+)$. The function $f(c) = u(c) - (c - \bar{c})u'(c) - u(\bar{c})$ has $f(\bar{c}) = 0$ and $f'(c) = -u''(c)(c - \bar{c})$, so $f > 0$ for $c \neq \bar{c}$; hence $c \to \bar{c}$, $w \to 0$, $S \to 0$, and $V'(B^{*+}) = -u'(\bar{c})$. The same argument on the left, where $V' = -u'(c)$, gives $V'(B^{*-}) = -u'(\bar{c})$.

Step 2: the exponents. Put $p := q^{\mathrm{rf}} - q$. The pricing equation gives $p' = q\Pi/w$ with $\Pi(B,q) := \xi(d^*)q - (1-d^*)(\lambda+\delta) \to \Pi_0 > 0$. Differentiating (19), using $V' = -qu'(c)$, $c = \mathcal{R} - w$, and the

envelope theorem $\mathcal{R}' = \mathcal{R}_B - \mathcal{R}_q p'$ with $\mathcal{R}_B = -(1-d^*)(\delta+\lambda) + qa(d^*)$ and $\mathcal{R}_q = a(d^*)B$,

$$w' = \mathcal{R}_B - \mathcal{R}_q p' + \frac{u'(c)}{|u''(c)|}\,\frac{\mathcal{R}_q q\Pi/w - (\mathcal{R}_B + \rho q)}{w} = \frac{M(x)}{w^2} + O(1/w),$$
$$M(x) \to M := \frac{u'(\bar{c})}{|u''(\bar{c})|}\,a(d_{\text{first}})B^*\Pi_0.$$

Since $w \to 0$, $w^2 w' = M + o(1)$ and $w^3 = 3Mx(1+o(1))$, that is $w = \beta x^{1/3}(1+o(1))$ with $\beta^3 = 3M$. Then $p' = \Pi_0(1+o(1))/w$ integrates to $p = \alpha x^{2/3}(1+o(1))$, and $S = -w/q$, $c = \mathcal{R} - w$ give the stated laws. The second derivative $V'' = -q'u'(c) - qu''(c)c'$ is dominated by $-|u''(\bar{c})|\beta x^{-2/3}/3$.

Step 3: the regular system (existence, uniqueness, and analyticity of the branch). In the variable $\chi = x^{1/3}$ the scaled unknowns $Q = (q^{\text{rf}} - q)/\chi^2$ and $\Sigma = -qS/\chi$ solve an analytic system of Briot–Bouquet type. With $p = \chi^2 Q$, $w = \chi\Sigma$, and $d/dx = (3\chi^2)^{-1} d/d\chi$,

$$\chi Q_\chi = \frac{3q\Pi}{\Sigma} - 2Q, \qquad \chi\Sigma_\chi = \frac{3u'}{|u''|}\,\frac{\mathcal{R}_q q\Pi}{\Sigma^2} - \Sigma - 3\chi\left[\frac{u'}{|u''|}\,\frac{\mathcal{R}_B + \rho q}{\Sigma} + \frac{\mathcal{R}_q q\Pi}{\Sigma}\right] + 3\chi^2\mathcal{R}_B,$$

with the coefficients evaluated at $c = \mathcal{R}(B^* + \chi^3, q^{\text{rf}} - \chi^2 Q) - \chi\Sigma$ and $(B^* + \chi^3, q^{\text{rf}} - \chi^2 Q)$. For CRRA utility and $d_{\text{first}} \in (0,1)$ all coefficients are analytic near $(0, \alpha, \beta)$. At $\chi = 0$ the right-hand sides vanish exactly at (α, β) and their Jacobian is $\begin{pmatrix} -2 & -3\Pi_0/\beta^2 \\ 0 & -3 \end{pmatrix}$ (the $(2,2)$ entry is $-6(u'/|u''|)a(d_{\text{first}})B^*\Pi_0/\beta^3 - 1 = -3$ by the definition of β). Neither eigenvalue is a positive integer, so the Briot–Bouquet theorem (Briot and Bouquet 1856; Hille 1976, Chapter 12) yields a unique solution analytic at $\chi = 0$ with the prescribed value. Since both eigenvalues are negative, every other solution of the linearized system behaves like χ^{-2} or χ^{-3} and is unbounded as $\chi \downarrow 0$, which is the sense in which the branch is unique. The nonlinear system leads there too: in $s = \ln\chi$ the Σ equation gives $d(\Sigma^3 - 3M)/ds = -3(\Sigma^3 - 3M) + O(\chi)$, so every branch with $\Sigma > 0$ that stays bounded as $\chi \downarrow 0$ converges to (α, β), where the linearization applies.

Step 4: the second-order coefficients. The analytic solution has $Q = \alpha + Q_1\chi + O(\chi^2)$ and $\Sigma = \beta + \Sigma_1\chi + O(\chi^2)$. In the coefficients of the system only $\psi(c)$, evaluated at $c = \bar{c} - \beta\chi + O(\chi^2)$, varies at order χ; the others differ from their values at (B^*, q^{rf}) by $O(\chi^2)$. Collecting the terms of order χ in the two equations, with $3\psi(\bar{c})a(d_{\text{first}})B^*\Pi_0 = \beta^3$, $1/\psi + \psi'/\psi = u'''/|u''|$, and $R_B + \rho = \rho - r + \Pi_0$ for $R_B := -(1-d_{\text{first}})(\delta+\lambda) + q^{\text{rf}}a(d_{\text{first}})$, gives Σ_1 as stated and $Q_1 = -\Pi_0\Sigma_1/\beta^2$. Finally $-S = w/q = \chi\Sigma\,(1 + O(\chi^2))$ and $q^{\text{rf}} - q = \chi^2 Q$. □

Proposition A.8 (existence of the no-default-basin equilibrium)**.** *In the deterministic model with the underwriting standard, let $F(0) < 1$ and $d_{\text{first}} < 1$, let u be real-analytic near $\bar{c}$, as CRRA utility is, let $\underline{q} < q^{\text{rf}}$ exceed the full-default prices, and let $B^* = B_{\text{pd}}(0, q^{\text{rf}})$ and $\bar{c} = 1 - rB^*$.*

(a) The left branch. *There is a unique drift $S > 0$ on $[0, B^*)$ with $S(B^*) = 0$ that solves $SS' = rS - (\rho - r)\psi(1 - rB + S)$, $\psi := u'/|u''|$. With $q \equiv q^{\text{rf}}$, $d \equiv 0$, $c = 1 - rB + S$, and V given by*

$V(B^*) = u(\bar{c})/\rho$ *and* $V' = -u'(c)$*, it satisfies the HJB and pricing equations on* $[0, B^*]$*, as long as* $S < \bar{S}$*.*

(b) The right branch. *The solution of* (16) *that leaves* B^* *along the cube-root expansion of Theorem 2 exists and is unique near* B^**. Suppose its maximal continuation keeps* $S^* < 0$ *and* $c^* > c_{\min}$ *up to the full-default threshold, and that beyond it buybacks are not optimal,* $-V_B \le qu'(c)$*. Then it extends to all of* $[B^*, \bar{B}]$*. Where the standard binds before full default, on* $\{q < \underline{q},\ \ell^* > 0\}$*, the system switches to the constrained regime of Proposition A.6(b),* $S = -a(d^{**})B$ *with* d^{**} *at the shadow price, and consumption steps down at the switch; beyond the full-default threshold the sovereign neither issues nor buys back, debt decays as* $B_t = B_0 e^{-at}$*, the price is the power* $q(B) \propto B^{-\xi(1)/a}$*, and* V *is given by quadrature.*

(c) The equilibrium. *Under the conditions of (a) and (b), the two branches together form a no-default-basin equilibrium on* $[0, \bar{B}]$ *in which debt reaches* B^* *from above in finite time, the only one whose right branch has the cube-root asymptotics of Theorem 2. In it* q *is continuous on* $[0, \bar{B}]$*, and* V *is* C^1 *below the point where the standard starts to bind.*

Proof of Proposition A.8. (a) In the parametrization $dB/d\vartheta = S$, $dS/d\vartheta = rS - (\rho - r)\psi(1 - rB + S)$ the basin equation is a regular system, and at $(B^*, 0)$ its vector field is $(0, -(\rho - r)\psi(\bar{c})) \neq 0$, since $\rho > r$. So exactly one trajectory passes through $(B^*, 0)$, and it crosses $S = 0$ downward: for $\vartheta < 0$ it has $S > 0$ and $B < B^*$. Followed backward in ϑ, B falls while $S > 0$, and S cannot return to zero, because on $S = 0$ the field points downward, so the trajectory could only reach $S = 0$ moving forward in ϑ. The system has no rest point with $S \ge 0$, so B reaches 0 at a finite ϑ, and the trajectory is the graph of a drift $S(B) > 0$ on $[0, B^*)$, the unique one with $S(B^*) = 0$. Define c and V as in the statement. Then $\frac{d}{dB}(\rho V - u(c) - SV') = -(\rho - r)u'(c) + Su''(c)(S' - r)$, which vanishes by the basin equation, since $S(S' - r) = -(\rho - r)u'/|u''|$. By construction, $\rho V(B^*) - u(\bar{c}) = 0$. So the HJB equation holds on $[0, B^*]$ with the drift S, which is the maximizer given $V' = -u'(c)$ and $q = q^{\mathrm{rf}}$ by concavity, as long as it is below the cap. Moreover, $d = 0$ is optimal there by Proposition 2, since $B \le B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$, and $S(0) > 0$ satisfies the state constraint. Finally, every path reaches B^* in finite time (Lemma 2) and stays, without default, so (9) gives $q \equiv q^{\mathrm{rf}}$. (b) Local existence and uniqueness near B^* are Step 3 of the proof of Theorem 2, which concerns the system (16) alone. Away from B^* the system (16) is a regular ODE as long as $S^* < 0$, $c^* > c_{\min}$, and $d^* \in (0, 1)$. The HJB equation, which is implicit in V' through c^*, can be solved for V' by the implicit function theorem because its derivative in V' is $S^* \neq 0$, and $(B, q) \mapsto d^*, \mathcal{R}^*$ are smooth there. So Picard–Lindelöf continues the branch uniquely from the cube-root start up to the first point where one of these fails or the standard starts to bind. On $\{q < \underline{q}\}$ with $\ell^* > 0$ the policy is the constrained one of Proposition A.6(b), $S = -a(d^{**})B < 0$, and the HJB equation $\rho V = u(c^{**}) - a(d^{**})BV'$ together with the pricing equation is again a regular system in (V, q), continued in the same way up to the full-default threshold. At the switch, consumption drops by about $q\ell^*$ and V' may have a kink. Beyond the full-default threshold, with $\underline{q}$ above the full-default prices the sovereign cannot issue, and by hypothesis it does not

buy back, so $S = -aB$. Then the pricing equation $\xi(1)q = -aBq'$ gives the power law, and $V(B) = \int_0^\tau e^{-\rho t}u((1-\gamma_0)F(0))dt + e^{-\rho\tau}V(B_f)$ with $\tau = a^{-1}\ln(B/B_f)$, where B_f is the debt at which the default rule, at the shadow price where the standard binds, reaches full default. Continuity of q on $(B^*, \bar{B}]$ follows from Proposition A.2, as the drift is negative throughout. (c) The value is C^1 at B^*: on the left $V'(B^{*-}) = -u'(\bar{c})$, and on the right $V' = -qu'(c) \to -u'(\bar{c})$, by Step 1 of the proof of Theorem 2, which uses only the branch. So the HJB equation holds at B^* with $S = 0$, and the pieced pair solves the HJB and pricing equations on $[0, \bar{B}]$, with the policies of the two branches attaining the maximum. It is of no-default-basin type by construction. Debt reaches B^* from above in finite time, since by the cube-root law the time to reach B^* from $B^* + x$ is of order $x^{2/3}$. The price is continuous at B^*, which is reached in finite time from both sides, and on each branch. Uniqueness: by Theorem 1 every no-default-basin equilibrium in which debt reaches B^* from above in finite time has its steady state at B^*. Its left branch is the unique trajectory of (a), and a right branch with the cube-root asymptotics is the unique branch of Step 3, continued uniquely as in (b). □

Lemma A.2 (local conditions at a partial-default steady state)**.** *In the deterministic model, suppose the equilibrium has a steady state B° in the interior of the partial-default region at which S and q are right-differentiable and V is twice right-differentiable, with $d^\circ = d^*(B^\circ, q^\circ) \in (0,1)$. Then $q^\circ = (1 - d^\circ)(\lambda + \delta)/\xi(d^\circ)$, the slope of the price is pinned by the impatience wedge,*

$$-\frac{q'(B^{\circ+})}{q^\circ} = \frac{\rho - r}{a(d^\circ)B^\circ},$$

and the slope of the drift follows from the pricing equation,

$$q'(B^{\circ+})\big(S'(B^{\circ+}) - \xi(d^\circ)\big) = (\lambda + \delta)(1 - \kappa q^\circ)\,\frac{d}{dB}d_{\text{int}}(B, q(B))\Big|_{B^{\circ+}},$$

so that the steady state is stable from above, $S'(B^{\circ+}) < 0$, if and only if the dilution term on the right exceeds $\xi(d^\circ)\,|q'(B^{\circ+})|$.

Proof of Lemma A.2. The first identity is (9) with constant d. Differentiating $\rho V = u(c) + SV'$ with $c = \mathcal{R}^* + qS$ gives $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q' + q'S + qS') + S'V' + SV''$. At $S = 0$ the two terms in S' cancel by the first-order condition $V' = -qu'(c)$, leaving $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q')$. With $V' = -qu'(c)$, $\mathcal{R}_q = a(d)B$, and $\mathcal{R}_B = -(1-d)(\delta + \lambda) + qa(d) = -rq$ by the first identity, this reads $-\rho q = -rq + a(d)Bq'$, the second identity. The third is l'Hôpital's rule applied to $q' = (\xi(d)q - (1-d)(\lambda + \delta))/S$. □

Proof of Proposition 3. At a steady state $S = 0$, so issuance is $\ell = S + a(d^\circ)B^\circ = a(d^\circ)B^\circ \geq aB^\circ > 0$, since $a(d) \geq a(1) = a > 0$. The standard (12) forbids $\ell > 0$ when $q < \underline{q}$, so $q^\circ \geq \underline{q}$. The state stays at B° with constant intensity d°, so (9) gives $q^\circ = q_r(d^\circ)$. Differentiating, $q_r'(d) = (\lambda + \delta)(\kappa(\lambda + \delta) - (r + \delta))/\xi(d)^2 < 0$, because $\kappa < 1$ and $\lambda = r$. Moreover, $d^\circ \geq d_{\text{first}}$, because the static rule never chooses an intensity in $(0, d_{\text{first}})$ (Proposition 2). Hence $\underline{q} \leq q_r(d^\circ) \leq q_r(d_{\text{first}})$.

The expression for the spread uses $\xi(d) - (1-d)(\lambda+\delta) = (1-\kappa)(\lambda+\delta)d$, which holds when $\lambda = r$. □

Proposition A.9 (the risk-neutral construction). *Let $u(c) = c$ and $\rho > r$ in the deterministic model, with issuance unrestricted in speed. Write $F = F(0)$ and define for $d \in [d_{\text{first}}, 1)$ the price $q_\rho(d) := (1-d)(\lambda+\delta)/(\rho + a(d))$ of a bond that defaults at rate d forever, discounted at the sovereign's rate, the debt $B(d)$, and the stability bound $\Omega(d)$ below. Here Θ of Proposition 4 is the output cost of the default per unit of the funds raised at the steady state:*

$$B(d) := \frac{\gamma_0 \gamma_1 F\, d^{\gamma_1 - 1}\, \xi(d)}{(1-\kappa)(\lambda+\delta)^2},$$
$$\Theta(d) = \frac{1 - \phi(d,0)}{q_r(d) B(d)},$$
$$\Omega(d) := \frac{s(d)\, a(d)}{(\gamma_1 - 1)\xi(d) - \kappa(\lambda+\delta)d},$$

with $\Omega(d) = \infty$ when the denominator is not positive.

(a) On any interval of debt through which the sovereign moves at a finite speed it is indifferent, $V' = -q$, and the price solves

$$q'(B) = -\frac{(\rho + a(d))\,\big(q - q_\rho(d)\big)}{a(d)B}, \qquad d = d^*(B, q),$$

with drift $S = a(d)B\,\xi(d)(q_r(d) - q)/\big((\rho + a(d))(q - q_\rho(d))\big)$; debt rises if and only if $q_\rho(d) < q < q_r(d)$.

(b) Every partial-default steady state is a point $(B(d), q_r(d))$, $d \in (d_{\text{first}}, 1)$, with $q_r(d) \geq \underline{q}$ under the standard. The one with intensity d is stable from above if and only if $\rho - r < \Omega(d)$.

(c) If $\Theta(d) \leq \rho - r < \Omega(d)$ and $q_r(d) \geq \underline{q}$, and the solution of (a) through $(B(d), q_r(d))$ can be continued above it up to $\bar{B}$, beyond the full-default threshold or where the standard binds by the full-default branch or the constrained regime of Proposition A.6(b), with negative drift and $-V' \leq q$ throughout, then there is an equilibrium in which the sovereign issues $B(d)$ at once from zero debt and stays there, withholding the share d forever.

(d) $\Omega(d_{\text{first}}) > s_{\text{first}}$ if and only if $(\gamma_1 - 1)\xi(d_{\text{first}}) < \delta$. When this holds and $q_r(d_{\text{first}}) > \underline{q}$, for every wedge $\rho - r$ in $(s_{\text{first}}, \Omega(d_{\text{first}}))$ there are intensities d slightly above d_{first} that satisfy the conditions of (c) other than the continuation condition.

Proof of Proposition A.9. With $u(c) = c$ the maximized Hamiltonian is $\mathcal{R}^*(B, q) + S\,(q + V')$, linear in the drift: where $q + V' > 0$ the sovereign makes a discrete issuance, where $q + V' < 0$ it reduces its debt as fast as it can, and a finite positive speed requires $q + V' = 0$.

(a) On an interval with $q + V' = 0$ the HJB equation reduces to $\rho V = \mathcal{R}^*(B, q(B))$. Differentiating and using the envelope theorem, $\mathcal{R}^*_q = a(d)B$, gives $-\rho q = \mathcal{R}^*_B + a(d)Bq'$ with

$\mathcal{R}_B^* = -(1-d)(\lambda+\delta) + qa(d)$, which is the stated equation because $(\rho+a)q - (1-d)(\lambda+\delta) = (\rho+a)(q-q_\rho)$. The pricing equation $\xi(d)q = (1-d)(\lambda+\delta) + Sq'$ gives $S = \xi(d)(q-q_r(d))/q'$, and substituting q' gives the drift. Since $\rho > r$, $q_\rho(d) < q_r(d)$, and the sign follows.

(b) At a steady state the price is $q_r(d^\circ)$ by (9), and d° is interior, so $(1-\kappa q)(\lambda+\delta)B = \gamma_0\gamma_1 F d^{\gamma_1-1}$ (Proposition 2). With $(1-\kappa q_r(d))/q_r(d) = (1-\kappa)/(1-d)$, this is $B = B(d)$. At $q = q_r(d)$ the equation in (a) has slope $q' = -q(\rho-r)/(a(d)B)$, the slope of Lemma A.2. Write $P(B,q) := \xi(d^*)q - (1-d^*)(\lambda+\delta)$, so that $S = P/q'$ along the solution and $S' = P_q + P_B/q'$ at the steady state. With $P_d = (1-\kappa q)(\lambda+\delta)$, $\partial_B d_{\text{int}} = d/((\gamma_1-1)B)$, and $\partial_q d_{\text{int}} = -\kappa d/((\gamma_1-1)(1-\kappa q))$, we have $P_B = (1-\kappa q)(\lambda+\delta)d/((\gamma_1-1)B)$ and $P_q = \xi(d) - \kappa(\lambda+\delta)d/(\gamma_1-1)$. If $P_q \le 0$ then $S' < 0$; otherwise $S' < 0$ if and only if $\rho - r < P_B a(d)B/(qP_q)$, which is $\Omega(d)$ after using $(1-\kappa q_r)(\lambda+\delta)d/q_r = s(d)$.

(c) Let $B^\circ = B(d)$ and $q^\circ = q_r(d)$. Set $q = q^\circ$ and $V(B) = V(B^\circ) + q^\circ(B^\circ - B)$ on $[0, B^\circ]$, with $\rho V(B^\circ) = \mathcal{R}^*(B^\circ, q^\circ)$, and above B° the solution of (a) and its continuation. From any $B < B^\circ$ the state moves at once to B° and stays, so the lenders' price there is q°, and the issuance is allowed because $q^\circ \ge \underline{q}$. On $[0, B^\circ]$, $q + V' = 0$, so the sovereign is indifferent among speeds and among discrete issuances that end at or below B°, and the HJB equation requires that standing still not be better: $D(B) := \rho V(B) - \mathcal{R}^*(B, q^\circ) \ge 0$. The function D is concave, because $\mathcal{R}^*(\cdot, q)$ is a maximum of functions affine in B. Moreover, $D(B^\circ) = 0$, and $D'(B^{\circ-}) = (\rho + a(d))\big(q_\rho(d) - q^\circ\big) < 0$. So $D \ge 0$ on $[0, B^\circ]$ if and only if $D(0) \ge 0$. Since $(1-d)(\lambda+\delta) - q^\circ a(d) = rq^\circ$, $\mathcal{R}^*(B^\circ, q^\circ) = \phi(d,0) - rq^\circ B^\circ$ and $D(0) = (\rho - r)q^\circ B^\circ - (1 - \phi(d,0))$, which is nonnegative if and only if $\rho - r \ge \Theta(d)$. A discrete issuance that ends above B°, at B' on the solution of (a), yields $q(B')(B' - B) + V(B')$, which does not exceed $V(B)$ because $V' = -q$ there and q decreases. Above B°, stability makes the drift negative near B°, and the continuation is the hypothesis. The closed form of Θ follows from $1 - \phi(d,0) = 1 - F + \gamma_0 F d^{\gamma_1}$, from $1 - F = (\gamma_1 - 1)\gamma_0 F d_{\text{first}}^{\gamma_1}$ by (14), and from $q_r(d)B(d) = \gamma_0\gamma_1 F d^{\gamma_1-1}(1-d)/((1-\kappa)(\lambda+\delta))$.

(d) Since $a(d) + \kappa(\lambda+\delta)d = \delta$, $\Omega(d_{\text{first}}) > s(d_{\text{first}})$ if and only if $\delta > (\gamma_1 - 1)\xi(d_{\text{first}})$, including the case of a nonpositive denominator. With $\Theta(d_{\text{first}}) = s_{\text{first}}$, from the closed form, continuity of Θ, Ω, and q_r gives, for a wedge in $(s_{\text{first}}, \Omega(d_{\text{first}}))$, an intensity d slightly above d_{first} with $\Theta(d) < \rho - r < \Omega(d)$ and $q_r(d) \ge \underline{q}$. □

Proof of Proposition 4. As in the proof of Proposition A.9, the sovereign makes a discrete issuance where $q + V' > 0$ and moves at a finite speed only where $q + V' = 0$.

(a) Along a path on which debt rises, the sovereign makes discrete issuances or moves along the solutions of Proposition A.9(a), and consecutive discrete issuances are one. Let B_e be the first debt level on the path at which $d^* > 0$. If the path reaches B_e along a solution of Proposition A.9(a), the price q_e there satisfies $q_e > q_\rho(0)$, for the drift to be positive just below, where $d = 0$, and $q_e \le q_r(d_{\text{first}})$, for it to be nonnegative just above, where $d = d_{\text{first}}$. So $q_\rho(0) < q_r(d_{\text{first}})$, which is $\rho - r > s_{\text{first}} = \Theta(d_{\text{first}})$. If the path reaches B_e by a discrete issuance from a point $B_O > 0$ of such a solution, the price is continuous at B_O, since a bond held just below B_O is carried into the issuance. So the landing price q_J equals the price at B_O and exceeds $q_\rho(0)$, while landing with a

nonnegative drift requires $q_J \le q_r(d_J) \le q_r(d_{\text{first}})$. Again, $\rho - r > \Theta(d_{\text{first}})$. If the discrete issuance is made at zero debt and lands at (B_J, q_J) with intensity $d_J \in [d_{\text{first}}, 1)$, then $V(0) = q_J B_J + V(B_J)$ with $\rho V(B_J) = \mathcal{R}^*(B_J, q_J)$, and staying at zero debt forever is worth $1/\rho$, so $\mathcal{R}^*(B_J, q_J) + \rho q_J B_J \ge 1$. As $q_J \le q_r(d_J)$, $\mathcal{R}^*(B_J, q_J) \le \phi(d_J, 0) - r q_J B_J$, hence $(\rho - r) q_J B_J \ge 1 - \phi(d_J, 0)$. By the first-order condition for d_J, $(1 - \phi(d_J, 0))/(q_J B_J) = (1 - \phi(d_J, 0))(1 - \kappa q_J)(\lambda + \delta)/\left(\gamma_0 \gamma_1 F d_J^{\gamma_1 - 1} q_J\right)$, which is decreasing in q_J and equals $\Theta(d_J)$ at $q_J = q_r(d_J)$. So $\rho - r \ge \Theta(d_J) \ge \Theta^*$. If the discrete issuance overshoots, landing above the steady state B° that the path then descends to along a solution of Proposition A.9(a), the price is nonincreasing above B° and $V' = -q$ there. So $V(0) = q_J B_J + V(B_J) \le q^\circ B^\circ + V(B^\circ)$ with $\rho V(B^\circ) = \phi(d^\circ, 0) - r q^\circ B^\circ$, and $V(0) \ge 1/\rho$ again gives $\rho - r \ge \Theta(d^\circ) \ge \Theta^*$.

(b) The definition of Θ gives $\Theta(d_{\text{first}}) = s(d_{\text{first}}) = s_{\text{first}}$. Write

$$\Theta(d) = \frac{(1 - \kappa)(\lambda + \delta)\, N(d)}{\gamma_1 (1 - d)}, \qquad N(d) = d + (\gamma_1 - 1) d_{\text{first}}^{\gamma_1} d^{1 - \gamma_1};$$

for $d \ge d_{\text{first}}$,

$$N'(d) = 1 - (\gamma_1 - 1)^2 (d_{\text{first}}/d)^{\gamma_1} \ge 1 - (\gamma_1 - 1)^2 \ge 0 \quad \text{when } \gamma_1 \le 2,$$

and $1/(1 - d)$ is increasing, so Θ is increasing.

(c) Set $\bar{\Omega} := \Omega(d_{\text{first}})$. Proposition A.9(d) gives intensities with $\Theta(d) < \rho - r < \Omega(d)$ and $q_r(d) \ge \underline{q}$, and Proposition A.9(c) gives the equilibrium under the continuation condition. □

A.4 Equilibria Along the Path from Zero Debt

We ask which equilibria of the deterministic model with the underwriting standard exist besides the no-default-basin equilibrium of Proposition A.8. We maintain the condition of Proposition 3,

$$q_r(d_{\text{first}}) < \underline{q}, \tag{20}$$

the stationary spread at the first default exceeds the spread at the floor, which holds at the calibrated parameters. Write $b_0 := B_{\text{pd}}(0, 0) > 0$ for the no-default threshold at a zero price. Since $B_{\text{pd}}(0, \cdot)$ is increasing, no equilibrium defaults at a debt below b_0 at any price, whether the standard is slack (Proposition 2) or binds (Proposition A.6(c), the thresholds at the shadow prices being at least b_0 as well). Paths are monotone, so from any debt level the state either stays or moves monotonically to a steady state, a zero of the drift at which it stays.

Lemma A.3 (steady states under the standard)**.** *Under* (20)*, every steady state $B^\circ > 0$ of every equilibrium has $d(B^\circ) = 0$, $q(B^\circ) = q^{\text{rf}}$, and $B^\circ \le B^* = B_{\text{pd}}(0, q^{\text{rf}})$.*

Proof. At a steady state issuance is $\ell = a(d^\circ) B^\circ \ge a B^\circ > 0$, which the standard permits only if $q(B^\circ) \ge \underline{q}$. The standard is then slack and the default decision is the static rule of Proposition 1: $d^\circ = 0$ or $d^\circ = d_{\text{int}} \ge d_{\text{first}}$. The state stays at B° with constant intensity, so $q(B^\circ) = q_r(d^\circ)$ by

(9). If $d^\circ \geq d_{\text{first}}$, then $q(B^\circ) \leq q_r(d_{\text{first}}) < \underline{q}$, since q_r is decreasing, a contradiction. So $d^\circ = 0$, $q(B^\circ) = q^{\text{rf}}$, and, $d = 0$ being optimal at (B°, q^{rf}), $B^\circ \leq B_{\text{pd}}(0, q^{\text{rf}})$. □

Suppose the sovereign borrows at zero debt, $S(0) > 0$. The path from zero debt then rises to a steady state B°, reached at a time $T \leq \infty$. Along it the sovereign issues, so $q(B_t) \geq \underline{q}$, the standard is slack, and the sovereign defaults at t if and only if $B_t > B_{\text{pd}}(0, q(B_t))$. The price along the path is continuous in t and satisfies

$$\frac{d}{dt} q(B_t) = \xi(d_t)\, q(B_t) - (1 - d_t)(\lambda + \delta), \tag{21}$$

which equals $(r + \delta)(q(B_t) - q^{\text{rf}}) < 0$ where $d_t = 0$ and $q < q^{\text{rf}}$. We assume that the default policy switches finitely often along the path.

Lemma A.4 (default along the path from zero debt)**.** *Under* (20)*, the set of dates at which the path from zero debt is in default is empty or an interval $[t_1, T)$ with $T < \infty$. In the second case $B^\circ = B^*$.*

Proof. No stretch ends before arrival. Suppose default on (t_a, t_2) and not at $t_2 < T$. If there is no default after t_2, then $q(B_{t_2}) = q^{\text{rf}}$ and the static tie at t_2 gives $B_{t_2} = B_{\text{pd}}(0, q^{\text{rf}}) = B^*$, which contradicts $B_{t_2} < B^\circ \leq B^*$. If default resumes at $t_3 > t_2$, then on (t_2, t_3) the price falls by (21) while debt rises, so $B_{\text{pd}}(0, q(B_t)) < B_{\text{pd}}(0, q(B_{t_2})) \leq B_{t_2} < B_t$ and the static rule prescribes default on (t_2, t_3), a contradiction. *No stretch lasts forever.* Suppose default at every $t \geq t_1$ with $T = \infty$. Then $d_t = d_{\text{int}} \geq d_{\text{first}}$ at every such date, since d_{int} equals d_{first} at the threshold and increases with debt, and Lemma A.1 gives $q(B_{t_1}) \leq q_r(d_{\text{first}}) < \underline{q}$, so the standard forbids issuance at B_{t_1} and the drift there is at most $-a(d)B < 0$, a contradiction. *Arrival.* So the stretch is $[t_1, T)$ with $T < \infty$ or empty. In the first case $B_t > B_{\text{pd}}(0, q(B_t))$ for $t < T$ and $q(B_t) \to q(B^\circ) = q^{\text{rf}}$ as $t \uparrow T$, so $B^\circ \geq B_{\text{pd}}(0, q^{\text{rf}}) = B^*$, and $B^\circ = B^*$ by Lemma A.3. □

Proposition A.10 (the two equilibria along the path from zero debt)**.** *Assume* (20)*, that the sovereign borrows at zero debt, and that the value is twice differentiable on the no-default part of the path from zero debt.*

(a) In every equilibrium the path from zero debt rises in finite time to a no-default steady state $B^\circ \leq B^*$ priced at q^{rf}, and one of two cases holds. Either the sovereign never defaults on the path and $q \equiv q^{\text{rf}}$ on $[0, B^\circ]$, with $B^\circ = B^*$ whenever B° is reached from above in finite time (Theorem 1). Or it defaults exactly on a terminal interval (B_2, B^*) of debt levels, with $B^\circ = B^*$, $B_2 = B_{\text{pd}}(0, q(B_2))$, $d = d_{\text{int}}(B, q(B)) \geq d_{\text{first}}$ on the interval, and $q < q^{\text{rf}}$ on $[0, B^*)$ with $q(B) = q^{\text{rf}} - e^{-(r+\delta)\tau(B)}(q^{\text{rf}} - q(B_2))$ on $[0, B_2]$, where $\tau(B)$ is the travel time from B to B_2: a pessimistic approach.

(b) Under the hypotheses of Proposition A.8, the branch of Theorem 2, analytic in $(B - B^*)^{1/3}$, continues

to $B < B^*$, where, with $x = B^* - B$,

$$S(B^* - x) = \beta x^{1/3}(1 + o(1)), \qquad q^{\mathrm{rf}} - q(B^* - x) = \alpha x^{2/3}(1 + o(1)),$$
$$c(B^* - x) = \bar{c} + \beta x^{1/3}(1 + o(1)), \qquad \tau(B^* - x) = \frac{3}{2\beta} x^{2/3}(1 + o(1)),$$

where τ is the time to reach B^*, and the sovereign is in partial default, $d = d_{\mathrm{int}}(B, q(B)) \geq d_{\mathrm{first}}$.

(c) Continued downward, the branch of (b) leaves the default region at a tie $B_2 = B_{\mathrm{pd}}(0, q(B_2)) > b_0$ and continues without default. If it keeps $S > 0$ and $c > c_{\min}$ down to $B = 0$, and $q \geq \underline{q}$ on $[B_2, B^*]$, then together with the right branch of Proposition A.8(b) it is an equilibrium, a pessimistic approach, and the only one whose branches stay bounded in the scaled variables of Theorem 2.

Proof. (a) Lemma A.3 gives the nature of B°. If the path never defaults, $q \equiv q^{\mathrm{rf}}$ along it by (9), the sovereign solves the no-default problem at the risk-free price, and the drift satisfies $SS' = rS - (\rho - r)\psi(c)$ by the proof of Lemma 2, so $S^2 \sim 2(\rho - r)\psi(\bar{c})(B^\circ - B)$ as $S \to 0$ and the arrival is in finite time. Otherwise Lemma A.4 gives the terminal stretch and $B^\circ = B^*$; $B_2 := B_{t_1}$ is a tie by continuity. On $[0, B_2]$ there is no default, so the Feynman–Kac value is q^{rf} less the loss at B_2 discounted at $r + \delta$ over the travel time, which is the formula stated, and $q < q^{\mathrm{rf}}$ on $[0, B^*)$ because of the loss on the stretch.

(b) Step 3 of the proof of Theorem 2 constructs the right branch as an analytic function of $\chi = (B - B^*)^{1/3}$ on a neighborhood of $\chi = 0$, as a solution of (16) with $d^* = d_{\mathrm{int}}$. The function is defined for $\chi < 0$, where $x = -\chi^3 > 0$, and the expansions $S = -\beta\chi - \Sigma_1\chi^2 + O(\chi^3)$ and $q^{\mathrm{rf}} - q = \alpha\chi^2 + Q_1\chi^3 + O(\chi^4)$ give the leading terms stated, with $c = \mathcal{R}^* + qS$. The system (16) encodes the first-order condition $V' = -qu'(c)$, which is sufficient for the drift because the objective is concave in S, and the static rule with $d^* = d_{\mathrm{int}}$. The only point to check on the left is that d_{int} is optimal, that is $B > B_{\mathrm{pd}}(0, q(B))$ (Proposition 2). The threshold is C^1 and increasing in the price near q^{rf}, with slope $C := \partial_q B_{\mathrm{pd}}(0, q^{\mathrm{rf}}) > 0$, so $B_{\mathrm{pd}}(0, q(B)) = B^* - C\alpha x^{2/3} + o(x^{2/3}) < B^* - x = B$ for small x. The standard, the drift cap, and the consumption floor are slack since $q \to q^{\mathrm{rf}} > \underline{q}$, $S \to 0$, and $c \to \bar{c}$, and V is obtained from $V' = -qu'(c)$ and $V(B^*) = u(\bar{c})/\rho$ as in Proposition A.8. The arrival time is $\int_0^x dy/S(B^* - y)$.

(c) Away from B^* the system (16) is a regular ODE while $S > 0$, $c > c_{\min}$, and the regime is fixed, as in Proposition A.8(b), so Picard–Lindelöf continues it. The default region is left where the static rule switches, at the first $B_2 < B^*$ with $\mathcal{R}(d_{\mathrm{int}}; B_2, q) = \mathcal{R}(0; B_2, q)$, which lies above b_0 because below b_0 defaulting is never optimal. At the switch $\mathcal{R}^*$, hence c, S, and V', are continuous, and on $[0, B_2]$ the system continues with the no-default resource function and $q' = (r + \delta)(q - q^{\mathrm{rf}})/S < 0$. Under the stated conditions every constraint is slack along the branch, the pieced (V, q) solves the HJB and pricing equations on $[0, \bar{B}]$ with the right branch of Proposition A.8(b), the price is continuous at B^*, which both sides reach in finite time, and the steady state is B^*. Uniqueness among branches bounded in the scaled variables is the uniqueness statement of Step 3 of Theorem 2, which does not depend on the sign of χ. □

The proposition takes $S(0) > 0$ as given. A sovereign that stays at zero debt, or at a steady state below capacity that the paths above it do not reach, faces a price discontinuity above it, and the argument does not cover that case. Figure 10 shows the two equilibria at the calibrated parameters near capacity; Appendix B.2 reports the computation.

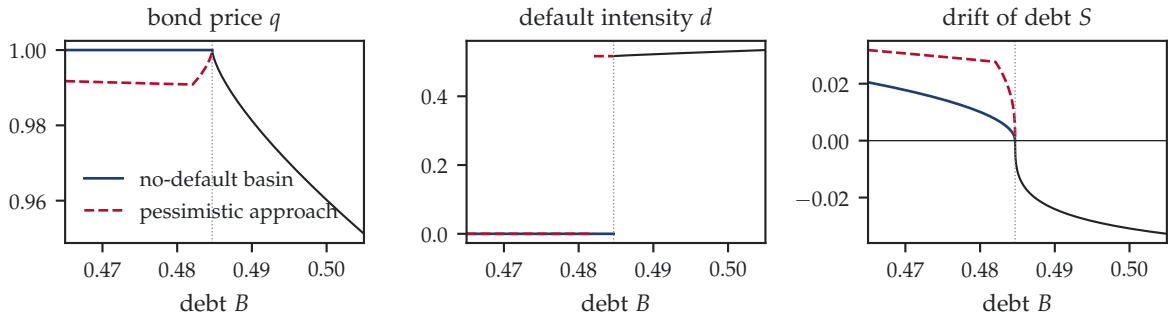


Figure 10: The two equilibria of Proposition A.10 near capacity at the calibrated parameters: the no-default basin (solid) and the pessimistic approach (dashed) below B^* (dotted vertical line), with the common cube-root branch above it.

A.5 Proofs for Section 3.4

Proof of Proposition 5. On $[0, \underline{B}]$ no state defaults and every path stays in the set, so (9) gives $q_j = q^{\mathrm{rf}}$ there. No default at that price requires $B \leq B_{\mathrm{pd}}(z_j, q^{\mathrm{rf}})$ in every state (Proposition 2), so $\underline{B} \leq \min_j B_{\mathrm{pd}}(z_j, q^{\mathrm{rf}})$. Near B° the price is $q^{\mathrm{rf}} = 1$ and no state defaults, so consumption in state k is $c_k = e^{z_k} - rB + S_k$, and the HJB equation of state j reads $\rho V_j = \max_S \{u(e^{z_j} - rB + S) + SV_j'\} + \sum_{k \neq j} \eta_{jk}(V_k - V_j)$. The maximand is strictly concave in S and the maximizer is interior, so with V_j twice differentiable at B° the envelope theorem gives, at B°, $\rho V_j' = -ru'(c_j) + S_j V_j'' + \sum_{k \neq j} \eta_{jk}(V_k' - V_j')$, and $S_j(B^\circ) = 0$ removes the second term. The first-order conditions $V_k' = -u'(c_k)$, valid at B° because each V_k is differentiable there and the cap and the floor are slack, turn this into (17). If c_j is the lowest consumption at B°, the right-hand side of (17) is at most zero while the left-hand side is positive, since $\rho > r$. □

Proposition A.11 (the best response preserves monotonicity in debt)**.** *In the general model, let q be nonincreasing in B and strictly positive, let the underwriting standard be slack so that the default rule is that of Proposition 1, and suppose the debt flow induced by the optimal policy under q is order preserving (two paths started at $B' < B$ with the same z-path satisfy $B_t' \leq B_t$). Then the price generated by those policies through (9) is again nonincreasing in B.*

Proof. By the monotonicity of d^* in B and in q (Section 3.1) and the monotonicity of q, $d^*(B, z) = d^*(B, z, q(B, z))$ is nondecreasing in B. With order preservation, $d^*(B_t', z_t) \leq d^*(B_t, z_t)$ pathwise, and Lemma A.1 gives the claim. □

Online Appendix to
"Debt Capacity and the Anatomy of Sovereign Crises"

by Sangdong Kim and Gabriel Mihalache

B Numerical Implementation

Discrete equations. Debt takes values on a grid B_i, $i = 0, \ldots, n_B - 1$, with step ΔB, and the log endowment on z_j, $j = 0, \ldots, n_z - 1$. One-sided differences are $D^+V_{ij} = (V_{i+1,j} - V_{ij})/\Delta B$ and $D^-V_{ij} = (V_{ij} - V_{i-1,j})/\Delta B$. At a node, for a given one-sided derivative V_B and price q, consumption is linear in the drift at fixed d, $c(S, d) = \mathcal{R}(d) + qS$. So the Hamiltonian $u(c(S, d)) + SV_B$ is concave in S with maximizer $S = (c^* - \mathcal{R}(d))/q$, $c^* = (u')^{-1}(-V_B/q)$, clamped to the admissible interval. The interval comes from $c \geq c_{\min}$, the cap $S \leq \bar{S}$ (applied also to $-S$), the sign restriction of the candidate, and the underwriting standard $S \leq -a(d)B$ where $q < \underline{q}$. Over d the maximized Hamiltonian is concave on $(0, 1]$ and is maximized by golden-section search (28 iterations, a tolerance of about 10^{-6} in d), with the candidates $d = 0$, $d = 1$, and the closed form (8) also evaluated. The consumption floor is $c_{\min} = 0.05\, e^z$ and the drift cap $\bar{S} = 1$, and neither binds at any reported solution. We form three candidates at every node: forward (D^+V, $S \geq 0$), backward (D^-V, $S \leq 0$), and zero drift. The one with the largest Hamiltonian is the policy. At B_0 the backward candidate is dropped and at B_{n_B-1} the forward one (state constraints). For a concave Hamiltonian this rule reduces to the standard upwind rule. With the fixed cost of default it is the monotone extension of that rule. We discretize the z-operator with upwinding of the exogenous drift and central second differences, so that every off-diagonal coefficient is nonnegative. With A the generator built from the policy, the discrete equilibrium is the root of

$$R_V \equiv \rho V - u(c) - AV = 0, \qquad R_q \equiv \xi(d)\, q - (1 - d)(\lambda + \delta) - Aq = 0, \tag{22}$$

and all convergence statements refer to $\|R_V\|_\infty$ and $\|R_q\|_\infty$ at the current (V, q) with the policy recomputed from them, never to step-to-step changes.

B.1 Time Stepping for the Finite-State Models

We solve the finite-state models by joint implicit upwind time stepping with the policy explicit from (V^k, q^k),

$$\Big[(\rho + 1/\Delta t)I - A^k\Big] V^{k+1} = u^k + V^k/\Delta t,$$
$$\Big[\operatorname{diag}\Big(\xi(d^k)\Big) + I/\Delta t - A^k\Big] q^{k+1} = (1 - d^k)(\lambda + \delta) + q^k/\Delta t,$$

with $\Delta t = 0.01$ and both solves sparse direct. We update the value and the price jointly at each implicit step, because the alternating iteration between them is unstable. The iteration starts from

the value of servicing the debt at the risk-free rate forever, $u(e^z - rB)/\rho$, and from the stationary price implied by the interior default policy at the risk-free price. It stops when $\|R_V\|_\infty < 10^{-7}$ and $\|R_q\|_\infty < 10^{-8}$. We place the upper end of the grid where the drift is negative in every state, so that the truncation is exact, and we check this placement on the solution. The pricing equation uses the same generator as the HJB equation, so its transport term is upwinded with the equilibrium drift: forward at B_0, where the drift is nonnegative, and backward at B_{n_B-1}, where the state constraint makes it nonpositive. The stationary distribution solves $A^\top g = 0$ with the converged generator. The three-state illustration of Section 3.4 uses 201 nodes on $[0, 1]$ and converges to residuals of 10^{-7} and 3×10^{-9}.

The three-state illustration. Figure 11 shows the equilibrium with three states, $z \in \{-0.08, 0, +0.08\}$. The low state rises to the middle one at the rate $\eta = 0.1$, and the middle and high states switch with each other at the same rate. Both upper states jump down to the low state only at the rare rate $\epsilon = 0.03$, once every three decades. Apart from these jumps the two upper states form a symmetric two-state chain, and η is half the mean-reversion rate $\mu = -\ln \rho_z$ of the diffusion model (Section 4.1), the rate at which such a chain matches the diffusion's autocorrelation. The rare jump is the device that lets a chain catch the sovereign on the wrong side of a threshold. We compute the solution with the standard at $\underline{q} = 0.6$ by the time stepping above. The default thresholds are $B = 0.35$, 0.475, and 0.615 in the low, middle, and high state, and the default intensity jumps to 0.46 at the first. The middle state borrows up to the node just below its own threshold, $B = 0.47$, and stops there. The rare jump is too unlikely to hold it back, so it is pinned by its own threshold, as in the deterministic model. The high state stops earlier, at the node just below the low state's threshold, $B = 0.345$, where the kink in its price schedule pins it. At that node the impatience term of (17) exceeds the precautionary term, 0.02 against 0.018, so at a flat price the high state would borrow further. The price there is within one percent of q^{rf}. From wherever it starts, the low state repays down to the node just below its own threshold, the same node. The ergodic distribution, shown by state in the last panel, has atoms at the two thresholds. It has mass 0.28 in the middle state at its own threshold, and mass 0.16 in the low state and 0.09 in the high state at the low state's threshold. It also has a continuous part in every state between the two thresholds, traced by the paths that follow a switch. Default happens along these paths. A jump from the middle state's atom lands the sovereign in partial default at $B = 0.47$, above the low state's threshold, and it withholds a share of its debt service while it repays. Default occurs on 0.07 of the ergodic mass, all of it in the low state, and mean debt is 0.41.

B.2 Deterministic Model

We compute the deterministic solutions at the calibrated parameters (Figures 3, 4, and 12, Table 3) by the time stepping of Appendix B.1 with a single endowment state, on a grid with 500 nodes per unit of debt on $[0, 1.5]$, to residuals of 10^{-7} and 5×10^{-9}. The computed steady state and first default agree with the closed forms (15) and (14) to within a grid step. The discrete system also

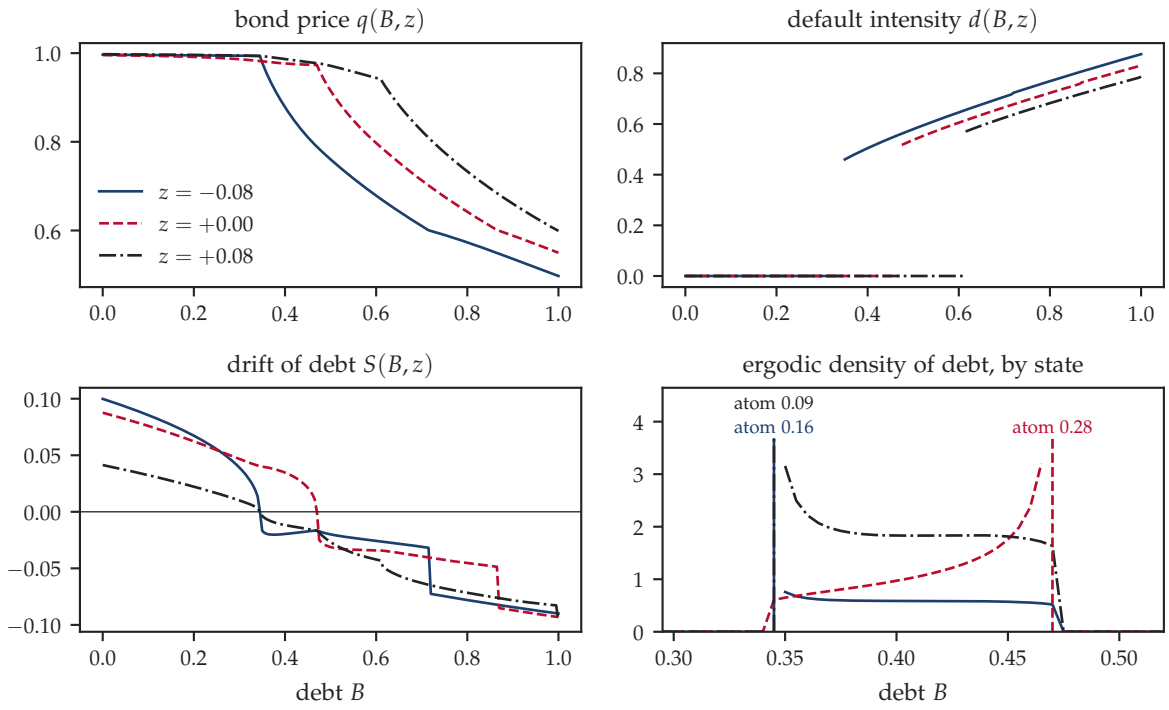


Figure 11: The chain model with three states: bond price, default intensity, and drift of debt by endowment state, and the ergodic density of debt by state (atoms are drawn as spikes). Parameters of Table 1, with $z \in \{-0.08, 0, 0.08\}$. The low state rises to the middle one, and the two upper states switch with each other, at the rate 0.1. Both upper states jump to the low state at the rate 0.03.

admits an exact solution by reverse shooting outward from the steady state, which we use as a check and to compute Figure 5. Nodes with $S < 0$ depend only on the node below, nodes with $S > 0$ only on the node above, and a zero-drift node on nothing. So once the steady-state node is known the system is triangular and can be solved node by node outward from it. (i) *Candidates.* At each node, all prices q^* solving $q^* = (1 - d^*)(\lambda + \delta)/\xi(d^*)$ with d^* the maximizer of $u(c(0, d))$ at that price are located by a scan of $(0, 1]$ with bisection; $V^* = u(c^*)/\rho$. (ii) *Local consistency.* We keep a candidate if for some root of the node above, solved in backward mode, the forward candidate at the steady-state node does not beat zero drift, and symmetrically below. (iii) *Node solve.* Given the upwind neighbor, for each trial q the HJB node equation is monotone in V and solved by bisection, and the price equation becomes a scalar function of q whose roots are all located. (iv) *Shooting outward.* Solve up and down from the candidate, keeping every root that is consistent with the previous node's choice once the new value is known. (v) *Global check.* Verify at every node that the chosen candidate attains the maximum Hamiltonian given both neighbors, and evaluate (22). A solution is accepted if the residuals are below 10^{-8}, exactly one candidate survives with no violation, and the solution on $[0, \bar{B}]$ is unchanged when $\bar{B}$ is increased. At the calibrated parameters every partial-default candidate has a price of at most $q_r(d_{\text{first}}) = 0.599$, just below the floor, so none is an equilibrium under the standard (Proposition 3). Reverse shooting cannot settle whether a candidate partial-default steady state is reached from zero debt (Section 3.3), because a whole family of solutions enters each candidate from below. We solve that problem, and the continuum of Section 3.3, by integrating forward from zero debt.

The cube-root law at the calibration. Figure 12 checks Theorem 2. On log-log axes it compares the computed drift and price gap to the right of the steady state, and the drift to its left, with the two-term expansions of the theorem and with the square-root law, the constants computed from the primitives, $\beta = 0.172$, $\alpha = 0.54$, $\Sigma_1 = -0.203$, and $Q_1 = 0.425$. The leading terms alone describe the solution only well inside the first grid step, because the second term is large: it is half the first already at $x = 0.1$, mostly through Π_0, since the first default is large here. With two terms the expansion matches the computed solution to within a few percent, except within a few grid steps of the steady state, where the comparison is limited by the grid's placement of the steady state, which the law itself says is off by $O(h)$. What the grid shows as a drop in consumption at the threshold is, to leading order, $\beta h^{1/3}$ for a grid step h.

The pessimistic approach of Proposition A.10 is computed by integrating (16) downward from the two-term expansion of Theorem 2 at $B^* - 10^{-6}$, in the form obtained by differentiating the HJB equation along the branch,

$$-\rho q = \mathcal{R}_B^* + \mathcal{R}_q^* q' + Sq\, c'/\psi(c), \qquad q' = \frac{\xi(d)q - (1-d)(\lambda+\delta)}{S}, \qquad S' = \frac{c' - \mathcal{R}_B^* - \mathcal{R}_q^* q' - q'S}{q},$$

with $\psi = u'/|u''|$; on the right of B^* the same system reproduces the time-stepped solution to grid accuracy. At the calibrated parameters the default stretch is $(0.482, 0.485)$, 0.26 percent of

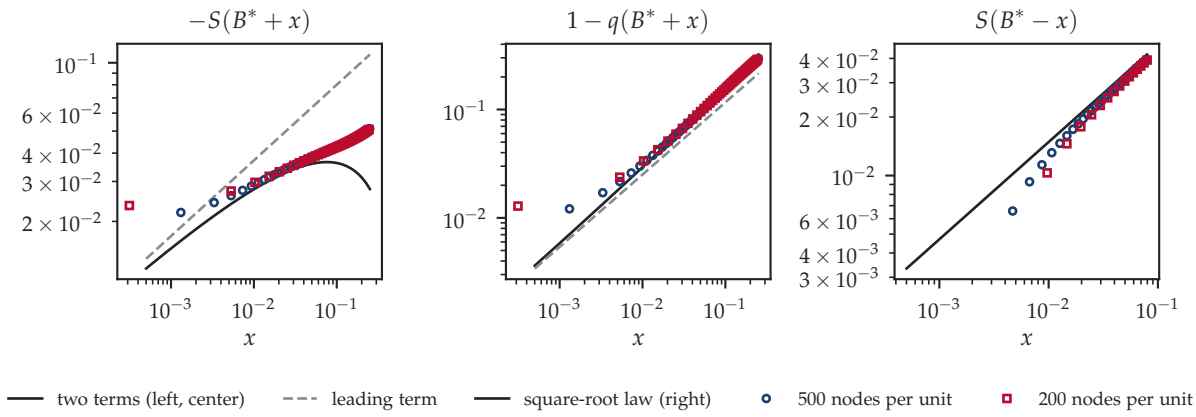


Figure 12: The cube-root law at the steady state. Markers: the computed drift and price gap at a distance x above the steady state (left and center) and the drift at a distance x below it (right), on two grids. Lines: the two-term expansions of Theorem 2 (solid) and their leading terms (dashed), with constants computed from the primitives. In the right panel, the line is the square-root law.

output, and takes 0.15 years; the price is lowest at its start, 0.991, a spread of 18 basis points; the drift stays positive down to zero debt, and the sovereign reaches capacity in 8.2 years instead of 9.8. On a grid of 4,251 nodes on $[0, 0.85]$ the pessimistic and the basin candidates have the same residuals under the scheme of this appendix, 5×10^{-6} in (22) for the value and 3×10^{-4} for the price away from B^*, and the policy kernel chooses partial default on the thirteen nodes of the stretch. Started from the pessimistic candidate, the time stepping converges to the basin equilibrium. The value at zero debt is lower in the pessimistic approach by 0.02 percent of consumption.

Comparative statics. Table 3 re-solves the deterministic model across variations of the default cost, the recovery, the discount rate, and the maturity around the calibrated point, on 200 nodes per unit of debt on $[0, 1.2]$, and compares the closed forms of Theorem 1 with the computed solution. The computed steady state and first default agree with the closed forms to within one or two grid steps in every case. The steady state rises with γ_0, κ, and γ_2 and falls with δ, as the closed form says, and it does not move with ρ, which changes only the price schedule beyond the steady state.

B.3 Diffusion Model

The diffusion model uses the same time stepping on the two-dimensional grid, with the z-operator discretized as above and Neumann conditions at $\pm\bar{z}$. We minimize the calibration loss by Nelder–Mead on a 25-node endowment grid, each evaluation a warm-started solve of 3,000 implicit steps of $\Delta t = 0.015$ that takes about a minute, and we report the best point at which the standard binds on less than one percent of the ergodic mass, re-solved on the 75-node grid, over which the moments are stable to the third digit. At the reported point the HJB residual is

Table 3: Comparative statics of the deterministic steady state

	Steady state		First default			Standard	Full default
Variation	$B_{\text{pd}}(0, q^{\text{rf}})$	computed	d_{first}	computed	$B_{\text{fd}}(0, q^{\text{rf}})$	binds from	from
Calibrated point	0.485	0.475	0.517	0.519	1.90	0.890	–
$\gamma_0 \times 0.8$	0.451	0.440	0.556	0.557	1.52	0.785	1.11
$\gamma_0 \times 1.2$	0.514	0.505	0.487	0.489	2.28	0.990	–
$\kappa = 0.45$	0.553	0.540	0.517	0.518	2.16	1.005	–
$\kappa = 0.65$	0.869	0.850	0.517	0.517	3.40	–	–
$\gamma_2 \times 0.5$	0.305	0.295	0.411	0.412	1.92	0.710	–
$\gamma_2 \times 1.5$	0.635	0.625	0.592	0.594	1.88	1.040	–
$\rho = 0.05$	0.485	0.475	0.517	0.519	1.90	0.940	–
$\rho = 0.07$	0.485	0.475	0.517	0.519	1.90	0.860	–
$\delta = 0.12$	0.584	0.575	0.517	0.519	2.28	1.020	–
$\delta = 0.22$	0.358	0.350	0.517	0.520	1.40	0.705	1.03

Notes: One parameter changed at a time from the calibrated point (Table 1), the endowment at its mean, $\underline{q} = 0.6$. "Computed" steady state: the last grid node with positive drift; "computed" first default: the default intensity at the first node with $d > 0$. "–": full default is not reached on the grid.

at most 9×10^{-7} and the pricing residual at most 10^{-3}, attained next to the zero-drift locus, and 2×10^{-5} elsewhere.[4] The impulse responses of Section 4.3 are expectations under the forward equation $\dot{g} = A^{\top} g$, started from the ergodic distribution pushed down in z by the shock (split linearly between neighboring nodes) and advanced with a Crank–Nicolson step of 0.01 years.

Moments. The model's moments follow the annual accounting of Arellano, Mateos-Planas, and Ríos-Rull (2023). Debt due is $(\delta + \lambda)B$ per year, debt service is $(1 - d)$ times debt due, defaulted coupons are d times debt due, and the annual partial default is defaulted coupons over debt due within the year, as in their equation (4). The debt level is B, its face value, which equals its risk-free value since $q^{\text{rf}} = 1$. Output and consumption are annual integrals, logged and linearly detrended. A year is in partial default if any coupon was defaulted on during it. At the calibrated point the instantaneous probability of being in default under the stationary distribution is 0.27, against an annual frequency of 0.38, a difference due to time aggregation. We compute moments that are functions of the state (debt, debt service, and debt due to output, the spread and its correlations) exactly from the stationary distribution of the discretized process, by solving the adjoint system $A^{\top} g = 0$. We compute moments that are path functionals under annual accounting (the partial default frequency, mean and standard deviation, the persistence and volatility of output, and consumption) from sixteen simulated paths of 25,000 years each, by Euler–Maruyama

4. These maxima exclude the nodes on the floor locus, where the price is within 0.02 of $\underline{q}$ or crosses it between neighboring debt levels. These nodes hold 3.8 percent of the ergodic mass, of which the standard binds on just under one percent. There the standard switches issuance on and off, which makes the discrete policy discontinuous in the price on the locus $q = \underline{q}$, a locus that no grid node hits exactly. The drift sign alternates between iterations, and the residuals stall at 3×10^{-3} for the HJB equation and 2×10^{-2} for the pricing equation. Phasing in issuance continuously over a narrow band of prices around $\underline{q}$ could reduce these errors.

steps with bilinearly interpolated policies, 100 steps per year, annual aggregation of flows, and reflecting boundaries for z. The same paths give the untargeted moments conditional on default.

A resolution artifact. The scheme obtains V_B by differencing, with an error of about $\frac{1}{2}|V_{BB}|\Delta B$, and the drift responds to that error with a gain of $1/(q^2|u''(c)|)$: consumption responds to the ratio V_B/q and the drift is the consumption gap divided by q again. Where the *resolution ratio*

$$\frac{\frac{1}{2}|V_{BB}|\,\Delta B}{q^2\,|u''(c)|\,|S|} \tag{23}$$

exceeds about one half, the scheme cannot determine the sign of the drift, and when the scheme loses the sign it tends to produce a positive drift, which carries the state to the top of the grid. The remedy is a grid whose spacing is proportional to q^2. We report with every solution the ergodic mass on unresolved nodes, those where the ratio exceeds one half, excluding stationary nodes ($|S| \leq 10^{-3}$).

B.4 Neural-Network Solution

Representation. We use one network with inputs $(2B/\bar{B} - 1,\ z/\bar{z})$, 3 hidden layers of 128 units with the hyperbolic tangent activation, and two outputs, $V = \bar{V} + s_V N_1$ and $q = \sigma(N_2)$ with σ the logistic function, so that the price is in $(0, 1)$ by construction and the value in a range set by the initial guess; 33,666 parameters in all. Given V_B and q at a point, the policies (d, S, c) are the maximizers of the Hamiltonian by the kernel of Appendix B: the drift from the first-order condition clamped to the admissible interval, including the standard's restriction $S \leq -a(d)B$ where $q < \underline{q}$ and the state constraints at $B = 0$ and $B = \bar{B}$, and the default intensity by golden-section search with the candidates $d = 0$, $d = 1$, and (8). The kernel is that of the finite-difference code, vectorized. At every grid node it reproduces the finite-difference policies to machine precision.

Training. We train the network by the implicit pseudo-time iteration of the grid scheme. At outer step k we compute the policies (d_k, S_k, c_k) from the current network, freeze them, and retrain the network, warm-started, to solve

$$V - \Delta t[u(c_k) + S_k V_B + \mathcal{L}_z V - \rho V] = V_k, \qquad q - \Delta t[(1 - d_k)(\lambda + \delta) + S_k q_B + \mathcal{L}_z q - \xi(d_k)q] = q_k,$$

in least squares, with V_k, q_k the previous network, over Sobol points of the state space drawn afresh at every step, 16,384 of them for the first 2,500 steps and 32,768 for the last 2,000, plus points on $B = 0$ and $B = \bar{B}$ where the state constraints enter the policies, and a penalty on $V_z^2 + q_z^2$ at $z = \pm\bar{z}$ for the reflecting boundaries. A quarter of the interior points are drawn instead within about 0.04 of the first-default locus of the current network, which is recomputed every 25 steps. Each outer step takes 50 Adam iterations, and the learning rate falls from 10^{-3} to

3×10^{-5} over the run. The pseudo time step grows from 0.02 by half every 100 steps to 0.20. In the last 1,125 of the 4,500 steps the collocation points are kept fixed, the policies are relaxed by one half between steps, and each step ends with a few iterations of L-BFGS. The iteration starts from the static initial guess of the grid scheme, fitted to the network by regression. We compute in double precision throughout.

The stationary residual. The usual loss of a physics-informed network, the mean squared residual of the stationary equations at random points, has minimizers that are not the equilibrium. In our experiments it converged, from a random initialization and from the grid solution itself, to smooth functions with average residuals smaller than those of the grid solution evaluated by automatic differentiation, which are dominated by the kinks, but with the wrong drift over large parts of the state space. Artificial viscosity and one-sided differences of the network did not restore the selection that the time stepping provides. The pseudo-time iteration, like the grid scheme, follows the time-dependent problem and selects its limit.

Accuracy. Figure 13 compares the two solutions at two endowment levels, and Table 4 compares them on the grid of the main solution. The errors are differences from the grid solution, and the moments come from each solution's policies by the procedure of Section 4.2. The network's value differs from the grid solution by 0.003 in root mean square, 10^{-3} of its range, or 0.02 percent of consumption; the price by 0.002 (at most 0.006). The first-default thresholds differ by 0.001 units of debt on average and 0.005 at most. The sign of the drift agrees on 98.2 percent of the grid, and on 98.6 percent of the nodes where the grid drift exceeds 10^{-3} in absolute value; the default region on 99.9 percent. The stationary residuals of the network, evaluated by automatic differentiation on the grid, are 2×10^{-3} for the HJB equation and 2×10^{-3} for the pricing equation in root mean square, against 4×10^{-5} and 2×10^{-4} for the discrete residuals of the grid solution. The largest HJB residuals sit at the extreme endowments, where the reflecting boundaries enter only through a penalty: the three outermost endowment nodes on each side, 8 percent of the grid, carry 49 percent of the squared residual. The pricing residuals concentrate at the kinks of the price, on the floor locus and next to the stationary debt, which together hold 12 percent of the grid and 45 percent of the squared residual, and at the default thresholds. Of the nodes where the drift has the wrong sign, 52 percent are within three grid steps of the stationary debt, where the grid drift itself is small, 0.003 in absolute value at the median.

The slope of the value. The ergodic distribution depends on the slope V_B more than on the level of the value. At low debt the drift is a few hundredths, and it responds to the slope with the gain of (23). The network's slope at debt below 0.3 is within 1 percent of the grid's in root mean square. A smaller network (three layers of 64 units, 2,048 points, 700 steps) came as close in the level of the value, within 0.3 percent of consumption, but its slope was off by about 13 percent there. Its drift had the wrong sign at low debt in the middle endowment states, so the sovereign

repaid instead of borrowing, and its ergodic mean debt was 0.045. With uniform collocation points, the same network, and the first 2,500 steps only, the slope error at low debt is 4 percent, mean debt is 0.255, and the calibration loss 0.129; the solution misses most of the atom at zero debt, 0.007 against 0.053.

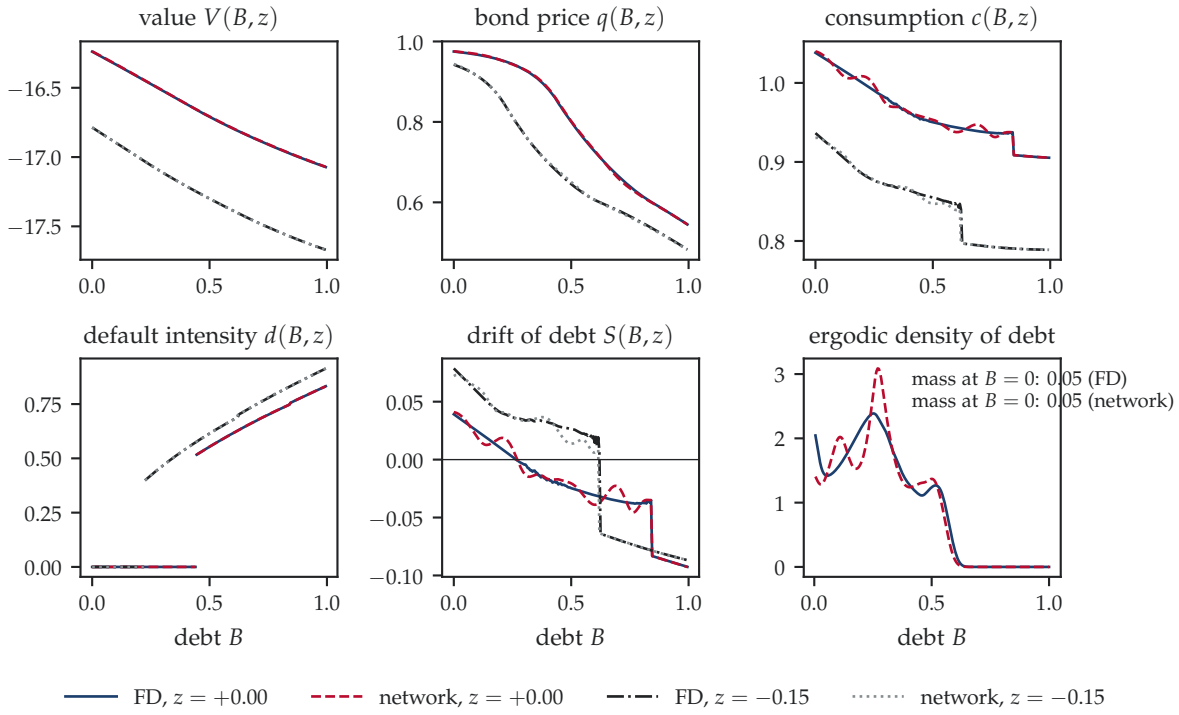


Figure 13: The diffusion model solved on a grid (FD) and on a neural network: value, bond price, consumption, default intensity, and drift of debt as functions of debt at two endowment levels, and the ergodic marginal density of debt under each solution's policies (the atoms at $B = 0$ are reported in the panel).

B.5 Self-Fulfilling Partial Default with Curvature and Shocks

Risk aversion. With CRRA utility the sovereign cannot jump, and it borrows into default gradually. Above a candidate steady state the price follows the solutions of Proposition A.9(a). From below, the state arrives along a square-root law with a nearly flat price, and away from the candidates risk-aversion slows the borrowing. We find the candidates reached from zero debt by integrating (16) forward from zero debt, over a grid of starting prices and consumption levels. Figure 5 shows a numerical example with $\nu = 2$ and a wedge of 210 basis points, computed by reverse shooting (Appendix B.2). Next to the no-default steady state at $B = 0.7$ there is a self-fulfilling partial-default steady state at $B = 1$, at which the sovereign withholds 28 percent of its debt service at a price of 0.9, a spread of 183 basis points. Facing those prices, the sovereign borrows from zero debt all the way into partial default. Default starts at $B = 0.62$, lower than under the optimistic price, because the threshold rises with the price. This steady state is one of

Table 4: Grid and neural-network solutions of the diffusion model

	Finite differences	Neural network
Moments		
Debt to output, mean	0.273	0.272
Debt to output, s.d.	0.189	0.183
Spread, s.d.	0.035	0.034
Partial default frequency	0.378	0.377
Partial default, mean	0.409	0.403
Spread, mean	0.029	0.028
Calibration loss	0.096	0.102
Mass at zero debt	0.053	0.052
Differences from the grid solution		
Value V, rms / max	–	0.003 / 0.008
Bond price q, rms / max	–	0.002 / 0.006
Drift S, rms	–	0.006
Default intensity d, rms	–	0.012
First-default threshold, mean / max	–	0.001 / 0.005
Drift sign agreement (percent)	–	98.2
Residuals and cost		
HJB residual, rms	4×10^{-5}	2×10^{-3}
Pricing residual, rms	2×10^{-4}	2×10^{-3}
Unknowns	15,075 nodes	33,666 parameters

Notes: Both solutions at the calibrated parameters with the standard at $\bar{q} = 0.6$. Moments by the procedure of Section 4.2 with each solution's policies on the 201×75 grid; the grid column recomputes the moments of Table 2 with the same seeds. Differences are over the grid nodes. Residuals: the discrete equations (22) for the grid solution; the continuous equations (18), with derivatives by automatic differentiation, for the network.

a continuum: at this wedge the steady states reached from zero debt fill the interval from 0.93 to 1.02. The left panel of Figure 14 shows the smallest wedge at which any is reached, as a function of the curvature. It is about 139 basis points for $\nu = 2$, against 47 for a risk-neutral sovereign. A sovereign that values smooth consumption gains less from front-loading, so it needs more impatience to borrow into default.

In every case we computed, a self-fulfilling partial-default steady state is reached from zero debt with curvature only above a threshold that exceeds the risk-neutral threshold Θ^* and rises with the curvature of utility. The threshold approaches Θ^* as the curvature vanishes, but slowly, so even a little curvature raises it markedly.

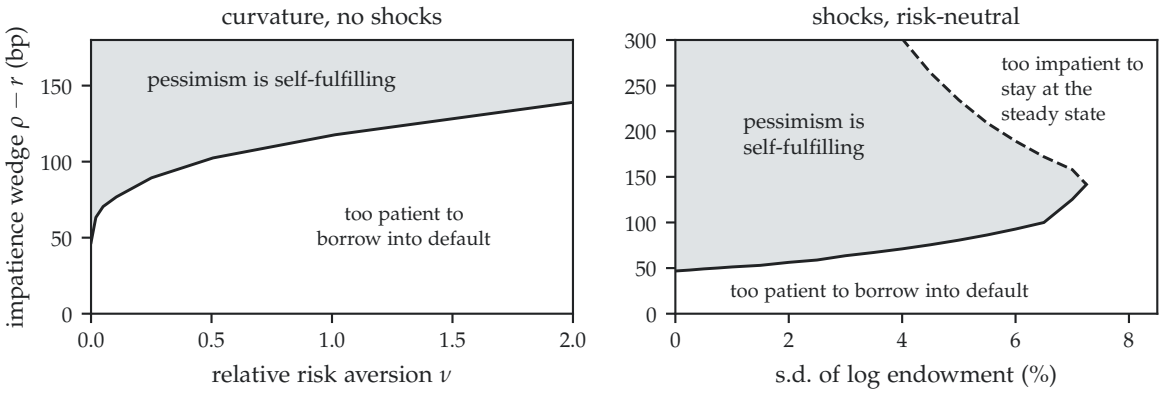


Figure 14: Where pessimism is self-fulfilling. Shaded: impatience wedges $\rho - r$ at which a self-fulfilling partial-default equilibrium exists, at the parameters of Figure 5. Both solid lines start at the risk-neutral threshold $s_{\text{first}} = 47$ basis points. The panels have different vertical scales. Left: the deterministic model with CRRA utility, against the relative risk aversion ν. The upper boundary is not computed. Right: the risk-neutral diffusion model with mean reversion $\mu = 0.225$, against the standard deviation of the log endowment. Above the dashed line the steady state is no longer stable.

Uncertainty. Shocks also work against self-fulfilling partial default. We study them with a risk-neutral sovereign and extend the construction of Proposition A.9 to the diffusion model, as described below. Lenders expect the sovereign to settle at a debt B° that does not depend on the endowment, and to stay there. The default intensity and the price at B° then vary with the endowment. A sovereign at zero debt makes a discrete issuance of B° when that beats staying debt-free, and above B° it is indifferent, as in Proposition A.9(a). The right panel of Figure 14 shows where such equilibria exist, at the parameters of Figure 5. Volatility narrows the range of supporting wedges from both sides. From below, the discrete issuance must beat staying debt-free in every endowment state, and in bad states the pessimistic steady state carries a deeper and costlier default. From above, the steady state must be stable in every state, and the dilution that holds the sovereign there is weaker in some states. At a standard deviation of the log endowment of about 7 percent the two boundaries meet, and no equilibrium of this kind remains.

The construction. We compute the right panel of Figure 14 with the following construction. The sovereign is risk-neutral, $u(c) = c$, and issuance is unrestricted in speed, as in Proposition A.9. The log endowment follows the diffusion of Section 4.1. The construction gives sufficient conditions for a self-fulfilling partial default, and we find the region numerically.

The steady debt. At B° the drift is zero and the default intensity is the static choice $d^\circ(z) = d^*(B^\circ, z, q^\circ(z))$. The price solves an ordinary differential equation in z,

$$\xi(d^\circ)\, q^\circ = (1 - d^\circ)(\lambda + \delta) + \mathcal{L}_z q^\circ,$$

and the value solves $\rho V^\circ = \mathcal{R}^*(B^\circ, z, q^\circ) + \mathcal{L}_z V^\circ$. With $\sigma = 0$ these reduce to $q^\circ = q_r(d^\circ)$ and $\rho V^\circ = \mathcal{R}^*$. We index the steady debts by the intensity d of the deterministic candidate, $B^\circ = B(d)$ of Proposition A.9, and follow the root of the price equation from $q_r(d)$.

Above the steady debt. On an interval through which the sovereign moves at a finite speed it is indifferent, $V_B = -q$. Differentiating $\rho V = \mathcal{R}^*(B, z, q) + \mathcal{L}_z V$ in B, with the envelope theorem, gives the price equation of Proposition A.9(a) with a diffusion term,

$$a(d) B\, q_B = (1 - d)(\lambda + \delta) - (\rho + a(d))\, q + \mathcal{L}_z q, \qquad d = d^*(B, z, q).$$

The equation is parabolic in B, so we integrate it upward from B°. The drift follows from the pricing equation, $S\, q_B = \xi(d) q - (1 - d)(\lambda + \delta) - \mathcal{L}_z q$. The steady debt is stable from above if $S_B < 0$ at $B^{\circ +}$ in every state. As in Lemma A.2, this slope follows from the two equations at B°. The continuation condition is that $S < 0$ and $q_B < 0$ up to the full-default region.

Below the steady debt. A sovereign below B° makes a discrete issuance that takes it to B° at the price $q^\circ(z)$. So $q(B, z) = q^\circ(z)$ and $V(B, z) = V^\circ(z) + q^\circ(z)(B^\circ - B)$. Standing still must not be better. As in the proof of Proposition A.9(c), the gap $\rho V - \mathcal{R}^* - \mathcal{L}_z V$ is concave in B and vanishes at B°. The condition is therefore that the gap be nonnegative at zero debt in every state,

$$\mathcal{R}^*(B^\circ, z, q^\circ) + \rho\, q^\circ B^\circ - e^z - B^\circ\, \mathcal{L}_z q^\circ \geq 0.$$

With $\sigma = 0$ this is $\rho - r \geq \Theta(d)$. Finally, issuance at B° requires $q^\circ \geq \underline{q}$ in every state.

Waiting and computation. A sovereign at zero debt could also wait for a better endowment before it makes the discrete issuance. We allow this by solving the corresponding optimal-stopping problem at zero debt and checking that the sovereign at B° does not prefer to buy back its debt and wait; waiting does not change the boundaries of the region. The endowment grid has 201 nodes on ± 3 stationary standard deviations with reflecting boundaries. We solve the price equation at B° by Newton's method, or by the monotone map $q \mapsto [\xi(d^*(q)) - \mathcal{L}_z]^{-1}(1 - d^*(q))(\lambda + \delta)$ where Newton's method stalls at the jump of d^*, and we integrate above B° in debt,

implicitly in $\mathcal{L}_z$. The region does not change when the grid is refined from 101 to 801 nodes, and with $\sigma = 0$ the construction reproduces the conditions of Proposition A.9. Its lower boundary depends on how far the endowment grid extends, because the condition on the discrete issuance binds in the worst states.

C Robustness

C.1 The Standard and the Grid

Without the standard. We solve the model at the calibrated parameters of Table 1 without the standard, on a debt grid extending to $\bar{B} = 2$. The sovereign then issues in full default in the lowest endowment states at prices up to 0.29. The ergodic mass there, however, is about 2×10^{-6}. The moments agree with those of the solution without the standard on $[0, 1]$ to within a few thousandths. At these parameters the standard lowers the spread volatility from 0.042 to 0.035 and raises the loss from 0.092 to 0.096. With the standard in place it binds in partial default on just under one percent of the ergodic mass, where the default intensity exceeds the market-price rule by two percent of that rule on average and by up to four percent.

Robustness of the calibrated point under the standard. Table 5 re-solves the calibrated point from a warm start on a finer debt grid, on a non-uniform grid of the kind of Appendix B.3, and with the barrier at $B = 1.5$. With the standard in place the moments move by amounts comparable to the sampling noise of the simulated moments, the top of the grid carries no mass, and the loss is unchanged to the second digit.

Table 5: The calibrated point under the standard on other grids

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Mass top	Unresolved
main grid	0.273	0.189	0.035	0.378	0.409	0.096	0.0000	0.015
debt step halved	0.272	0.188	0.035	0.376	0.410	0.096	0.0000	0.007
non-uniform debt grid	0.273	0.188	0.035	0.375	0.410	0.096	0.0000	0.005
barrier at $B = 1.5$	0.273	0.189	0.035	0.378	0.409	0.096	0.0000	0.015

Notes: Each row re-solves the calibrated point from a warm start. Main grid: 201 debt nodes (step 0.005) on $[0, 1]$ and 75 endowment nodes; debt step halved: 401 debt nodes; non-uniform: 572 debt nodes with steps from 0.0007 to 0.004, proportional to the square of the lowest price at each debt level; all rows use 75 endowment nodes. Mass top: ergodic mass in the top tenth of the debt grid; unresolved: ergodic mass on nodes where the resolution ratio (23) exceeds one half, excluding stationary nodes ($|S| \leq 10^{-3}$).

C.2 Saving

As in Arellano, Mateos-Planas, and Ríos-Rull (2023), the main text restricts the sovereign to borrowing, $B \geq 0$. Here we relax the restriction at the calibrated parameters of Table 1, without recalibrating, as a check on the model's behavior. Below zero debt the sovereign holds a risk-free

asset at the price q^{rf} and has nothing to default on. When its debt crosses zero, it retires its outstanding bonds, or issues new ones, at the market price, so bonds trade only at $B \geq 0$ and the bond price has the same form as without saving. We solve on 401 debt nodes on $[-1, 1]$, with the step of the main grid and 75 endowment nodes, warm-started from the main solution; away from the floor locus the value residual is below 2×10^{-5}, and the bottom of the grid carries about 0.01 percent of the ergodic mass. The moments use gross debt, $\max\{B, 0\}$, and in states with assets the spread is the one at which the sovereign would issue at zero debt, as at the atom at zero debt in the main solution.

Figure 15 compares the ergodic density with and without saving. The difference is confined to the left of the distribution. The atom at zero debt, which holds 5 percent of the ergodic mass without saving, becomes a ridge of assets in good times: above about 0.7 stationary standard deviations of the endowment the stationary debt is negative, and 19 percent of the mass holds assets, about a fifth of output on average. On positive debt the density keeps its stationary debt, the bend where the price reaches the floor, its mode, and its tail into default. Table 6 reports the moments: saving does not change the depth and dispersion of partial default, the defaulted coupons, the volatility of the spread, or the output process. Partial default is somewhat less frequent and debt somewhat lower on average, because time that the sovereign spends at zero debt without saving is spent holding assets with it, and consumption is a little smoother relative to output. The mechanisms of Sections 3 and 4 operate on positive debt and are not affected.

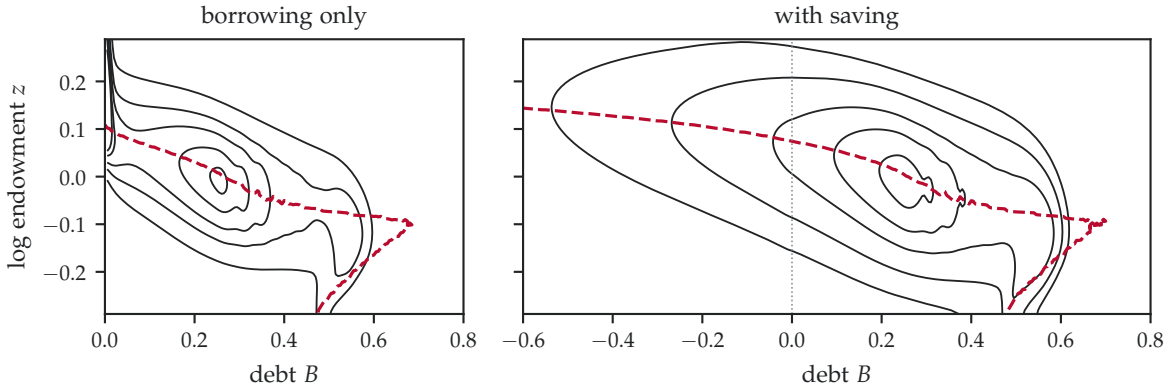


Figure 15: The ergodic density of debt and the endowment at the calibrated parameters, when the sovereign only borrows (left, as in Figure 7) and when it can also save (right). Dashed: the zero-drift locus. The panel widths are proportional to the range of debt they show.

Table 6: Moments when the sovereign only borrows and when it can also save

Moment	Borrowing only	With saving
Partial default: frequency	0.38	0.35
Partial default: mean $\mid d > 0$	0.41	0.41
Partial default: s.d. $\mid d > 0$	0.22	0.22
Debt to output: mean	0.27	0.24
Debt to output: s.d.	0.19	0.20
Debt service to output: mean	0.037	0.032
Debt service to output: s.d.	0.020	0.022
Debt due to output: mean	0.052	0.046
Spread: s.d.	0.035	0.034
Output: persistence	0.88	0.88
Output: s.d.	0.11	0.11
Defaulted coupons to output: mean $\mid d > 0$	0.041	0.041
Defaulted coupons to output: s.d. $\mid d > 0$	0.028	0.028
Spread: mean	0.029	0.027
Spread: correlation with output	-0.82	-0.79
Spread: correlation with debt	0.88	0.87
Consumption s.d. relative to output s.d.	0.87	0.85

Notes: The calibrated parameters of Table 1, not recalibrated. The moments above the line are the targets of Table 2, and those below it are its untargeted moments, computed in the same way.